Supplementary Material for: Boundary and Interface Classes for Universal Chern Hamiltonians via Doubling and Jamming

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November 8, 2026

Abstract

This document collects the secondary results, full appendix proofs, all robustness variants, hexagonal and square benchmark symbols, the symbolic verification appendix, and the full numerical record for the main paper "Boundary and Interface Classes for Universal Chern Hamiltonians via Doubling and Jamming". The main results, abstract, introduction, and headline theorems are stated in the main paper; cross-references to the main paper use the `M-` prefix. The structure of the supplement is: full proofs of the finite-M Schur-complement estimates (Section 1); the sign ledger (Section 6.1); two-band hexagonal and square benchmark symbols and their proofs (Section 6.3); the robustness variants for finite defects, finite corners, tail-split and annular families (Sections 6.5 to 6.7); the Section 5 variants (Section 6.4); the Roe-algebra interface construction variants (Sections 5 and 6.8); the symbolic verification appendix (Section 6.9); the detailed appendix proofs of every body result (Sections 6 and 6.111); the full numerical experiments (Section 7); and the reproducibility appendix (Sections 7.1 and 7.2). This supplement and the companion code are distributed through the project GitHub repository (https://github.com/yspennstate/Doubling-and-Jamming-A-Universal-Chern-Hamiltonian-on-Triangulated-Surfaces-with-Boundary).

1 Full proofs of finite-dimensional Schur-complement estimates

This section collects the full proofs of the finite-dimensional Schur-complement estimates whose statements are recorded in the main paper, Section 3. The proofs themselves are housed in the detailed-proofs section, Sections 6 and 6.24, and are reached through the cross-references below. Each result of the main paper has a dedicated proof subsection in Section 6 matching the appendix-label convention used throughout the supplement.

- Schur-complement resolvent formula (Theorem 3.1): Section 6.24.
- Compressed resolvent convergence (Theorem 3.2): Section 6.25.
- Boundary Hamiltonian as the large-jam limit (Theorem 3.3): Section 6.26.
- Degenerate full block-resolvent limit (Theorem 3.4): Section 6.27.

- First-order Schur expansion (Theorem 3.5): Section 6.29.
- General contour-projection comparison (Theorem 3.6): Section 6.30.
- Bounded-energy spectral localization (Theorem 3.7): Section 6.31; comparison with the relative boundary Hamiltonian (Theorem 3.8): Section 6.32.
- First-order eigenvalue localization (Theorem 3.9): Section 6.33.
- Copy-side leakage bound (Theorem 3.10): Section 6.34.
- Inertia, monotonicity, and no-copy-state bounds (Theorems 3.11 to 3.13): Sections 6.35 to 6.37.
- Inherited gap and direct-sum jam estimates (Theorems 3.14 and 3.15): Sections 6.38 and 6.39.
- Internal bulk spectral equivalence (Theorem 3.16): Section 6.40.
- Routine Schur-complement corollaries: Section 6.28.

The proofs themselves now follow the appendix layout used throughout the rest of the supplement.

2 Setup remarks

The fullness hypothesis on $\partial\mathcal{K}$ used throughout the main paper rules out two doubling collisions:

(i) *Face collision.* If a face $\{v_0, v_1, v_2\}$ has all three vertices on $\partial\mathcal{K}$, its reflected copy has the same vertex set, so $D(\mathcal{K})$ is not an abstract simplicial complex.

(ii) *Edge collision and valence failure.* If an interior edge $[v_0, v_1]$ has both endpoints on $\partial\mathcal{K}$, the reflected copy has the same vertex set and cannot be distinguished from the original, and each face incident to $[v_0, v_1]$ in $\mathcal{K}$ produces a reflected face also incident to $[v_0, v_1]$, so the edge ends up incident to four or more faces in $D(\mathcal{K})$. The simplex-count formula $|D(\mathcal{K})_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|$ also fails.

The fullness condition rules both out and holds automatically after one barycentric subdivision. All refinement families used in the main paper satisfy the condition.

We also record here the closed-case cap-product convention used in Section 2. For the fundamental cycle $|X\rangle = \sum_{\tau\in\mathcal{K}_2} |\tau\rangle$, and the dual cochains δ_ξ, the convention is $\delta_\xi \frown [v_0, \dots, v_2] = \delta_\xi([v_{2-p}, \dots, v_2])\,[v_0, \dots, v_{2-p}]$, which gives the Higson–Prodan operator P. The relation $BS = -SB^\dagger$ does *not* by itself imply $(B + B^\dagger)S + S(B + B^\dagger) = 0$; this is why the body anticommutation statements appear either as hypotheses or as measured finite-matrix defects.

Corollary 2.1 (Dimension of the doubled simplicial Hilbert space)**.** $\dim \mathcal{H}(D(\mathcal{K})) = 2\dim \mathcal{H}(\mathcal{K}) - 2|\partial\mathcal{K}_0|$.

Proof. See Section 6.14.

Proposition 2.2 (Euler characteristic)**.** $\chi(D(\mathcal{K})) = 2\chi(\mathcal{K})$.

Proof. See Section 6.15.

Corollary 2.3 (Genus formula)**.** *Let $\mathcal{K}$ be connected, orientable, genus g, with $b \geq 1$ boundary components. Then $g_{\text{double}} = 2g + b - 1$.*

Proof. See Section 6.16.

Corollary 2.4 (Quotient equality of the relative and doubled boundary models)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular refinement family. With $H_\pm^{\mathrm{PL},\infty} = \bigoplus_n H_\pm^{\mathrm{PL}}(\mathcal{K}_n)$ and $H_\pm^{\partial,\infty} = \bigoplus_n H_\pm^{\partial}(\mathcal{K}_n)$ on the original-side Hilbert space, $H_\pm^{\partial,\infty} - H_\pm^{\mathrm{PL},\infty} \in C^*(Y \subset X)$, and the relative duality defect $\bigoplus_n (B_{\mathcal{K}_n} S_{\mathcal{K}_n}^{\mathrm{PL}} + S_{\mathcal{K}_n}^{\mathrm{PL}} B_{\mathcal{K}_n}^\dagger)$ belongs to $C^*(Y \subset X)$. Continuous functional calculus passes both models to the same image in $\mathcal{A}/\mathcal{I}$.*

Proof. See Section 6.20.

Corollary 2.5 (Asymptotic bulk spectral equivalence)**.** *With $d_n = \dim \mathcal{H}(\mathcal{K}_n)$ and $r_n = \dim \ell^2(N_{R_{\mathrm{PL}}}(\partial\mathcal{K}_n))$, the spectral distribution functions of $H_\pm^{\mathrm{PL}}(\mathcal{K}_n)$ and $H_\pm^{\partial}(\mathcal{K}_n)$ satisfy $\sup_t |d_n^{-1} N_{\pm,n}^{\partial}(t) - d_n^{-1} N_{\pm,n}^{\mathrm{PL}}(t)| \le r_n/d_n$. For families with $r_n/d_n \to 0$ the normalized spectral distribution functions converge to the same limits.*

Proof. See Section 6.23.

3 Restatements of main-paper body results referenced in the appendix proofs

The appendix proofs below refer back to the body lemma/theorem statements of the main paper. For convenience and to make the supplement self-contained, the body statements whose *statements* were compressed in the main paper (kept as informal restatements rather than as labelled environments) are repeated here with their original labels.

Definition 3.1 (Propagation / locality radius)**.** A bounded operator T on $\mathcal{H}(\mathcal{K}) = \ell^2(X(\mathcal{K}))$ has *propagation at most R* if $\langle \sigma, T\tau \rangle = 0$ whenever $\mathrm{dist}_{X(\mathcal{K})}(\sigma, \tau) > R$. We also call such an operator R-local.

Section 2 body labels (closed-case and orientation conventions)

The labels 3 and 3 are used by the appendix to refer to the closed-case cap-product definition and the orientation/sign conventions; in this supplement they point at this restatement subsection. The conventions themselves are stated in Section 2 of the main paper.

Equation-label aliases used by the appendix proofs

The label keys 3, 3, 3, 3, 3, 3, 3 used by the supplement's appendix proofs are defined here as section-local aliases. The equations they correspond to are the explicit resolvent and leakage bounds collected in Theorems 3.2 and 3.10 and in the relative-boundary comparison Theorem 2.8; the aliases let the supplement's appendix proofs cite them with the original label keys without xr round-trips.

Numerical-table aliases

The certificate tables (Tables 1 and 2) are produced by the companion script and appear in this supplement as Section 7; the body refers to them through these label aliases.

Table 1: Periodic triangular Bloch certificate (FHS check; reproduced from the companion script).

Hamiltonian	Grid	FHS Chern	min $\lvert E\rvert$	min link	max plaquette
H_+	11×11	-1.000000	0.550897	0.905137	0.265005
H_+	21×21	-1.000000	0.540655	0.970926	0.080163
H_+	31×31	-1.000000	0.540954	0.986311	0.037594
H_+	51×51	-1.000000	0.540291	0.994870	0.014056
H_-	11×11	1.000000	0.550897	0.905137	0.265005
H_-	21×21	1.000000	0.540655	0.970926	0.080163
H_-	31×31	1.000000	0.540954	0.986311	0.037594
H_-	51×51	1.000000	0.540291	0.994870	0.014056

Table 2: Periodic triangular extension checks (FHS).

Diagnostic	Model	Grid	expected	FHS	min $\lvert E\rvert$	min link
positive mass	$H_{+,m}$, $m = 0.5$	31×31	-1	-1.000000	0.287664	0.959784
positive mass	$H_{-,m}$, $m = 2$	31×31	1	1.000000	0.806343	0.993537
finite cover	$H_{+,A}$, $A = \mathrm{diag}(2, 1)$	31×31	-2	-2.000000	0.540954	0.945837
finite cover	$H_{-,A}$, $A = \mathrm{diag}(2, 1)$	31×31	2	2.000000	0.540954	0.945837
orientation-reversing cover	$H_{+,A}$, $A = \mathrm{diag}(1, -1)$	31×31	1	1.000000	0.540954	0.986311
cell enlargement	H_+, 2×1 folded cell	21×21	-1	-1.000000	0.540655	0.965480
cell enlargement	H_-, 2×1 folded cell	21×21	1	1.000000	0.540655	0.965480

4 Auxiliary Roe-algebra lemmas, stability, and cross-checks

The following auxiliary Roe-algebra lemmas, stability statements, and cross-check lemmas are referenced by the body proof of Theorem 4.9 in the main paper. Their statements live here in the supplement; the corresponding proof bodies appear in Section 6.111 below.

Lemma 4.1 (Local analytic properties of $H_\pm(K)$)**.** *For a bounded-geometry triangulation K, the operators $H_\pm(K)$ are self-adjoint and have finite hopping range in the barycentric graph metric. Along a bounded-geometry family the hopping range and operator norms are bounded uniformly.*

Proof. See Section 6.112.

Lemma 4.2 (Combes–Thomas decay for the closed-bulk Fermi projection)**.** *Assume the uniform hopping and gap constants in Theorems 4.1 and 4.4. Then there are constants $C, c > 0$, independent of n, such that $\left|\langle \delta_v, P_{\pm,n}\delta_w\rangle\right| \le Ce^{-c\,d(v,w)}$ for all barycentric vertices $v, w \in X(D(\mathcal{K}_n))$.*

Proof. See Section 6.116.

Lemma 4.3 (Boundary map for the interface projection)**.** *The map ∂_Y in Theorem 4.7 is the standard K-theory connecting map for the Roe extension. For the lift $h(J^\infty_{\pm,M})$ of p_J, $\partial_Y([p_J]) = [u_{J,h}] \in K_1(C^*(Y \subset X))$, and the class is independent of the admissible cutoff h.*

Proof. See Section 6.118.

Definition 4.4 (Ewert–Meyer pairing for the original/copy partition)**.** The Ewert–Meyer bulk–interface pairing is the coarse Mayer–Vietoris pairing of a bulk $K_0(\mathcal{A}/\mathcal{I})$-class with the interface cocycle obtained from the partition $(X^+, X^-; Y)$ [29]. Here this pairing is evaluated on the class $[p_J]$ through the connecting map ∂_Y above. The compatibility assertion is not an extra hypothesis: it is the statement that, for the doubled-jammed original/copy partition, Ewert–Meyer's coarse boundary map is represented by the same Roe extension and the same cutoff unitary $u_{J,h}$.

5 A coarse interface class construction

We place the doubled-and-jammed refinement family in a Roe-algebra construction that produces the interface class. In the headline application, the required closed-bulk gap is supplied by Higson–Prodan's closed theorem. We use the quotient extension

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{A} \xrightarrow{\pi} \mathcal{A}/\mathcal{I} \longrightarrow 0, \qquad \mathcal{A} := C^*(X), \qquad \mathcal{I} := C^*(Y \subset X),$$

and define the interface class from a canonical quotient projection associated to a sidewise realization.

Here $C^*(X)$ denotes the scalar Roe algebra: the norm closure in $\mathcal{B}(\ell^2(X))$ of locally compact finite-propagation operators. We do not use the uniform Roe algebra in this paper. In Theorem 4.9 of Section 4, the Ewert–Meyer Mayer–Vietoris boundary map for the original/copy partition is identified internally with the Roe connecting map used below. The HP–Ewert–Meyer terminology below fixes the convention for that pairing; it is not an additional input for the original/copy headline theorem.

Uniform locality, boundedness, and a closed-bulk gap give the quotient projection and the interface K_1-class. In the HP-covered case, Higson–Prodan supplies the gap and Chern value; in the periodic triangular case, Section 5 supplies them directly. The original/copy K_1-class identity is Theorem 4.9; the integer winding evaluation is internal in the Toeplitz periodic case (Theorem 4.10) and conditional in general (Theorem 4.11).

In the present scalar-coefficient setting, the Roe algebra $C^*(X)$ is *unital*: the identity operator on $\ell^2(X)$ has propagation 0 and is locally compact (since every finite-support compression is finite rank), hence belongs to $C^*(X)$. Similarly, every diagonal $\{0,1\}$-valued multiplication operator on $\ell^2(X)$ belongs to $C^*(X)$. The quotient $\mathcal{A}/\mathcal{I}$ is therefore also unital, and the continuous functional calculus in both algebras is the standard unital one.

This convention is the scalar, ample-representation version of the Roe algebra. Ewert–Meyer's Mayer–Vietoris boundary map is used below only through the induced connecting map for the coarsely excisive pair (X^+, X^-) and its compatible pairing with the closed bulk Chern class. Passing between this unital scalar presentation and stabilized or nonunital presentations changes the bookkeeping, not the connecting class; Theorem 4.9 identifies the connecting class used here with the Ewert–Meyer convention for the original/copy partition.

Remark 5.1 (Notation and conventions). Throughout this section, whenever T is a bounded self-adjoint operator on $\ell^2(X)$, we write

$$\operatorname{spec}_{\mathrm{op}}(T) := \operatorname{spec}_{\mathcal{B}(\ell^2(X))}(T)$$

for its operator spectrum. Because $C^*(X) \subset \mathcal{B}(\ell^2(X))$ is a unital inclusion, the $C^*(X)$-spectrum and the operator spectrum of any $T \in C^*(X)$ coincide. All spectral-gap statements for concrete operators refer to $\operatorname{spec}_{\mathrm{op}}$.

We define the canonical quotient projection via *continuous* cutoff functions that agree with $\chi_{(-\infty,0)}$ on the relevant bounded spectral windows. This avoids applying the discontinuous function $\chi_{(-\infty,0)}$ across the quotient map, since the spectrum of a quotient-algebra element can differ from the operator spectrum of its lift. In the present unital scalar Roe algebra this is ordinary unital functional calculus; in nonunital variants one should use compactly supported cutoffs on the same bounded spectral windows.

5.1 Uniformly regular refinement families

Let $(\mathcal{K}_n)$ be a refinement sequence of a fixed compact oriented surface with boundary. For each n, write

$$X_n := X\big(D(\mathcal{K}_n)\big)$$

for the set of barycenters of all simplices in the doubled triangulation, equipped with the barycentric graph metric from Definition 3.1.

Definition 5.2 (Uniformly regular refinement family)**.** The refinement sequence $(\mathcal{K}_n)$ is *uniformly regular* if the metric spaces X_n have uniformly bounded geometry: for every $r > 0$ there exists $N(r) < \infty$ such that

$$|B_{X_n}(x, r)| \le N(r) \qquad \text{for all } n \text{ and all } x \in X_n.$$

A sufficient concrete condition is a uniform bound on simplex incidences under refinement.

For each n, let $X_{n,\mathrm{orig}}$ be the subset of simplex barycenters lying on the original side together with the glued boundary simplices, and let $X_{n,\mathrm{copy}}$ be the subset of simplex barycenters strictly on the reflected side. Let $\Gamma_n \subset X_n$ denote the set of simplex barycenters whose simplices meet the glued boundary subcomplex. If $R > 0$, define the R-collar

$$Y_n^{(R)} := N_R(\Gamma_n) \subset X_n.$$

Definition 5.3 (Coarse disjoint unions and the interface ideal)**.** Let

$$X := \bigsqcup_{n=1}^{\infty} X_n$$

be a coarse disjoint union equipped with a metric that restricts to the given metric on each component and separates distinct components by distances tending to infinity; concretely, for every $r > 0$ there is N_r such that $\operatorname{dist}(X_n, X_m) > r$ whenever $n \neq m$ and $\max\{n, m\} > N_r$. For fixed $R \ge 0$, let

$$Y^{(R)} := \bigsqcup_{n=1}^{\infty} Y_n^{(R)} \subset X.$$

The Roe algebra $C^*(X)$ is the norm closure of bounded finite-propagation operators on $\ell^2(X)$ that are locally compact. The interface ideal $C^*(Y^{(R)} \subset X)$ is the norm closure of bounded finite-propagation operators supported within a bounded neighborhood of $Y^{(R)}$.

Lemma 5.4 (Roe ideal property)**.** *For every $R \ge 0$, the algebra $C^*(Y^{(R)} \subset X)$ is a closed two-sided ideal in $C^*(X)$.*

Proof. See Section 6.76.

Proposition 5.5 (Independence of fixed collar radius)**.** *Let*

$$\Gamma := \bigsqcup_{n=1}^{\infty} \Gamma_n \subset X, \qquad Y^{(R)} := \bigsqcup_{n=1}^{\infty} N_R(\Gamma_n).$$

For any fixed $R, R' \geq 0$,

$$C^*(Y^{(R)} \subset X) = C^*(Y^{(R')} \subset X).$$

Consequently the interface ideal used below is independent of the particular fixed collar radius.

Proof. See Section 6.77.

Proposition 5.6 (Coarse invariance of the interface ideal)**.** *Let $Y_0, Y_1 \subset X$ be subsets at finite Hausdorff distance: there is $R < \infty$ such that*

$$Y_0 \subset N_R(Y_1), \qquad Y_1 \subset N_R(Y_0).$$

Then

$$C^*(Y_0 \subset X) = C^*(Y_1 \subset X).$$

Proof. See Section 6.78.

Remark 5.7. For background on coarse spaces and Roe algebras, see Roe [35]. Here the only Roe-algebra input is the concrete quotient extension

$$0 \to C^*(Y \subset X) \to C^*(X) \to C^*(X)/C^*(Y \subset X) \to 0,$$

together with the standard K_1-class of a unitary in $\mathbf{1} + \mathcal{I}$.

5.2 Uniform locality and the bulk Fermi projection

For each n, write

$$H_{\pm,n} := H_{\pm}(D(\mathcal{K}_n)).$$

Assumption 5.8 (Uniform cap-product locality input)**.** There exist constants $R_S, N_S, c_S > 0$, independent of n, such that for every n and every pair of simplex barycenters $\sigma, \tau \in X_n$:

(i) if $d_{X_n}(\sigma, \tau) > R_S$, then $S_n(\sigma, \tau) = 0$;

(ii) each row and each column of the matrix of S_n has at most N_S nonzero entries;

(iii) $|S_n(\sigma, \tau)| \leq c_S$.

Remark 5.9. Assumption 5.8 is the uniform bounded-geometry locality input used for the Higson–Prodan cap-product operator on the closed doubled family. For the exact Higson–Roe/Higson–Prodan duality operator on a uniformly regular closed family, this is an imported consequence of geometric control: the cap-product operator is a geometrically controlled map in the sense of Higson–Roe [11], hence has uniformly bounded propagation and uniformly bounded matrix coefficients. Equivalently, it may be kept as an explicit hypothesis for abstract variants of the construction. Concretely, the required input is a uniform finite row and column support bound and a uniform entry bound for the cap-product matrix. In the local Higson–Prodan formula this comes from the fact that each term pairs only simplices in a uniformly bounded barycentric neighborhood, with coefficients drawn from a fixed finite set of incidence signs and phase factors; the lemma below records the resulting Schur-test estimate used here.

Lemma 5.10 (Uniform locality and norm bound for S_n)**.** *Assume that the Higson–Prodan cap-product operators S_n satisfy Assumption 5.8. Then*

$$\operatorname{prop}(S_n) \le R_S, \qquad \|S_n\| \le N_S c_S$$

for all n.

Proof. See Section 6.79.

Proposition 5.11 (Uniform locality and norm bound)**.** *Let $(\mathcal{K}_n)$ be a uniformly regular refinement family, and assume 5.8. Then there exist constants $R_{\mathrm{loc}}, B_{\mathrm{bulk}} > 0$ such that for all n:*

(a) $H_{\pm,n}$ has propagation at most R_{loc} on $\ell^2(X_n)$;

(b) $\|H_{\pm,n}\| \le B_{\mathrm{bulk}}$.

Proof. See Section 6.80.

Remark 5.12 (Structure of the gap condition). The operator S_n is degree-reversing, so its $\mathcal{H}_0 \leftrightarrow \mathcal{H}_2$ block is a rectangular matrix of dimensions $|D(\mathcal{K}_n)_2| \times |D(\mathcal{K}_n)_0|$. On a closed surface with Euler characteristic χ, the relation $3|K_2| = 2|K_1|$ gives $|K_0| - |K_2| = \chi - |K_2|/2$, so generically $|K_0| \neq |K_2|$ and S_n has $\big||K_0| - |K_2|\big|$ zero eigenvalues from the dimension mismatch. Hence the sufficient condition $\operatorname{dist}(0, \operatorname{spec}(S_n)) > 0$ from Proposition 2.2 is *not* available for typical triangulations (it holds only when $|K_0| = |K_2|$, e.g., the tetrahedron).

Nevertheless, $H_{\pm,n}$ can still be gapped because the term $(B_n + B_n^\dagger)$ may be nonzero on $\ker(S_n)$. More precisely:

$$\langle \psi, H_{\pm,n}^2 \psi \rangle = \|(B_n + B_n^\dagger)\psi\|^2 + \|S_n\psi\|^2 \pm \langle \psi, (B_n^\dagger S_n + S_n B_n)\psi \rangle$$

for each ψ. On $\ker(S_n)$, $\|S_n\psi\|^2 = 0$ and the cross term vanishes as a quadratic form: $\langle \psi, (B_n^\dagger S_n + S_n B_n)\psi \rangle = \langle B_n\psi, S_n\psi \rangle + \langle S_n\psi, B_n\psi \rangle = 0$ since $S_n\psi = 0$. So the gap on $\ker(S_n)$ is controlled by $(B_n + B_n^\dagger)^2$. On $\ker(S_n)^\perp$, the positive term S_n^2 also contributes.

The uniform gap therefore depends on a joint spectral property of $B_n + B_n^\dagger$ and S_n together, not on S_n alone. The Higson–Roe controlled-duality machinery supplies the relevant locality and Poincaré-duality input for the closed model, and the HP-covered headline application uses Higson–Prodan's clean-gap theorem for the closed doubles. The numerics (Figure 14) report the measured gap on the tested refinement levels, but they are not used as a proof of a refinement-uniform gap.

The uniform bulk hypothesis used by the coarse interface theorem is the following.

Assumption 5.13 (Uniform bulk hypotheses)**.** The refinement family $(\mathcal{K}_n)$ is uniformly regular and there exist constants $R_{\mathrm{loc}}, B_{\mathrm{bulk}}, g_{\mathrm{bulk}} > 0$ such that for all n:

(a) $H_{\pm,n}$ has propagation at most R_{loc} on $\ell^2(X_n)$;

(b) $\|H_{\pm,n}\| \le B_{\mathrm{bulk}}$;

(c) $\operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(H_{\pm,n})\big) \ge g_{\mathrm{bulk}}$.

Remark 5.14. Parts (a) and (b) are reduced in Proposition 5.11 to uniform regularity together with the standard cap-product locality input recorded in Assumption 5.8. Part (c) is the spectral input not proved by the elementary locality argument. As explained in Remark 5.12, this is a joint spectral condition on $B_n + B_n^\dagger$ and S_n that does not reduce to invertibility of S_n alone. For the HP-covered closed doubles used in the headline theorem, Higson–Prodan supplies (c).

Assumption 5.15 (Higson–Prodan closed-bulk input)**.** The original refinement family $(\mathcal{K}_n)_n$ is uniformly regular, and the closed doubled refinement family $L_n = D(\mathcal{K}_n)$ is covered by the Higson–Prodan closed universal-model theorem. Thus the following data are available:

(a) the Higson–Roe/Higson–Prodan controlled Poincaré-duality operator S_n with the uniform locality, row-support, and coefficient bounds of Assumption 5.8;

(b) constants $B_{\mathrm{HP}}, g_{\mathrm{HP}} > 0$, independent of n, such that

$$\|H_{\pm,n}\| \le B_{\mathrm{HP}}, \qquad \operatorname{dist}\bigl(0, \operatorname{spec}_{\mathrm{op}}(H_{\pm,n})\bigr) \ge g_{\mathrm{HP}};$$

(c) the Higson–Prodan/Bellissard–Connes bulk Chern pairing Ch for which the associated closed-bulk Fermi classes have Chern number ± 1.

Remark 5.16. Assumption 5.15 records the cited closed-bulk theorem used for the jammed boundary model. The geometric input is the controlled Poincaré-duality construction of Higson–Roe: bounded-geometry oriented combinatorial manifolds give analytically controlled Hilbert–Poincaré complexes [11, Theorem 3.14 and Proposition 4.1]. Higson and Prodan specialize this controlled construction to the tight-binding Hamiltonians $H_\pm = (B + B^\dagger) \pm S$ and identify the closed-bulk Chern classes in their universal Chern model [8, 9]. The present paper cites that closed theory exactly as it cites the standard background theorems; after the HP gap and Chern conclusions are in place, the doubled-and-jammed interface conclusions below are proved internally.

Theorem 5.17 (Closed-bulk consequences of the Higson–Prodan input)**.** *If Assumption 5.15 holds, then Assumptions 5.8 and 5.13 hold. Moreover the bulk Fermi projection*

$$P_\pm = \chi_{(-\infty,0)}(H_\pm^\infty),$$

constructed below in Proposition 5.18, is the Higson–Prodan closed-bulk Fermi projection for the refinement family of doubles, and its imported Chern pairing satisfies

$$\mathrm{Ch}([P_\pm]) = \pm 1.$$

Proof. See Section 6.81.

Let

$$H_\pm^\infty := \bigoplus_{n=1}^{\infty} H_{\pm,n} \qquad \text{on} \qquad \ell^2(X) = \bigoplus_{n=1}^{\infty} \ell^2(X_n).$$

Proposition 5.18 (Bulk Roe-algebra class)**.** *Assume 5.13. Then $H_\pm^\infty \in C^*(X)$ is bounded, self-adjoint, and finite-propagation. Let*

$$P_\pm := \chi_{(-\infty,0)}\bigl(H_\pm^\infty\bigr)$$

denote the negative operator spectral projection of $H_\pm^\infty$ on $\ell^2(X)$. Then $P_\pm \in C^*(X)$ and therefore defines a class

$$[P_\pm] \in K_0\bigl(C^*(X)\bigr).$$

Proof. See Section 6.82.

Proposition 5.19 (Gap-homotopy invariance of the bulk Fermi class)**.** *Let $s \mapsto H_s$, $s \in [0,1]$, be a norm-continuous path of self-adjoint elements of $C^*(X)$. Assume that there are constants $B, g > 0$ such that*

$$\mathrm{spec}_{\mathrm{op}}(H_s) \subset [-B, -g] \cup [g, B] \qquad \text{for every } s \in [0,1].$$

Set

$$P_s := \chi_{(-\infty,0)}(H_s).$$

Then $s \mapsto P_s$ is a norm-continuous path of projections in $C^*(X)$. In particular,

$$[P_s] \in K_0(C^*(X))$$

is independent of s.

Proof. See Section 6.83.

Proposition 5.20 (Quantitative stability of a spectral projection)**.** *Let $H = H^*$ be a bounded self-adjoint operator, let $V = V^*$ be bounded, and let Γ be a positively oriented contour with*

$$\delta_\Gamma := \mathrm{dist}(\Gamma, \mathrm{spec}(H)) > 0.$$

If

$$\|V\| < \delta_\Gamma,$$

then $\Gamma \subset \rho(H+V)$ and

$$\|P_\Gamma(H+V) - P_\Gamma(H)\| \leq \frac{|\Gamma|}{2\pi} \frac{\|V\|}{\delta_\Gamma(\delta_\Gamma - \|V\|)}.$$

Proof. See Section 6.84.

Small-norm Fermi and positive-side interface stability is a variant statement collected in Appendix 6.8.

5.3 A collar-locality theorem

For each n, let $P_{\mathrm{orig},n}$ and $P_{\mathrm{copy},n}$ be the projections associated to the decomposition

$$\ell^2(X_n) = \mathcal{H}_{\mathrm{orig},n} \oplus \mathcal{H}_{\mathrm{copy},n}.$$

Define the split operator

$$H_{\pm,n}^{\mathrm{split}} := P_{\mathrm{orig},n} H_{\pm,n} P_{\mathrm{orig},n} + P_{\mathrm{copy},n} H_{\pm,n} P_{\mathrm{copy},n}.$$

Proposition 5.21 (Finite collar-locality)**.** *Assume that $H_{\pm,n}$ has propagation at most R_{loc} for every n, and set*

$$Y_n := Y_n^{(R_{\mathrm{loc}})} = N_{R_{\mathrm{loc}}}(\Gamma_n).$$

Then

$$\chi_{X_n \setminus Y_n} \left(H_{\pm,n} - H_{\pm,n}^{\mathrm{split}} \right) \chi_{X_n \setminus Y_n} = 0. \tag{5.1}$$

Moreover, for every $M > 0$,

$$\chi_{X_n \setminus Y_n} \left(H_{\pm,M,n} - \left(H_{\pm,n}^{\mathrm{split}} + M P_{\mathrm{copy},n} \right) \right) \chi_{X_n \setminus Y_n} = 0. \tag{5.2}$$

Proof. See Section 6.86.

Corollary 5.22 (Support near the coarse interface)**.** *Assume 5.13. Let*

$$H_{\pm}^{\mathrm{split},\infty} := \bigoplus_{n=1}^{\infty} H_{\pm,n}^{\mathrm{split}}, \qquad P_{\mathrm{copy}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\mathrm{copy},n}, \qquad J_{\pm,M}^{\infty} := \bigoplus_{n=1}^{\infty} H_{\pm,M,n}.$$

Then

$$H_{\pm}^{\infty} - H_{\pm}^{\mathrm{split},\infty} \in C^*(Y \subset X), \qquad J_{\pm,M}^{\infty} - \big(H_{\pm}^{\mathrm{split},\infty} + MP_{\mathrm{copy}}^{\infty}\big) \in C^*(Y \subset X),$$

where $Y = Y^{(R_{\mathrm{loc}})}$.

Proof. See Section 6.87.

For later use define

$$P_{\mathrm{orig}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\mathrm{orig},n}, \qquad H_{\pm,\mathrm{copy}}^{\infty} := \bigoplus_{n=1}^{\infty} P_{\mathrm{copy},n} H_{\pm,n} P_{\mathrm{copy},n},$$

and for each n,

$$H_{\pm,n}^{\mathrm{copy}} := P_{\mathrm{copy},n} H_{\pm,n} P_{\mathrm{copy},n}\big|_{\mathcal{H}_{\mathrm{copy},n}}.$$

For each fixed sign $\pm$, define the uniform negative copy-side threshold

$$B_{\mathrm{copy},-} := \sup_n \max\{0,\ -\lambda_{\min}(H_{\pm,n}^{\mathrm{copy}})\}.$$

By Assumption 5.13(b), this quantity is finite and satisfies

$$B_{\mathrm{copy},-} \le B_{\mathrm{bulk}}.$$

For each $M > 0$, define the auxiliary negative-side comparison operator

$$\widetilde{T}_{\pm,M}^{-} := P_{\mathrm{orig}}^{\infty} + H_{\pm,\mathrm{copy}}^{\infty} + MP_{\mathrm{copy}}^{\infty}.$$

5.4 The jammed side is topologically trivial

Proposition 5.23 (Uniform positivity of the auxiliary negative-side model)**.** *Assume 5.13. If*

$$M > B_{\mathrm{copy},-},$$

then the auxiliary comparison operator

$$\widetilde{T}_{\pm,M}^{-} = P_{\mathrm{orig}}^{\infty} + H_{\pm,\mathrm{copy}}^{\infty} + MP_{\mathrm{copy}}^{\infty}$$

is bounded, self-adjoint, local, and strictly positive as an operator on $\ell^2(X)$*. Moreover,*

$$\mathrm{spec}_{\mathrm{op}}(\widetilde{T}_{\pm,M}^{-}) \subset \{1\} \cup [M - B_{\mathrm{copy},-}, \infty),$$

so

$$\mathrm{dist}\big(0, \mathrm{spec}_{\mathrm{op}}(\widetilde{T}_{\pm,M}^{-})\big) \ge \min\{1, M - B_{\mathrm{copy},-}\}.$$

In particular, its negative operator spectral projection vanishes:

$$\chi_{(-\infty,0)}\big(\widetilde{T}_{\pm,M}^{-}\big) = 0.$$

The stronger condition $M > B_{\mathrm{bulk}}$ *is sufficient but not necessary.*

Proof. See Section 6.88.

Remark 5.24. The bare operator $H^{\infty}_{\pm,\mathrm{copy}} + MP^{\infty}_{\mathrm{copy}}$ is positive on the copy sector but has a zero block on the original-side sector when viewed as an operator on all of $\ell^2(X)$. It is therefore *not* gapped at 0 on the full Hilbert space. The added term $P^{\infty}_{\mathrm{orig}}$ is inserted solely to remove that spurious zero block while preserving the negative-side agreement away from the interface. Any fixed positive scalar multiple $\mu P^{\infty}_{\mathrm{orig}}$ with $\mu > 0$ could be used in place of $P^{\infty}_{\mathrm{orig}}$.

5.5 Quotient decomposition away from the interface

For each n, define

$$X_n^+ := X_{n,\mathrm{orig}} \cup Y_n, \qquad X_n^- := X_{n,\mathrm{copy}} \cup Y_n, \qquad Y_n := Y_n^{(R_{\mathrm{loc}})}.$$

Set

$$X^+ := \bigsqcup_{n=1}^{\infty} X_n^+, \qquad X^- := \bigsqcup_{n=1}^{\infty} X_n^-.$$

Then $X = X^+ \cup X^-$ and $X^+ \cap X^- = Y$.

Proposition 5.25 (Coarse excision of the doubled decomposition)**.** *Under Assumption 5.13, the decomposition $X = X^+ \cup X^-$ is coarsely excisive: for every $r > 0$ there exists $s(r) > 0$ such that*

$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y).$$

Proof. See Section 6.89.

Let

$$\mathcal{A} := C^*(X), \qquad \mathcal{I} := C^*(Y \subset X), \qquad \pi : \mathcal{A} \to \mathcal{A}/\mathcal{I}$$

be the quotient map. Also set

$$Z^+ := X^+ \setminus Y, \qquad Z^- := X^- \setminus Y.$$

Let

$$Q_+ := \chi_{Z^+}, \qquad Q_- := \chi_{Z^-}, \qquad Q_Y := \chi_Y$$

be the corresponding diagonal multiplication operators on $\ell^2(X)$.

Remark 5.26 (Strict-side projections in $C^*(X)$). Because $C^*(X)$ is unital in the scalar-coefficient setting, the diagonal $\{0,1\}$-valued operators Q_+, Q_-, and Q_Y all belong to $\mathcal{A} = C^*(X)$: each has propagation 0 and is locally compact. They are pairwise orthogonal projections summing to $\mathbf{1}$ and they preserve $\mathcal{I}$ under left and right multiplication.

Proposition 5.27 (Quotient splitting by strict-side compressions)**.** *Assume 5.13. With the notation above:*

(i) for every algebraic Roe operator T of propagation at most R,

$$Q_Y T,\ TQ_Y,\ Q_+TQ_-,\ Q_-TQ_+ \in \mathcal{I};$$

(ii) if

$$e_+ := \pi(Q_+), \qquad e_- := \pi(Q_-),$$

then e_+ and e_- are orthogonal central projections in $\mathcal{A}/\mathcal{I}$ with

$$e_+ + e_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}};$$

(iii) the formulas

$$E_+(\pi(a)) := \pi(Q_+ a), \qquad E_-(\pi(a)) := \pi(Q_- a) \qquad (a \in \mathcal{A})$$

equivalently $E_\pm(b) = e_\pm b$, define well-defined $*$-homomorphisms

$$E_\pm : \mathcal{A}/\mathcal{I} \longrightarrow \mathcal{A}/\mathcal{I}$$

such that

$$E_\pm^2 = E_\pm, \qquad E_+E_- = E_-E_+ = 0, \qquad E_+ + E_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}},$$

and

$$E_\pm(b_1)b_2 = b_1 E_\pm(b_2) \qquad \text{for all } b_1, b_2 \in \mathcal{A}/\mathcal{I}.$$

Consequently,

$$\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I})$$

as a direct sum of closed two-sided ideals.

Proof. See Section 6.90.

Proposition 5.28 (Quotient-side description)**.** *Assume 5.13. Let J, T^+, T^- be bounded self-adjoint local operators on $\ell^2(X)$. Since X is a discrete bounded-geometry scalar Roe space, these operators belong to $\mathcal{A} = C^*(X)$. Define*

$$P^\pm := \chi_{(-\infty,0)}(T^\pm),$$

and assume:

(a) $0 < \gamma_\pm := \operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^\pm)\big)$*;*

(b) there exists $R_{\mathrm{ag}} > 0$ such that

$$\chi_{X^+\setminus N_{R_{\mathrm{ag}}}(Y)}(J - T^+)\chi_{X^+\setminus N_{R_{\mathrm{ag}}}(Y)} = 0,$$

$$\chi_{X^-\setminus N_{R_{\mathrm{ag}}}(Y)}(J - T^-)\chi_{X^-\setminus N_{R_{\mathrm{ag}}}(Y)} = 0.$$

Then:

(i)

$$\pi(J) = E_+(\pi(T^+)) + E_-(\pi(T^-));$$

(ii) for every $0 < \gamma < \min\{\gamma_+, \gamma_-\}$, every $B \geq \max\{\|T^+\|, \|T^-\|\}$, and every $h \in C_c(\mathbb{R}, [0,1])$ satisfying

$$h(t) = 1 \quad \text{for } t \in [-B, -\gamma], \qquad h(t) = 0 \quad \text{for } t \in [\gamma, B],$$

one has

$$h(\pi(J)) = E_+(\pi(P^+)) + E_-(\pi(P^-));$$

(iii) the element

$$p_J := E_+(\pi(P^+)) + E_-(\pi(P^-))$$

is a projection in $\mathcal{A}/\mathcal{I}$, independent of the choice of γ and h, and for every such h

$$p_J = h(\pi(J)) = \pi(h(J)).$$

Proof. See Section 6.91.

Remark 5.29 (Relation to the literal quotient spectral projection). Since $\mathcal{A}/\mathcal{I}$ is unital, the projection p_J from Proposition 5.28 coincides with $\chi_{(-\infty,0)}(\pi(J))$ whenever $0 \notin \operatorname{spec}_{\mathcal{A}/\mathcal{I}}(\pi(J))$. The continuous-cutoff formulation is used because it avoids the need to verify that last spectral condition. The compactly supported version above is also the form compatible with nonunital or stabilized Roe-algebra presentations.

Definition 5.30 (Sidewise class-A interface)**.** Let J, T^+, and T^- be bounded self-adjoint local operators on $\ell^2(X)$. We say that $(J; T^+, T^-)$ is a *sidewise class-A interface along* Y if:

(i) 0 lies in a gap of the operator spectra $\operatorname{spec}_{\mathrm{op}}(T^{\pm})$;

(ii) there exists $R > 0$ such that

$$\chi_{X^+\setminus N_R(Y)}(J - T^+)\chi_{X^+\setminus N_R(Y)} = 0, \qquad \chi_{X^-\setminus N_R(Y)}(J - T^-)\chi_{X^-\setminus N_R(Y)} = 0;$$

(iii) the operator Fermi projections

$$P^+ := \chi_{(-\infty,0)}(T^+), \qquad P^- := \chi_{(-\infty,0)}(T^-)$$

belong to $\mathcal{A} = C^*(X)$.

5.6 Boundary class of a liftable quotient projection

Proposition 5.31 (Boundary class of a liftable quotient projection)**.** *Let $p \in M_n(\mathcal{A}/\mathcal{I})$ be a projection that admits a self-adjoint lift $\widetilde{p} \in M_n(\mathcal{A})$. Then*

$$\exp(2\pi i\,\widetilde{p}) - \mathbf{1} \in M_n(\mathcal{I}),$$

so $\exp(2\pi i\,\widetilde{p}) \in \mathbf{1} + M_n(\mathcal{I})$ defines a class in $K_1(\mathcal{I})$. This class depends only on p, not on the chosen self-adjoint lift. We denote it by

$$\partial_Y(p) \in K_1(\mathcal{I}).$$

This fixes our sign convention for the connecting map; some references use the inverse class.

Moreover, if $a \in M_n(\mathcal{A})$ is self-adjoint and $h \in C(\mathbb{R}, [0,1])$ satisfies

$$h|_{\operatorname{spec}(\pi(a))} = \chi_{(-\infty,0)}|_{\operatorname{spec}(\pi(a))},$$

then $\pi(h(a))$ is a projection in $M_n(\mathcal{A}/\mathcal{I})$ and

$$\partial_Y\big(\pi(h(a))\big) = \big[\exp(2\pi i\,h(a))\big] \in K_1(\mathcal{I}).$$

Proof. See Section 6.92.

Proposition 5.32 (Homotopy invariance of positive-side boundary classes)**.** *Assume 5.13, so that the quotient-side map E_+ is defined. Let $s \mapsto P_s$, $s \in [0,1]$, be a norm-continuous path of projections in $\mathcal{A} = C^*(X)$. Then*

$$\partial_Y\big(E_+(\pi(P_s))\big) \in K_1(\mathcal{I})$$

is independent of s.

Proof. See Section 6.93.

Definition 5.33 (Interface class of a sidewise realization)**.** If $(J; T^+, T^-)$ is a sidewise class-A interface along Y, let

$$P^+ := \chi_{(-\infty,0)}(T^+), \qquad P^- := \chi_{(-\infty,0)}(T^-),$$

and define the canonical quotient projection

$$p_J := E_+(\pi(P^+)) + E_-(\pi(P^-)) \in \mathcal{A}/\mathcal{I}.$$

By Proposition 5.28, if

$$0 < \gamma < \min\Big\{\operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^+)\big), \operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^-)\big)\Big\}$$

and, for some $B \geq \max\{\|T^+\|, \|T^-\|\}$, the function $h \in C_c(\mathbb{R}, [0,1])$ satisfies

$$h(t) = 1 \ \text{ for } t \in [-B, -\gamma], \qquad h(t) = 0 \ \text{ for } t \in [\gamma, B],$$

then

$$p_J = h(\pi(J)) = \pi(h(J)).$$

Its *interface class* is

$$\operatorname{Ind}_{\mathrm{int}}(J; T^+, T^-) := \partial_Y(p_J) \in K_1(\mathcal{I}).$$

Equivalently, for every such cutoff h,

$$\operatorname{Ind}_{\mathrm{int}}(J; T^+, T^-) = \big[\exp(2\pi i\, h(J))\big] \in K_1(\mathcal{I}),$$

and this class is independent of the choice of h.

Sidewise interface-supported changes, finite-component changes, norm-vanishing component defects, sidewise homotopies, and positive copy-confinement variants are collected in Appendix 6.8.

Remark 5.34. The interface class is built from the quotient projection p_J and the genuine extension

$$0 \to C^*(Y \subset X) \to C^*(X) \to C^*(X)/C^*(Y \subset X) \to 0.$$

5.7 Interface class for the jammed family

Theorem 5.35 (Doubled-and-jammed interface class)**.** *Assume 5.13. Let*

$$J^{\infty}_{\pm,M} = \bigoplus_{n=1}^{\infty} H_{\pm,M,n}$$

be the direct-sum jammed family on the coarse disjoint union X, and let

$$Y = Y^{(R_{\mathrm{loc}})}$$

be the coarse interface from Corollary 5.22. If

$$M > B_{\mathrm{copy},-},$$

then with

$$T^+ := H^{\infty}_{\pm}, \qquad T^- := \widetilde{T}^{-}_{\pm,M},$$

the triple $(J^{\infty}_{\pm,M}; T^{+}, T^{-})$ *is a sidewise class-A interface along* Y *in the sense of Definition 5.30. Its canonical quotient projection is*

$$p_{J^{\infty}_{\pm,M}} = E_{+}\big(\pi(P_{\pm})\big), \tag{5.3}$$

where

$$P_{\pm} := \chi_{(-\infty,0)}(H^{\infty}_{\pm}) \in C^{*}(X)$$

is the bulk operator Fermi projection from Proposition 5.18. Equivalently, if

$$0 < \gamma < \min\Big\{ g_{\mathrm{bulk}},\ 1,\ M - B_{\mathrm{copy},-} \Big\}$$

and, for some $B \geq \max\{\|T^{+}\|, \|T^{-}\|\}$*, the function* $h \in C_c(\mathbb{R}, [0,1])$ *satisfies*

$$h(t) = 1 \quad \text{for } t \in [-B, -\gamma], \qquad h(t) = 0 \quad \text{for } t \in [\gamma, B],$$

then

$$\pi(h(J^{\infty}_{\pm,M})) = E_{+}\big(\pi(P_{\pm})\big).$$

Consequently,

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^{+}, T^{-}) = \partial_{Y}\big(E_{+}(\pi(P_{\pm}))\big) \in K_{1}\big(C^{*}(Y \subset X)\big). \tag{5.4}$$

In particular, the jammed side contributes no negative-energy quotient projection.

Proof. See Section 6.100.

This theorem uses the closed-bulk gap in Assumption 5.13(c), together with the uniform locality and boundedness hypotheses stated there. In the headline application, these hypotheses are supplied by Theorem 5.17. The finite-dimensional decoupling estimates of Section 3 do not use a refinement-limit bulk gap; they require only their stated finite-matrix separation hypotheses.

Large-jam-scale independence is recorded with the other Roe variants in Appendix 6.8.

5.8 Identification of the bulk class

The Hamiltonians $H_{\pm,n} = H_{\pm}(D(\mathcal{K}_n))$ are exactly the Higson–Prodan universal Chern Hamiltonians on the closed oriented surfaces $D(\mathcal{K}_n)$. For closed doubles covered by the Higson–Prodan theorem, the bulk Fermi projection $P_{\pm} = \chi_{(-\infty,0)}(H^{\infty}_{\pm})$ is the corresponding Higson–Prodan Fermi projection.

In this subsection Ch denotes the numerical bulk Chern pairing supplied by the Higson–Prodan theorem for the closed refinement family. We do not construct a new Chern cocycle on an arbitrary Roe algebra here. Theorem 4.9 uses the Higson–Prodan bulk pairing associated with the closed doubled family and identifies the original/copy coarse interface pairing internally.

Theorem 5.36 (Bulk Chern number of the doubled family, imported from Higson–Prodan)**.** *Assume 5.13. Assume moreover that the refinement family of closed doubles* $(D(\mathcal{K}_n))_n$ *is an admissible Higson–Prodan refinement family satisfying the hypotheses of the Higson–Prodan bulk theorem, and that the Chern pairing* Ch *used here is the corresponding Higson–Prodan bulk Chern pairing. Then the* K_0*-class* $[P_{\pm}] \in K_0(C^{*}(X))$ *is the Higson–Prodan bulk class for* $(D(\mathcal{K}_n))_n$*. Consequently, by the Higson–Prodan theorem [8, 9],*

$$\mathrm{Ch}([P_{\pm}]) = \pm 1.$$

Proof. See Section 6.102.

The next definition records the Ewert–Meyer convention used to phrase the restatement of Theorem 4.9 in bulk–interface language. The intermediate Chern-± 1 and HP corollaries are collected in Appendix 6.8.

Definition 5.37 (HP–Ewert–Meyer boundary convention)**.** A boundary refinement family $(\mathcal{K}_n)_n$ has the *HP–Ewert–Meyer boundary convention* when $D(\mathcal{K}_n)$ satisfies the Higson–Prodan closed-bulk input in Assumption 5.15, and the associated original/copy coarse boundary data

$$(X; X^+, X^-; Y)$$

fit the Roe-algebraic bulk–interface framework of Ewert–Meyer [29]. Concretely, the required data are:

(a) $X = X^+ \cup X^-$ is a coarse partition with interface Y; in particular, for every $r > 0$ there is $s(r) > 0$ such that
$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y).$$

(b) Away from Y, the positive side agrees with the Higson–Prodan bulk model and the negative side agrees with a trivial gapped comparison model.

(c) The connecting map
$$\partial_Y : K_0(C^*(X)/C^*(Y \subset X)) \longrightarrow K_1(C^*(Y \subset X))$$
is paired with the Ewert–Meyer interface cocycle using the sign convention of Theorem 4.9; a quotient class with Chern number c_+ on the positive side and c_- on the negative side has winding number $c_+ - c_-$.

Convention checks and tail variants are collected in Appendix 6.8.

The Ewert–Meyer input used here is the coarse Mayer–Vietoris bulk–interface boundary map and its identification with the bulk–edge map [29, Propositions 3.2.1–3.2.2 and Corollary 3.2.3]. Kubota's controlled framework and the Ludewig–Thiang edge-following theorem give closely related formulations [28, 32]; the restatement below uses the Ewert–Meyer convention fixed in Definition 5.37.

Status of the interface K_1-class identity and integer winding. For the doubled-jammed original/copy partition, Section 4 proves the K_1-class identity internally. The restatement below records the same K_1-class identity in Ewert–Meyer language, with Higson–Prodan supplying the closed-bulk gap and Chern value. The integer winding evaluation is internal in the straight periodic Toeplitz case (Theorem 4.10) and is the conditional corollary Theorem 4.11 for general partitions.

Theorem 5.38 (Ewert–Meyer restatement of the Roe interface theorem)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular boundary refinement family whose closed doubles satisfy the Higson–Prodan universal-model theorem, and use the original/copy coarse partition. Then, for every $M > B_{\mathrm{copy},-}$,*

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^+, T^-) = \partial_Y\big(E_+(\pi(P_\pm))\big) \in K_1(C^*(Y \subset X)).$$

The projection $P_\pm$ is the Higson–Prodan closed-bulk Fermi projection and

$$\mathrm{Ch}([P_\pm]) = \pm 1.$$

In the straight periodic Toeplitz case (Theorem 4.10), the integer interface winding pairing is internal and evaluates to

$$\mathrm{Wind}\big(\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M};T^{+},T^{-})\big)=\pm 1.$$

For general partitions, the integer interface winding pairing is recorded as a conditional corollary (Theorem 4.11) requiring an integer-valued Chern-jump pairing on $K_1(\mathcal{I})$, whose construction outside the periodic class is not carried out in this paper. The K_1-class identity itself is unconditional modulo HP. The same class is independent of the jam scale M for all $M > B_{\mathrm{copy},-}$.

Proof. See Section 6.108.

Relabeling, orientation reversal, and robustness of the compatible winding are stated in Appendix 6.8.

Remark 5.39 (From bulk Chern number to numerical interface index). Theorem 4.9 is the headline winding result: for the doubled-jammed original/copy partition it supplies the Ewert–Meyer identification internally and uses the Higson–Prodan closed-bulk theorem for the gap and Chern value. Theorem 5.38 records the same conclusion in Ewert–Meyer language. This identification is not a boundary-gap statement and does not assert that the negative quotient of the closed doubled bulk projection vanishes. In the corresponding six-term exact sequence

$$K_0(\mathcal{I})\to K_0(\mathcal{A})\to K_0(\mathcal{A}/\mathcal{I})\xrightarrow{\partial} K_1(\mathcal{I})\to K_1(\mathcal{A})\to K_1(\mathcal{A}/\mathcal{I})$$

the connecting map is evaluated by the interface winding pairing. The positive side has Chern number ± 1, and the jammed negative side is trivial, so the Chern jump is ± 1.

Remark 5.40 (What the global theorem does and does not say). Theorem 5.35 is a *global* statement about the interface class supported near the entire glued boundary. It does not by itself imply that each boundary component carries its own local index or local boundary map. A componentwise statement requires an additional localized pairing. In the continuum Hall setting, Drouot and Zhu prove a curved-interface formula in which the local edge conductance is an intersection number times the bulk Hall conductance [34, 33]. We view that as the correct model for any future local theorem in the present lattice setting.

6 Detailed appendix proofs

6.1 Sign ledger

The table records the orientation conventions used in the periodic triangular crossing and in the Toeplitz edge map. It is a ledger for the proof of Theorem 5.5, not a separate numerical input.

Explicit sign chain. The sign convention used in the headline corollary $\mathrm{wind} = \mp \det[t\ n]$ for the periodic triangular case is fully determined by composing the following four signed steps; a referee can verify each step by referring to the row of the table below and the cited proposition. (i) *Bloch torus orientation.* The Bloch torus carries the orientation $dk_1 \wedge dk_2$; the Chern integer $\mathrm{Ch}(P) = \frac{1}{2\pi i}\int_{\mathbb{T}^2}\mathrm{tr}(P\,dP\wedge dP)$ uses this orientation, giving $\mathrm{Ch}(E_-(H_+^{\triangle})) = -1$ and $\mathrm{Ch}(E_-(H_-^{\triangle})) = +1$ by Theorem 5.5. (ii) *Boundary map sign.* The Higson–Roe K-theory connecting map ∂_Y is fixed by Convention (C3) of the main paper: for any self-adjoint lift $\tilde{P}$ of a quotient projection $\pi(P)$, $\partial[\pi(P)] = [\exp(2\pi i\,\tilde{P})]$, with no sign flip. (iii) *Toeplitz extension sign.* The half-space Toeplitz extension in the positive-normal direction n is fixed by Theorem 5.7: the winding-number pairing

satisfies $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P])) = \varepsilon(t,n)\,\mathrm{Ch}(P)$ where $\varepsilon(t,n) = \det[t\ n] \in \{\pm 1\}$ records the change-of-basis Jacobian from the canonical (k_1,k_2) frame to the tangential/positive-normal frame (κ,η). (iv) *Composite winding sign.* Composing (ii) and (iii) gives $\mathrm{wind} = \varepsilon(t,n)\,\mathrm{Ch}(E_-(H_\pm^{\triangle})) = \mp\varepsilon(t,n) = \mp\det[t\ n]$, in agreement with both Theorem 4.10 and Theorem 5.9.

If any one of the four sign choices above were flipped, the displayed winding would be the opposite integer; for instance, replacing the Bloch orientation by $dk_2 \wedge dk_1$ would swap the Chern signs of step (i) (and hence flip the right-hand side in step (iv)), while replacing n by $-n$ in step (iii) would flip $\varepsilon(t,n)$. Because all four choices are fixed once for all in Convention (C3) and in the table below, the cited final integer is uniquely determined.

Object	Orientation convention	Sign consequence
Bloch torus	$dk_1 \wedge dk_2$	FHS plaquettes use the same ordered coordinate square.
Crossing space	$dq_1 \wedge dq_2 \wedge d\mu$	The local degree is the sign of $\det Du$.
Self-dual basis	e_1, e_2, e_3, e_4 oriented by $e_1 \wedge e_2 \wedge e_3 \wedge e_4$	The self-dual compression has degree -1.
Mass reversal	$k \mapsto -k$ on $\mathbb{T}^2$	The base map preserves orientation; complex conjugation reverses c_1.
Toeplitz edge	Tangential/positive-normal basis (t,n)	Winding is $\det[t\ n]\,\mathrm{Ch}(P)$.

6.2 Routine perturbation proofs

The two auxiliary finite-dimensional perturbation statements below are retained for reference and kept out of the body. Their proofs are included with the other appendix proof blocks.

6.3 Two-band hexagonal and square benchmark symbols

Remark 6.1 (Status of the two-band benchmark symbols). The two-band hexagonal and square symbols collected in this subsection are auxiliary benchmark classes; they are not derivations of the full Higson–Prodan universal symbol for arbitrary hexagonal or square triangulations. They feed only the cross-check Remark 4.18, whose role is a sanity check against the periodic triangular calculation in Section 5. The reader interested only in the headline Theorem 4.9 may skip this subsection on first reading. The underlying proofs are collected in Sections 6.67 to 6.70 and 6.73 to 6.75.

Hexagonal symbol. With $k = (k_1,k_2) \in \mathbb{T}^2$, $a_1(k) = \sin k_1$, $a_2(k) = \sin k_2$, $a_3^{\mathrm{hex}}(k) = \cos k_1 + \cos k_2 - 1$:

Definition 6.2 (Hexagonal two-band Bloch representative)**.** For $m \in \mathbb{R}$, $H_{\mathrm{hex}}(k;m) = \begin{pmatrix} m\,a_3^{\mathrm{hex}}(k) & a_1(k) - ia_2(k) \\ a_1(k) + ia_2(k) & -m\,a_3^{\mathrm{hex}}(k) \end{pmatrix}$.

Remark 6.3 (Massless hexagonal closure). $H_{\mathrm{hex}}(0,0;0)$ is gapless at zero (direct substitution; see Section 6.65 for the one-line verification).

Lemma 6.4 (Hexagonal determinant and bulk gap)**.** *For $m \neq 0$ and $k \in \mathbb{T}^2$, $|\det H_{\mathrm{hex}}(k;m)| \geq c_{\mathrm{hex}}(m) := \min\{1/4, (\sqrt{3}-1)^2 m^2\}$, hence $\mathrm{dist}(0, \mathrm{spec} H_{\mathrm{hex}}(k;m)) \geq \gamma_{\mathrm{hex}}(m) := \sqrt{c_{\mathrm{hex}}(m)}$.*

Proof. See Section 6.66.

Lemma 6.5 (Hexagonal Chern integer for $m > 0$)**.** *For $m > 0$, with orientation $dk_1 \wedge dk_2$, $\mathrm{Ch}(E_-(H_{\mathrm{hex}}(\cdot; m))) = -1$.*

Proof. See Section 6.67.

Lemma 6.6 (Hexagonal mass reversal)**.** *For $m < 0$, $\mathrm{Ch}(E_-(H_{\mathrm{hex}}(\cdot; m))) = +1$.*

Proof. See Section 6.68.

Lemma 6.7 (Hexagonal Toeplitz winding)**.** *Let $P_{\mathrm{hex},m} = \chi_{(-\infty,0)}(H_{\mathrm{hex}}(\cdot; m))$; for a primitive tangential/positive-normal lattice basis (t, n), $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_{\mathrm{hex},m}])) = \det[t\ n]\ \mathrm{Ch}(P_{\mathrm{hex},m})$, which gives -1 for $m > 0$ and $+1$ for $m < 0$.*

Proof. See Section 6.69.

Theorem 6.8 (Hexagonal two-band bulk-edge correspondence)**.** *For every $m \neq 0$, $H_{\mathrm{hex}}(k; m)$ is uniformly gapped at zero with gap at least $\gamma_{\mathrm{hex}}(m)$, $\mathrm{Ch}(P_{\mathrm{hex},m}) = -\mathrm{sgn}(m)$, and for every primitive straight periodic edge with basis (t, n), $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_{\mathrm{hex},m}])) = -\det[t\ n]\,\mathrm{sgn}(m)$.*

Proof. See Section 6.70.

Square symbol. With $b_1(k) = \sin k_1$, $b_2(k) = \sin k_2$, $b_3^{\square}(k) = 1 + \cos k_1 + \cos k_2$:

Definition 6.9 (Square two-band Bloch representative)**.** For $m \in \mathbb{R}$,

$$H_{\square}(k; m) = \begin{pmatrix} m\, b_3^{\square}(k) & b_1(k) - i b_2(k) \\ b_1(k) + i b_2(k) & -m\, b_3^{\square}(k) \end{pmatrix}.$$

Remark 6.10 (Massless square closure). $H_{\square}(0, 0; 0)$ is gapless at zero (direct substitution; see Section 6.71 for the verification).

Lemma 6.11 (Square determinant and bulk gap)**.** *For $m \neq 0$ and $k \in \mathbb{T}^2$, $|\det H_{\square}(k; m)| \geq c_{\square}(m) := \min\{1/4, (\sqrt{3} - 1)^2 m^2\}$, hence $\mathrm{dist}(0, \mathrm{spec} H_{\square}(k; m)) \geq \gamma_{\square}(m) := \sqrt{c_{\square}(m)}$.*

Proof. See Section 6.72.

Lemma 6.12 (Square Chern integer)**.** *For $m > 0$, with orientation $dk_1 \wedge dk_2$, $\mathrm{Ch}(E_-(H_{\square}(\cdot; m))) = +1$.*

Proof. See Section 6.73.

Lemma 6.13 (Square Toeplitz winding)**.** *Let $P_{\square,m} = \chi_{(-\infty,0)}(H_{\square}(\cdot; m))$; for a primitive tangential/positive-normal lattice basis (t, n), $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_{\square,m}])) = \det[t\ n]\ \mathrm{Ch}(P_{\square,m})$, which gives $+1$ for $m > 0$ and -1 for $m < 0$.*

Proof. See Section 6.74.

Theorem 6.14 (Square two-band bulk-edge correspondence)**.** *For every $m \neq 0$, $H_{\square}(k; m)$ is uniformly gapped at zero with gap at least $\gamma_{\square}(m)$, $\mathrm{Ch}(P_{\square,m}) = \mathrm{sgn}(m)$, and for every primitive straight periodic edge with basis (t, n), $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_{\square,m}])) = \det[t\ n]\,\mathrm{sgn}(m)$.*

Proof. See Section 6.75.

6.4 Section 5 variants

The periodic triangular section in the body keeps the closed-bulk calculation, the Chern-sign calculation, and the primitive straight Toeplitz edge map. The following subsections collect the variant statements that use the same inputs with extra quotient, tail, finite-defect, finite-corner, or annular hypotheses.

6.5 Robustness variants: finite defects

Corollary 6.15 (Robust straight-periodic Toeplitz winding)**.** *In the setting of Corollary 5.9, the Toeplitz winding values are unchanged by the following operations, provided the stated sidewise comparison gaps remain open:*

(a) replacing the closed periodic triangular bulk by $H^{\triangle}_{\pm,m}$ with fixed $m > 0$;

(b) adding a self-adjoint finite-range periodic bulk perturbation V to $H^{\triangle}_{\pm,m}$ with $\sup_k \|V(k)\| < g_{\pm,m}$, and in particular adding V to $H^{\triangle}_{\pm}$ with $\sup_k \|V(k)\| < 2^{-13}$;

(c) adding any bounded finite-propagation self-adjoint perturbation supported in a fixed straight-edge collar, including periodic edge or collar terms that are not norm-small;

(d) replacing scalar jamming by a bounded componentwise scalar jam above the positivity threshold, or by a sufficiently positive bounded finite-propagation copy-side confining potential;

(e) changing finitely many initial components, adding norm-vanishing component defects, or passing through a norm-continuous sidewise homotopy with uniformly open comparison gaps.

(f) replacing the straight interface subset Y by Y' at finite Hausdorff distance from Y, with the same sidewise realization outside a fixed neighborhood of $Y \cup Y'$;

(g) adding a bounded self-adjoint finite-propagation perturbation to the jammed operator, or to the sidewise data with the comparison gaps still open, supported in a fixed edge collar and in a finite tangential region;

(h) after deleting finitely many coarse-disjoint-union components, making arbitrary bounded finite-propagation changes on the deleted components when the tail has a split, finite-corner, or K_1-trivial-corner Toeplitz description.

The same conclusion holds componentwise for every split finite family covered by Theorem 6.28 and for every finite-corner split family covered by Theorem 6.17. The finite-defect and tail-split cases have the same winding values as the associated straight periodic tail.

Proof. See Section 6.61.

Theorem 6.16 (Bounded-distance rough and eventually periodic collars)**.** *Let Y_0 be a primitive straight periodic interface for a finite-range periodic bulk projection P, and let $Y_1 \subset X$ be another interface subset at finite Hausdorff distance from Y_0. Then*

$$C^*(Y_0 \subset X) = C^*(Y_1 \subset X) =: \mathcal{I}.$$

Suppose $(J_i; T_i^+, T_i^-)$, $i = 0, 1$, are sidewise class-A interfaces along Y_i, and that

$$J_1 - J_0, \qquad T_1^+ - T_0^+, \qquad T_1^- - T_0^- \in \mathcal{I}$$

after viewing both realizations on the same Roe space. Then the quotient Fermi data agree:

$$\pi(\chi_{(-\infty,0)}(T_1^{\pm})) = \pi(\chi_{(-\infty,0)}(T_0^{\pm})), \qquad p_{J_1} = p_{J_0}.$$

Consequently

$$\mathrm{Ind}_{\mathrm{int}}(J_1; T_1^+, T_1^-) = \mathrm{Ind}_{\mathrm{int}}(J_0; T_0^+, T_0^-) \in K_1(\mathcal{I}),$$

and the rough-edge winding is the straight Toeplitz winding

$$\det[t\ n]\ \mathrm{Ch}(P)$$

for the primitive tangential/positive-normal basis (t, n) of the straight tail.

If the difference between the rough collar and the straight periodic collar is supported in a fixed normal collar and in a finite tangential interval, then the displayed differences are finite rank. Thus eventually periodic straight or bounded-distance rough collars with a finite transition region have the same quotient class and the same winding as their periodic tail, provided the sidewise comparison gaps remain open.

Proof. See Section 6.62.

The fixed-collar robustness checks in Figures 16 and 17 and table 10 are finite-matrix checks of the same ideal-stability mechanism; the winding equality here is the quotient-ideal argument, not a separate finite-volume index.

6.6 Robustness variants: finite corners

Theorem 6.17 (Finite-corner split Toeplitz additivity)**.** *Let H and P be as in Theorem 5.7. Suppose an interface ideal $\mathcal{I}_{\mathrm{poly}}$ contains a finite-dimensional corner ideal $\mathcal{F}$ and fits into an exact sequence*

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}_{\mathrm{poly}} \xrightarrow{q} \bigoplus_{j=1}^{r} \mathcal{I}_j \longrightarrow 0,$$

where each $\mathcal{I}_j$ is identified with the straight periodic Toeplitz edge ideal $C(S^1, \mathcal{K})$ for a primitive tangential/normal lattice basis (t_j, n_j). Assume that, after quotienting by $\mathcal{F}$, the corresponding quotient extension is the direct sum of these r Toeplitz extensions. If

$$\partial_{\mathrm{poly}}([P]) \in K_1(\mathcal{I}_{\mathrm{poly}})$$

is the boundary class for the polygonal or cornered interface extension, then

$$q_* \partial_{\mathrm{poly}}([P]) = \bigoplus_{j=1}^{r} \partial_{\mathrm{T},t_j,n_j}([P]).$$

Moreover $q_* : K_1(\mathcal{I}_{\mathrm{poly}}) \to K_1(\bigoplus_j \mathcal{I}_j)$ is injective. Thus the finite corner ideal contributes no K_1-coordinate, and the j-th component winding is

$$\mathrm{wind}_j\big(\partial_{\mathrm{poly}}([P])\big) = \det[t_j\ n_j]\ \mathrm{Ch}(P).$$

Proof. See Section 6.57.

Corollary 6.18 (K1-trivial corner ideals)**.** *In Theorem 6.17, the finite-dimensional hypothesis on $\mathcal{F}$ may be replaced by the weaker condition*

$$K_1(\mathcal{F}) = 0.$$

Under the same quotient-extension hypothesis, q_* is injective on K_1, the finite-corner class is determined by its straight-edge quotient coordinates, and

$$\mathrm{wind}_j\big(\partial_{\mathrm{poly}}([P])\big) = \det[t_j\ n_j]\ \mathrm{Ch}(P).$$

This applies, in particular, when $\mathcal{F}$ is finite-dimensional, a c_0-direct sum of finite-dimensional matrix algebras, or a bounded product of finite-dimensional matrix algebras.

Proof. See Section 6.58.

Corollary 6.19 (Collar-separation criterion for finite-corner quotients)**.** *Let $Y = C \cup Y^{(1)} \cup \cdots \cup Y^{(r)}$ be a cornered interface collar in a bounded-geometry Roe space X. Suppose:*

(i) the corner-supported ideal $\mathcal{F}_C$, obtained as the closure of operators supported in bounded neighborhoods of C, has $K_1(\mathcal{F}_C) = 0$;

(ii) the edge pieces separate off the corner set: for every propagation bound R there is $S(R)$ such that the sets $N_R(Y^{(i)}) \setminus N_{S(R)}(C)$ and $N_R(Y^{(j)}) \setminus N_{S(R)}(C)$ are disjoint whenever $i \neq j$;

(iii) outside a bounded neighborhood of C, each $Y^{(j)}$ is identified with a primitive straight periodic collar, and restriction to that collar identifies its edge quotient ideal with the Toeplitz ideal $\mathcal{I}_j \cong C(S^1, \mathcal{K})$.

Then the interface ideal $\mathcal{I}_Y = C^*(Y \subset X)$ fits into an exact sequence

$$0 \longrightarrow \mathcal{F}_C \longrightarrow \mathcal{I}_Y \xrightarrow{q} \bigoplus_{j=1}^{r} \mathcal{I}_j \longrightarrow 0.$$

If the quotient extension is the corresponding direct sum of primitive straight Toeplitz extensions, the component winding formula of Corollary 6.18 applies.

Proof. See Section 6.59.

Remark 6.20. The split hypotheses in Theorems 6.28 and 6.17 are genuine extra localization inputs. They hold for finite direct sums of independent straight periodic half-space collars and for cornered models where removing a K_1-trivial corner ideal leaves such a direct sum, for instance under the collar-separation criterion in Corollary 6.19. They are not asserted for arbitrary curved or nonperiodic boundary components.

6.7 Robustness variants: tail-split and annular

Corollary 6.21 (Finite covers and finite-index cell enlargements)**.** *Let $A \in M_2(\mathbb{Z})$ have $\det A \neq 0$, and define the finite-cover pullback*

$$H^{\triangle}_{\pm,A}(\kappa) := H^{\triangle}_{\pm}(A\kappa), \qquad \kappa \in \mathbb{T}^2.$$

Then $H^{\triangle}_{\pm,A}$ is gapped at zero and

$$\mathrm{Ch}\big(E_-(H^{\triangle}_{+,A})\big) = -\det A, \qquad \mathrm{Ch}\big(E_-(H^{\triangle}_{-,A})\big) = \det A.$$

The same conclusion holds for the positive-mass symbols $H^{\triangle}_{\pm,m}(A\kappa)$, $m > 0$. Small self-adjoint finite-range perturbations with norm below the corresponding pulled-back gap preserve these Chern numbers.

If the same periodic triangular operator is only rewritten with respect to a finite-index sublattice of periods, equivalently with a larger real-space fundamental cell and the folded Bloch Hamiltonian, the zero gap and the Chern signs -1 for $H^{\triangle}_{+}$ and $+1$ for $H^{\triangle}_{-}$ are unchanged.

Proof. See Section 6.47.

Corollary 6.22 (Finite periodic triangular tori)**.** *Every finite $N_1 \times N_2$ periodic quotient obtained from the symbol above has a zero gap bounded below by 2^{-13}, uniformly in N_1, N_2. For quotients large enough to be embedded as simplicial tori, this is the corresponding finite periodic triangular torus. Its occupied finite-volume Bloch states are samples of the same rank-three Bloch bundle.*

Proof. See Section 6.48.

The next interface statement uses the internal periodic Bloch gap and Chern sign from Theorems 5.2 and 5.5. Straight periodic winding values are evaluated by the Toeplitz extension inside the paper; nonstraight local pairings are not asserted in this periodic block.

Corollary 6.23 (Periodic annular closed-bulk input and interface class)**.** *Let $(\mathcal{K}_N)_N$ be a sequence of triangular-lattice cylinders whose doubles are the periodic triangular tori of Corollary 6.22. Then the closed-bulk gap and Chern value needed for the doubled-and-jammed boundary construction are supplied by Theorems 5.2 and 5.5, not by the Higson–Prodan closed-bulk theorem. Hence, for every sufficiently large jam scale M, the doubled-and-jammed interface class is*

$$\partial_Y(E_+(\pi(P_\pm))) \in K_1(C^*(Y \subset X)).$$

For primitive straight periodic components, the Toeplitz pairing gives winding signs -1 for $H^{\triangle}_{+}$ and $+1$ for $H^{\triangle}_{-}$.

Proof. See Section 6.49.

The FHS computation in Table 1 independently confirms the Bloch Chern signs in Theorem 5.5 on the displayed grids, and Table 2 checks the finite-cover, folded-cell, and fixed-positive-mass variants used below.

Corollary 6.24 (Robust periodic annular input)**.** *The conclusion of Corollary 6.23 remains valid if the closed periodic triangular bulk Hamiltonian on the doubled cylinders is replaced by one of the following finite-range periodic bulk models:*

(i) $H^{\triangle}_{\pm,m}(k) = B(k) + B(k)^* \pm mS(k)$, *for any fixed $m > 0$;*

(ii) $H^{\triangle}_{\pm,m}(k) + V(k)$, *where $m > 0$ is fixed, V is self-adjoint finite-range periodic, and $\sup_k \|V(k)\| < g_{\pm,m}$ in the notation of Corollary 5.6;*

(iii) $H^{\triangle}_{\pm}(k) + V(k)$, *where V is self-adjoint, finite-range periodic, and*

$$\sup_{k\in\mathbb{T}^2} \|V(k)\| < 2^{-13}.$$

(iv) the finite-cover pullbacks and finite-index cell enlargements covered by Corollary 6.21.

The coarse interface collar Y is enlarged, if necessary, by the finite propagation radius of the chosen periodic bulk model. For each such model, the closed-bulk gap and the closed-bulk Chern sign are supplied internally by Proposition 5.4 and Corollary 5.6. Therefore, for every jam scale above the corresponding copy-side positivity threshold, the doubled-and-jammed family has the interface class

$$\partial_Y(E_+(\pi(P_\pm^{\mathrm{per}}))) \in K_1(C^*(Y \subset X)),$$

where $P_\pm^{\mathrm{per}}$ is the negative Fermi projection of the chosen periodic closed-bulk model. This class is independent of the large scalar jam scale and is unchanged if the scalar jam is replaced by any sufficiently positive bounded finite-propagation copy-side confining potential. For primitive straight periodic components, the Toeplitz winding value is the corresponding periodic Bloch Chern number supplied by Theorem 5.5 and Corollary 6.21.

Proof. See Section 6.50.

Corollary 6.25 (Periodic boundary families with internal bulk input)**.** *Let $(\mathcal{K}_n)_n$ be any sequence of triangular-lattice cylinders whose doubles are finite periodic triangular tori, with arbitrary periods tending to infinity along a coarse disjoint union. The following choices give boundary families for which the closed-bulk locality, gap, and Chern-sign inputs are verified internally in this paper:*

(a) the unperturbed periodic triangular Hamiltonians $H_\pm^{\triangle}$;

(b) the positive-duality-strength Hamiltonians $H_{\pm,m}^{\triangle}$, for any fixed $m > 0$;

(c) the perturbed Hamiltonians $H_{\pm,m}^{\triangle} + V$, for any fixed $m > 0$ and any self-adjoint finite-range periodic V satisfying $\sup_k \|V(k)\| < g_{\pm,m}$;

(d) the explicit-radius perturbations $H_\pm^{\triangle} + V$, for any self-adjoint finite-range periodic V satisfying $\sup_k \|V(k)\| < 2^{-13}$;

(e) finite-cover pullbacks of the triangular symbols, whose Chern numbers are multiplied by the signed covering degree, and finite-index cell enlargements, whose Chern signs are unchanged.

For every family in (a)–(e), every sufficiently large scalar jam scale, every bounded componentwise scalar jam scale bounded below by the corresponding positivity threshold, and every sufficiently positive bounded finite-propagation copy-side confining potential produces the same interface class

$$\partial_Y(E_+(\pi(P_\pm^{\mathrm{per}}))) \in K_1(C^*(Y \subset X)),$$

after enlarging the coarse interface collar by the propagation radius of the selected periodic bulk model and copy-side confinement. The class is also unchanged by altering finitely many initial components of the cylinder sequence, or by passing through any norm-continuous sidewise homotopy whose comparison gaps stay uniformly open. For primitive straight periodic components, its Toeplitz winding value is the corresponding periodic Bloch Chern number from Theorem 5.5 and Corollary 6.21.

Proof. See Section 6.51.

Corollary 6.26 (Finite-cover primitive Toeplitz formula)**.** *Let H and P be as in Theorem 5.7, and let $A \in M_2(\mathbb{Z})$ have $\det A \neq 0$. Set*

$$H_A(\kappa) := H(A\kappa), \qquad P_A(\kappa) := P(A\kappa).$$

For every primitive tangential/positive-normal basis (t, n) in the cover coordinates,

$$\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_A])) = \det[t\ n]\ \det A\ \mathrm{Ch}(P).$$

For the triangular symbols this gives

$$\mathrm{wind}(H^{\triangle}_{+,A}) = -\det[t\ n]\ \det A, \qquad \mathrm{wind}(H^{\triangle}_{-,A}) = \det[t\ n]\ \det A,$$

with the same formula for positive-mass symbols and for the small periodic perturbations covered by Corollary 5.6.

Proof. See Section 6.54.

Corollary 6.27 (Gapped periodic homotopies and Toeplitz winding)**.** *Let $s \mapsto H_s(k)$, $0 \le s \le 1$, be a norm-continuous family of finite-range self-adjoint periodic Bloch Hamiltonians with a common finite hopping range. Assume that*

$$\inf_{s,k} \mathrm{dist}\big(0, \mathrm{spec} H_s(k)\big) > 0, \qquad P_s(k) := \chi_{(-\infty,0)}(H_s(k)).$$

Then, for every primitive tangential/positive-normal basis (t, n),

$$\partial_{\mathrm{T},t,n}([P_s]) \in K_1(C(S^1, \mathcal{K}))$$

is independent of s, and

$$\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_s])) = \det[t\ n]\ \mathrm{Ch}(P_0) = \det[t\ n]\ \mathrm{Ch}(P_1).$$

Thus any finite-range periodic model connected to one of the internally verified triangular families by such a zero-gapped homotopy has the same straight-edge Toeplitz winding.

Proof. See Section 6.55.

Theorem 6.28 (Split componentwise Toeplitz winding)**.** *Let H and P be as in Theorem 5.7. Suppose a periodic straight-edge model has finitely many interface components $Y^{(1)}, \dots, Y^{(r)}$, and suppose the edge ideal splits as a C^*-direct sum of the component ideals,*

$$C^*(Y \subset X) \cong \bigoplus_{j=1}^{r} C^*(Y^{(j)} \subset X),$$

with the j-th summand identified with the Toeplitz edge ideal $C(S^1, \mathcal{K})$ for a primitive tangential/normal lattice basis (t_j, n_j). Assume also that the quotient extension is the direct sum of these component Toeplitz extensions. Then the global boundary class decomposes as

$$\partial_Y([P]) = \bigoplus_{j=1}^{r} \partial_{\mathrm{T},t_j,n_j}([P]) \quad \text{in} \quad K_1(C^*(Y \subset X)).$$

Consequently the j-th component winding is

$$\mathrm{wind}_j\big(\partial_Y([P])\big) = \det[t_j\ n_j]\ \mathrm{Ch}(P).$$

If each component is oriented by the induced boundary orientation, the tangential coordinate is reversed exactly when the normal sign is reversed, so each induced-orientation component winding records the same bulk Chern number $\mathrm{Ch}(P)$.

Proof. See Section 6.56.

Theorem 6.29 (Tail split component winding maps)**.** *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union of a straight-periodic boundary family from Corollary 6.25. Suppose that, after deleting finitely many initial components, the tail interface ideal has either*

(i) a split direct-sum Toeplitz decomposition as in Theorem 6.28, or

(ii) a finite-corner or K_1-trivial-corner quotient decomposition as in Theorem 6.17 and Corollary 6.18.

Let (t_j, n_j), $1 \le j \le r$, be the primitive tangential/positive-normal bases of the tail components, and let Ω_j be the component map obtained by deleting the finite initial components, passing to the j-th Toeplitz quotient coordinate, and applying the usual winding homomorphism. Then the maps Ω_j are well defined on the original interface class and

$$\Omega_j\big(\mathrm{Ind}_{\mathrm{int}}(J; T^+, T^-)\big) = \det[t_j \ n_j]\ \mathrm{Ch}(P).$$

For a finite-cover pullback by A, the right-hand side is

$$\det[t_j \ n_j]\ \det A\ \mathrm{Ch}(P).$$

The same formulas are unchanged by arbitrary bounded finite-propagation changes on finitely many initial components.

Proof. See Section 6.63.

Corollary 6.30 (Internal straight-periodic classes)**.** *For the following boundary/interface families, with the split, quotient, tail, and sidewise-gap hypotheses stated in the cited results, no Higson–Prodan closed-bulk input and no Ewert–Meyer pairing assumption are used. Their zero gap, Chern value, interface class, and Toeplitz winding are supplied by the internal periodic calculation and the Toeplitz evaluation above:*

(a) primitive straight periodic edges of the unperturbed triangular models $H_{\pm}^{\triangle}$;

(b) primitive straight periodic edges of $H_{\pm,m}^{\triangle}$, for every fixed $m > 0$;

(c) primitive straight periodic edges of $H_{\pm,m}^{\triangle} + V$, for every fixed $m > 0$ and every self-adjoint finite-range periodic perturbation satisfying $\sup_k \|V(k)\| < g_{\pm,m}$; in particular this includes $H_{\pm}^{\triangle} + V$ with $\sup_k \|V(k)\| < 2^{-13}$;

(d) finite-cover pullbacks of the families in (a)–(c), with Chern numbers multiplied by the signed covering degree, and finite-index cell enlargements of the same operators, with the original Chern signs unchanged;

(e) any finite split direct sum of the families in (a)–(d), with component ideals and quotient extension splitting as in Theorem 6.28;

(f) any finite-corner or K_1-trivial-corner split sum of the families in (a)–(d), with quotient extension as in Theorem 6.17, Corollary 6.18, or Corollary 6.19;

(g) finite-defect, bounded-distance straight-edge, and tail-split variants covered by Theorem 6.16, Corollary 6.15, and Theorem 6.29;

(h) the same families after replacing scalar jamming by any bounded componentwise scalar jam uniformly above the positivity threshold, or by any sufficiently positive bounded finite-propagation copy-side confinement;

(i) the same families after adding norm-vanishing component defects, with the sidewise comparison gaps still open.

For the triangular families before applying a finite cover, the winding values are -1 *for* H_+ *and* $+1$ *for* H_-*, up to the orientation factor* $\det[t_j\ n_j]$ *on each primitive component. For a finite-cover pullback, these values are further multiplied by the signed covering degree. Fixed-collar finite-propagation edge perturbations, norm-vanishing component defects, and norm-continuous sidewise homotopies preserve these values whenever the sidewise comparison gaps remain open; this last gap condition is a stability condition for the perturbation path, not an imported bulk–interface theorem.*

Proof. See Section 6.64.

6.8 Roe interface variants

This subsection records stability, convention, relabeling, tail, and confinement variants for the Roe interface K_1-class. Where an integer winding value appears, it refers to the internal Toeplitz integer pairing in the straight periodic case (Theorem 4.10); for general partitions the integer pairing is the conditional statement Theorem 4.11.

Corollary 6.31 (Small-norm stability of Fermi and interface classes)**.** *Let* $H = H^* \in C^*(X)$ *satisfy*

$$\operatorname{spec}_{\mathrm{op}}(H) \subset [-B, -g] \cup [g, B]$$

for some $B, g > 0$*. Let* $V = V^* \in C^*(X)$ *with*

$$\|V\| < g.$$

Then

$$H + V$$

is gapped at zero, with

$$\operatorname{spec}_{\mathrm{op}}(H + V) \cap \big(-(g - \|V\|),\, g - \|V\|\big) = \emptyset,$$

and

$$\big[\chi_{(-\infty,0)}(H + V)\big] = \big[\chi_{(-\infty,0)}(H)\big] \qquad \text{in } K_0(C^*(X)).$$

If, in addition, $H = H^\infty_\pm$ *under Assumption 5.13 and* $V = V^* \in C^*(X)$ *satisfies*

$$\|V\| < g_{\mathrm{bulk}},$$

then, with

$$P_V := \chi_{(-\infty,0)}(H^\infty_\pm + V).$$

one has

$$\partial_Y\big(E_+(\pi(P_V))\big) = \partial_Y\big(E_+(\pi(P_\pm))\big) \qquad \text{in } K_1(C^*(Y \subset X)).$$

Proof. See Section 6.85.

Proposition 6.32 (Interface-supported sidewise changes)**.** *Let $(J;T^+,T^-)$ be a sidewise class-A interface along Y. Let*

$$J' = J + V, \qquad T'^{+} = T^+ + V^+, \qquad T'^{-} = T^- + V^-,$$

where $V, V^+, V^- \in \mathcal{I}$ are self-adjoint. Assume that $(J';T'^{+},T'^{-})$ is again a sidewise class-A interface along Y. Then

$$p_{J'} = p_J, \qquad \mathrm{Ind}_{\mathrm{int}}(J';T'^{+},T'^{-}) = \mathrm{Ind}_{\mathrm{int}}(J;T^+,T^-).$$

The special case $V^+ = V^- = 0$ gives invariance under perturbing only the interface Hamiltonian J by a self-adjoint element of $\mathcal{I}$.

Proof. See Section 6.94.

Proposition 6.33 (Stability under finite-component changes)**.** *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union, and assume each Y_n is nonempty. Let $V = V^*$ be a bounded finite-propagation operator supported on only finitely many components: there is $N < \infty$ such that*

$$V\chi_{X_n} = \chi_{X_n}V = 0 \qquad \text{for all } n > N.$$

Then $V \in C^*(Y \subset X)$. Consequently, changing any of the sidewise data J, T^+, T^- on finitely many initial components does not change the interface class, provided the sidewise gap hypotheses remain valid after the change.

Proof. See Section 6.95.

Proposition 6.34 (Norm-vanishing component defects)**.** *Let $X = \bigsqcup_n X_n$ be the coarse disjoint union and assume each Y_n is nonempty. Let*

$$V = \bigoplus_{n=1}^{\infty} V_n$$

be a bounded self-adjoint block-diagonal operator with $V_n \in \mathcal{B}(\ell^2(X_n))$ and $\|V_n\| \to 0$. Then

$$V \in C^*(Y \subset X).$$

Consequently, adding such norm-vanishing component defects to J, T^+, or T^- does not change the sidewise interface class, provided the resulting comparison operators remain gapped and the sidewise class-A hypotheses remain valid.

Proof. See Section 6.96.

Proposition 6.35 (Homotopy invariance of the sidewise interface class)**.** *Let*

$$s \longmapsto (J_s; T_s^+, T_s^-), \qquad s \in [0,1],$$

be a family of sidewise class-A interfaces along the same interface Y. Assume that $s \mapsto T_s^{\pm}$ are norm-continuous and that there is a uniform comparison gap

$$\mathrm{dist}(0, \mathrm{spec}_{\mathrm{op}}(T_s^{\pm})) \geq \gamma > 0$$

for all $s \in [0,1]$. Then

$$\mathrm{Ind}_{\mathrm{int}}(J_s; T_s^+, T_s^-)$$

is independent of s.

Proof. See Section 6.97.

Corollary 6.36 (Topological independence of the confining copy potential)**.** *Assume 5.13. Let $s \mapsto V_s$, $s \in [0,1]$, be a norm-continuous family of bounded self-adjoint finite-propagation operators on $\ell^2(X)$ such that*

$$V_s = P^{\infty}_{\mathrm{copy}} V_s P^{\infty}_{\mathrm{copy}}$$

and, on the copy sector,

$$V_s \geq M_0 \mathbf{1} \qquad \text{for a constant } M_0 > B_{\mathrm{copy},-}.$$

Define

$$J_s := H^{\infty}_{\pm} + V_s, \qquad T^{+}_s := H^{\infty}_{\pm}, \qquad T^{-}_s := P^{\infty}_{\mathrm{orig}} + H^{\infty}_{\pm,\mathrm{copy}} + V_s.$$

Then the sidewise interface class of $(J_s; T^{+}_s, T^{-}_s)$ is independent of s.

Proof. See Section 6.98.

Theorem 6.37 (Interface theorem for general confining copy potentials)**.** *Assume 5.13. Let $V = V^*$ be a bounded finite-propagation operator on $\ell^2(X)$ such that*

$$V = P^{\infty}_{\mathrm{copy}} V P^{\infty}_{\mathrm{copy}}$$

and, on the copy sector,

$$V \geq M_0 \mathbf{1} \qquad \text{for some } M_0 > B_{\mathrm{copy},-}.$$

Define

$$J_V := H^{\infty}_{\pm} + V, \qquad T^{+} := H^{\infty}_{\pm}, \qquad T^{-}_V := P^{\infty}_{\mathrm{orig}} + H^{\infty}_{\pm,\mathrm{copy}} + V.$$

Then $(J_V; T^{+}, T^{-}_V)$ is a sidewise class-A interface along Y. Its canonical quotient projection is

$$p_{J_V} = E_{+}\big(\pi(P_{\pm})\big),$$

the negative-side comparison has the quantitative positivity margin

$$\operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^{-}_V)\big) \geq \min\{1, M_0 - B_{\mathrm{copy},-}\},$$

and hence

$$\operatorname{Ind}_{\mathrm{int}}(J_V; T^{+}, T^{-}_V) = \partial_Y\big(E_{+}(\pi(P_{\pm}))\big) \in K_1(C^*(Y \subset X)).$$

In particular, the interface class is unchanged by replacing scalar jamming by any sufficiently positive bounded finite-propagation copy-side confining potential.

Proof. See Section 6.99.

Corollary 6.38 (Independence of the interface class for large jam scales)**.** *Assume 5.13. If $M_1, M_2 > B_{\mathrm{copy},-}$, then*

$$\operatorname{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M_1}; T^{+}_1, T^{-}_1) = \operatorname{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M_2}; T^{+}_2, T^{-}_2) = \partial_Y\big(E_{+}(\pi(P_{\pm}))\big),$$

where

$$T^{+}_j = H^{\infty}_{\pm}, \qquad T^{-}_j = \widetilde{T}^{-}_{\pm,M_j}.$$

Proof. See Section 6.101.

Combining Theorems 5.35 and 5.36 with Corollary 6.38, we obtain the following explicit statement.

Corollary 6.39 (Interface class from a Chern-± 1 bulk projection)**.** *Assume 5.13, the additional Higson–Prodan hypotheses from Theorem 5.36, and $M > B_{\mathrm{copy},-}$. Then*

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^{+}, T^{-}) = \partial_Y\big(E_+(\pi(P_\pm))\big) \in K_1\big(C^*(Y \subset X)\big),$$

where $P_\pm$ is the Higson–Prodan bulk Fermi projection with $\mathrm{Ch}([P_\pm]) = \pm 1$. The interface class depends only on the bulk projection and the quotient decomposition, not on the jam scale M.

Proof. See Section 6.103.

Corollary 6.40 (Higson–Prodan input implies the doubled-and-jammed interface class)**.** *Assume that the closed doubled refinement family satisfies the Higson–Prodan closed-bulk input in Assumption 5.15. Then Assumptions 5.8 and 5.13 hold. Consequently, for every $M > B_{\mathrm{copy},-}$,*

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^{+}, T^{-}) = \partial_Y\big(E_+(\pi(P_\pm))\big) \in K_1\big(C^*(Y \subset X)\big),$$

where $P_\pm$ is the Higson–Prodan bulk Fermi projection and

$$\mathrm{Ch}([P_\pm]) = \pm 1.$$

The construction of this K_1-class is internal to the present paper; in the original/copy partition the K_1-class identity is Theorem 4.9. The integer winding evaluation is internal in the straight periodic Toeplitz case (Theorem 4.10) and is the conditional corollary Theorem 4.11 for general partitions.

Proof. See Section 6.104.

Proposition 6.41 (Ewert–Meyer convention criterion)**.** *Let $(\mathcal{K}_n)_n$ be a uniformly regular full-boundary refinement family. Suppose that the closed doubled family $D(\mathcal{K}_n)$ satisfies the Higson–Prodan closed-bulk input in Assumption 5.15. Then the original/copy coarse data satisfy the HP–Ewert–Meyer boundary convention of Definition 5.37, with the Roe/Ewert–Meyer K_1-class identity supplied by Theorem 4.9. No boundary spectral gap is required.*

Proof. See Section 6.105.

Corollary 6.42 (HP-covered boundary refinement families)**.** *Let $(\mathcal{K}_n)_n$ be a refinement sequence of a fixed compact oriented PL surface with full boundary subcomplex. Suppose the closed doubles $D(\mathcal{K}_n)$, after the standard rescaling used in the closed Higson–Prodan refinement limit, are covered by the Higson–Prodan closed universal-model theorem:*

(i) the closed doubles have the HP closed-bulk gap and Chern value;

(ii) the original/copy decomposition gives the coarse partition of Definition 5.37.

Then the doubled-and-jammed original/copy family has the interface K_1-class of Theorem 4.9. The following common refinement schemes are natural HP-covered families when they satisfy the bounded-geometry hypotheses of the closed theorem:

(a) iterated barycentric refinements;

(b) bounded-valence stellar or edge refinements;

(c) quasi-uniform shape-regular triangulations used in finite-element refinements.

For any such family, and every $M > B_{\mathrm{copy},-}$, the doubled-and-jammed interface K_1-class is given by Theorem 4.9; in the straight periodic Toeplitz case the integer winding is ± 1 internally (Theorem 4.10), and for general partitions the integer winding is the conditional statement Theorem 4.11.

Proof. See Section 6.106.

Remark 6.43 (Only the tail of a refinement family matters). Suppose that, after deleting finitely many initial refinements, the tail $(\mathcal{K}_n)_{n\geq N_0}$ is HP-covered in the sense of Corollary 6.42. Extend the tail by any finite set of initial components with bounded local sidewise data for which the comparison operators are gapped at zero. Then the doubled-and-jammed interface class and its Ewert–Meyer winding value are the same as those of the tail. In particular, the HP-covered condition only has to hold eventually along a refinement sequence; see Section 6.107 for the verification.

Remark 6.44 (Relabeling and orientation reversal). The HP–Ewert–Meyer boundary convention, the interface class, and the Ewert–Meyer winding value are invariant under orientation-preserving boundary-compatible simplicial isomorphisms of refinement families. If the surface orientation is reversed, then the roles of H_+ and H_- are interchanged and the winding sign is reversed; see Section 6.109.

Corollary 6.45 (Robustness of the interface class and winding)**.** *Under the hypotheses of Theorem 5.38, the interface K_1-class is unchanged by the following operations, whenever the stated sidewise comparison gaps remain open. In the straight periodic Toeplitz case (Theorem 4.10), the resulting integer winding is also unchanged.*

(a) replacing the scalar jam scale M by any other scalar $M' > B_{\mathrm{copy},-}$;

(b) replacing scalar jamming by a bounded finite-propagation copy-side confinement V satisfying

$$V = P^{\infty}_{\mathrm{copy}} V P^{\infty}_{\mathrm{copy}}, \qquad V \geq M_0 \mathbf{1}, \qquad M_0 > B_{\mathrm{copy},-};$$

this includes bounded componentwise scalar jam scales $V = \bigoplus_n M_n P_{\mathrm{copy},n}$ with $B_{\mathrm{copy},-} < \inf_n M_n \leq \sup_n M_n < \infty$;

(c) changing the sidewise data by self-adjoint interface-supported operators;

(d) changing the boundary/copy coupling by uniformly bounded finite-propagation perturbations supported in any fixed boundary collar;

(e) replacing the chosen fixed collar model for Y by another fixed-radius collar;

(f) changing the operators on finitely many initial refinement components;

(g) adding norm-vanishing component defects;

(h) passing through a norm-continuous sidewise homotopy with a uniform comparison gap.

Proof. See Section 6.110.

6.9 Symbolic verification appendix

The companion script `periodic_triangular_bulk_certificate.py` in the project repository implements the certificate (GitHub project repository). The script numerically evaluates the closed-bulk Chern number on the triangular fundamental cell at four grid sizes and writes the certificate table `periodic_triangular_certificate_table.tex`. The minimal SymPy snippet below is illustrative: it builds the Bloch boundary operator, the cap-product blocks, and the Hermitian duality block $S(k)$ using the same convention as the companion script ($S[4:6, 0:1] = \frac{i}{2}(P_2^* + P_0)$, $S[1:4, 1:4] = \frac{i}{2}(P_1^* - P_1)$, $S[0:1, 4:6] = -\frac{i}{2}(P_0^* + P_2)$), and prints the Hermitian Hamiltonians at a sample point. The Chern-sign verification itself uses the numerical Fukui–Hatsugai–Suzuki procedure in the companion script, which produces the integer table reproduced as Table 1 below.

```
# Minimal symbolic check. Builds B(k), P_p(k), S(k) and the two Hermitian
# Bloch Hamiltonians H_pm(k) = B(k) + B(k)^* +- S(k). Same conventions as
# the companion script periodic_triangular_bulk_certificate.py.
import sympy as sp
q1, q2 = sp.symbols('q1 q2', real=True)
i = sp.I

# Ordered basis: v, a, b, c, f, g.
z1 = sp.exp(i*q1)
z2 = sp.exp(i*q2)
d1 = sp.Matrix([[z1-1, z2-1, z1*z2-1]])
d2 = sp.Matrix([[1, -z2], [z1, -1], [-1, 1]])

# Bloch boundary operator B(k) (degree-lowering).
B = sp.zeros(6)
B[0:1, 1:4] = d1
B[1:4, 4:6] = d2

# Cap-product blocks (same convention as the companion script).
P0 = sp.Matrix([[sp.conjugate(z1*z2)], [sp.conjugate(z2)]])
P1 = sp.Matrix([[0, sp.conjugate(z1), 0], [0, 0, 0],
                [-sp.conjugate(z2), 0, 0]])
P2 = sp.Matrix([[1, 1]])

# Hermitian Bloch duality block S(k), single consistent assignment per block:
#   S[4:6, 0:1] =  (i/2) * (P2^* + P0)
#   S[1:4, 1:4] =  (i/2) * (P1^* - P1)
#   S[0:1, 4:6] = -(i/2) * (P0^* + P2)
S = sp.zeros(6)
S[4:6, 0:1] = (i/2) * (P2.conjugate().T + P0)
S[1:4, 1:4] = (i/2) * (P1.conjugate().T - P1)
S[0:1, 4:6] = (-i/2) * (P0.conjugate().T + P2)

H_plus  = B + B.conjugate().T + S
H_minus = B + B.conjugate().T - S

# Hermiticity sanity check at a sample point.
```

```
sample = {q1: sp.Rational(1, 7), q2: sp.Rational(2, 9)}
herm_plus = sp.simplify(H_plus.subs(sample) - H_plus.subs(sample).H)
print("H_+(k) Hermiticity defect at sample k:")
print(herm_plus)
```

Note on this listing. The script `periodic_triangular_bulk_certificate.py` in the project repository (GitHub project repository) is the load-bearing implementation: it runs the Fukui–Hatsugai–Suzuki Chern computation on a finite k-grid and writes the certificate table (Table 1). The SymPy snippet above is an illustrative excerpt showing the convention; it builds the Hermitian Bloch matrices $H_{\pm}(k)$ and is short enough to run unmodified. The proof of the Chern signs is the analytic harmonic-compression and crossing-orientation argument in Section 6.45, not the symbolic computation.

The proof of each result in the main text is collected here in the order in which the results appear.

6.10 Proof of Result 2.2

Proof of Theorem 2.2. Expand $H_{\pm}^2$ and use the assumed anticommutation to eliminate the cross term. Since $(B + B^\dagger)^2 \succeq 0$ and $S^2 \succeq \mu_S^2 \mathbf{1}$, we obtain $H_{\pm}^2 \succeq \mu_S^2 \mathbf{1}$. □

6.11 Proof of Result 2.3

Proof of Theorem 2.3. Since

$$H_{\pm}^2 = Q^2 + S^2 \pm (QS + SQ),$$

we have $Q^2 \succeq 0$ and $S^2 \succeq \mu_S^2 \mathbf{1}$. Also,

$$\pm(QS + SQ) \succeq -\|QS + SQ\| \, \mathbf{1} = -\varepsilon_{\mathrm{ac}} \mathbf{1}.$$

Hence $H_{\pm}^2 \succeq (\mu_S^2 - \varepsilon_{\mathrm{ac}})\mathbf{1}$. The spectral-gap conclusion follows immediately. □

6.12 Proof of Result 2.4

Proof of Theorem 2.4. We verify that $D(\mathcal{K})$ is a well-defined abstract simplicial complex. A simplex is determined by its vertex set, so we must check that no two distinct simplices (of any dimension) in the doubled complex share the same vertex set.

Faces. Let $F = \{v_0, v_1, v_2\}$ be a face of $\mathcal{K}$. By the fullness condition in Definition 2.1, F has at least one interior vertex v_0. Every reflected face contains at least one copy-side vertex $\bar{w} \in \overline{\mathcal{K}}_0^{\mathrm{int}}$, and no original face contains any copy-side vertex. Hence no original face coincides with any reflected face. Since faces within each copy are distinct, all 2-simplices of $D(\mathcal{K})$ have distinct vertex sets.

Edges. Interior edges of $\mathcal{K}$ have at least one interior endpoint (by fullness: both endpoints on $\partial\mathcal{K}$ would force the edge to be a boundary edge). Hence every reflected interior edge acquires at least one copy-side vertex and has a different vertex set from the original. Boundary edges are identified (not duplicated), so they appear once. Thus all 1-simplices of $D(\mathcal{K})$ have distinct vertex sets.

Edge valences. Every interior edge of $\mathcal{K}$ is incident to exactly two faces in $\mathcal{K}$. Its reflected copy (with a different vertex set) is incident to exactly two reflected faces. So each such pair contributes two edges, each incident to two faces. Every boundary edge is incident to one original face and one reflected face, hence to exactly two faces. Thus every edge of $D(\mathcal{K})$ is incident to exactly two faces.

At an interior vertex of either copy, the link is already a cycle. At a boundary vertex v, the link in $\mathcal{K}$ is a path and the link in $\overline{\mathcal{K}}$ is a second path with the opposite boundary orientation. After the boundary identification these two paths glue end-to-end to form a cycle. Thus every vertex link in $D(\mathcal{K})$ is a cycle, so $D(\mathcal{K})$ is a closed triangulated surface.

Finally, orient the faces in the original copy by the given surface orientation and orient the faces in the reflected copy by the opposite orientation. Along every glued boundary edge, the two induced edge orientations are opposite, so the face orientations are compatible across the gluing. Hence $D(\mathcal{K})$ is oriented. □

6.13 Proof of Result 2.5

Proof of Theorem 2.5. Faces are duplicated and never identified (each reflected face has a distinct vertex set by the fullness condition), so $|D(\mathcal{K})_2| = 2|\mathcal{K}_2|$. Every non-boundary edge has at least one interior endpoint (by fullness), so its reflected copy has a distinct vertex set; non-boundary edges are therefore duplicated. Boundary edges are identified, not duplicated. Hence $|D(\mathcal{K})_1| = 2|\mathcal{K}_1^{\mathrm{int}}| + |\partial\mathcal{K}_1| = 2(|\mathcal{K}_1| - |\partial\mathcal{K}_1|) + |\partial\mathcal{K}_1| = 2|\mathcal{K}_1| - |\partial\mathcal{K}_1|$. The same reasoning applies to vertices: interior vertices are duplicated, boundary vertices are identified, giving $|D(\mathcal{K})_0| = 2|\mathcal{K}_0| - |\partial\mathcal{K}_0|$. Since $\partial\mathcal{K}$ is a disjoint union of cycles, each connected component has as many vertices as edges, so $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. □

6.14 Proof of Result 2.1

Proof of Theorem 2.1. Use

$$\dim \mathcal{H}(\mathcal{K}) = |\mathcal{K}_0| + |\mathcal{K}_1| + |\mathcal{K}_2|, \qquad \dim \mathcal{H}(D(\mathcal{K})) = |D(\mathcal{K})_0| + |D(\mathcal{K})_1| + |D(\mathcal{K})_2|,$$

insert Theorem 2.5, and simplify with $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. □

6.15 Proof of Result 2.2

Proof of Theorem 2.2. Insert the identities from Theorem 2.5 into $\chi = |\mathcal{K}_0| - |\mathcal{K}_1| + |\mathcal{K}_2|$ and use $|\partial\mathcal{K}_0| = |\partial\mathcal{K}_1|$. □

6.16 Proof of Result 2.3

Proof of Theorem 2.3. For an orientable surface with boundary, $\chi(\mathcal{K}) = 2 - 2g - b$ [15]. By Proposition 2.2,

$$\chi(D(\mathcal{K})) = 4 - 4g - 2b.$$

Since a closed orientable surface of genus G has Euler characteristic $2 - 2G$, one gets $G = 2g + b - 1$. □

6.17 Proof of Result 2.6

Proof of Theorem 2.6. For a p-cochain α, identified with a p-chain by the simplex basis, the standard cap-product boundary formula [15, Section 3.3] is

$$\partial(c \cap \alpha) = (-1)^p\big((\partial c) \cap \alpha - c \cap \delta\alpha\big).$$

Taking $c = [\mathcal{K}, \partial\mathcal{K}]$, whose boundary is zero in the relative chain complex, gives (3). Applying j_{1-p} to that identity and subtracting from $B_{\mathcal{K},2-p} j_{2-p} P_p^{\mathrm{rel}}$ gives (3).

The operator $B_{\mathcal{K},r}j_r - j_{r-1}\partial_r^{\mathrm{rel}}$ is zero on every relative simplex whose codimension-one faces are all represented by non-boundary simplices. When it is nonzero, its output is a boundary simplex, and its input simplex meets $\partial\mathcal{K}$. Since the cap-product maps P_p^{rel} have a fixed finite propagation radius, a nonzero coefficient in (3) can occur only with source and target inside a uniformly bounded collar of $\partial\mathcal{K}$. The operator $S_{\mathcal{K}}^{\mathrm{PL}}$ is obtained from $\widehat{P}$ and $\widehat{P}^\dagger$ by fixed degree phases and Hermitian symmetrization; taking adjoints preserves the same support condition. Hence the displayed duality defect is collar-supported. □

6.18 Proof of Result 2.7

Proof of Theorem 2.7. With the coefficient convention in Section 3, a matrix coefficient of either cap-product operator is a signed count of oriented faces for which the source simplex appears as the prescribed suffix and the target simplex appears as the complementary prefix. If both simplices are outside the $R_{\frown}$-collar, every face contributing to the coefficient is an original-side face disjoint from the glued boundary. On such faces the closed fundamental cycle of $D(\mathcal{K})$ restricts exactly to the relative fundamental chain $[\mathcal{K}, \partial\mathcal{K}]$, and the quotient map q does not remove the complementary prefix. Hence the two signed coefficient sums are identical. □

6.19 Proof of Result 2.8

Proof of Theorem 2.8. Self-adjointness is built into the Hermitian symmetrization. Locality follows from the chain-level cap product: a matrix coefficient of P^{rel} can be nonzero only when the two simplices are incident inside a common face or star. The representative map j removes boundary representatives but does not enlarge support, so $S_{\mathcal{K}}^{\mathrm{PL}}$, and hence $H_{\pm}^{\mathrm{PL}}(\mathcal{K})$, has finite propagation.

The relative duality defect is collar-supported by Lemma 2.6.

The $B + B^\dagger$ part of $H_{\pm}^{\partial}(\mathcal{K})$ agrees exactly with $B_{\mathcal{K}} + B_{\mathcal{K}}^\dagger$. Compressing the doubled boundary operator to the original side keeps the original faces, edges, and vertices and removes only reflected simplices not contained in $\partial\mathcal{K}$. Thus the difference between $H_{\pm}^{\partial}(\mathcal{K})$ and $H_{\pm}^{\mathrm{PL}}(\mathcal{K})$ is entirely in the cap-product term.

Lemma 2.7 gives coefficient-by-coefficient equality between the doubled cap-product operator, compressed to the original side, and the lifted relative cap-product operator away from a fixed boundary collar. Thus any discrepancy can occur only when the local cap-product calculation involves a simplex meeting $\partial\mathcal{K}$, or when the relative quotient has discarded a boundary representative. Since the cap-product rule has a fixed finite propagation radius, all such discrepancies have both source and target inside a fixed boundary collar. Defining

$$R_{\partial,\pm} := H_{\pm}^{\partial}(\mathcal{K}) - H_{\pm}^{\mathrm{PL}}(\mathcal{K})$$

gives (2.3). The difference of two self-adjoint operators is self-adjoint, and the collar-support statement follows from the preceding support analysis. The uniform statement follows because the cap-product propagation radius is fixed over a uniformly regular refinement family. □

6.20 Proof of Result 2.4

Proof of Theorem 2.4. The difference on the n-th component is $R_{\partial,\pm,n}$, supported in the fixed boundary collar from Theorem 2.8. The direct sum has uniformly bounded propagation and uniformly bounded matrix coefficients by uniform regularity and the fixed local cap-product convention. Hence it belongs to the algebraic interface ideal, and therefore to $C^*(Y \subset X)$. The quotient

equality follows immediately. The same theorem gives the fixed collar support of the relative duality defect, so its direct sum is in the same ideal. Finally, quotient maps commute with continuous functional calculus for bounded self-adjoint elements. □

6.21 Proof of Result 2.9

Proof of Theorem 2.9. Since $R = P_W R P_W$, $\operatorname{rank} R \le r$. Weyl's rank inequality for spectral distribution functions [17, Chapter III] gives

$$N_A(t) - \operatorname{rank} R \le N_{A+R}(t) \le N_A(t) + \operatorname{rank} R,$$

which proves the first estimate. The interval estimate follows by subtracting the distribution functions at a and b. The final statement is Theorem 2.8. □

6.22 Proof of Result 2.10

Proof of Theorem 2.10. Let $h_n \to 0$ be the quasi-uniform mesh size. Uniform regularity gives, for each fixed R, a constant C_R such that every radius-R ball in the barycentric graph contains at most C_R simplices. Hence the R-collar of the boundary is covered by radius-R balls centered at boundary vertices and boundary edges, so

$$r_n(R) \le C_R\big(|(\partial\mathcal{K}_n)_0| + |(\partial\mathcal{K}_n)_1|\big).$$

Since $\partial\mathcal{K}_n$ is a disjoint union of cycles, $|(\partial\mathcal{K}_n)_0| = |(\partial\mathcal{K}_n)_1|$. Boundary edge lengths are comparable to h_n, while the total boundary length of the fixed PL surface is finite and independent of n. Therefore

$$|(\partial\mathcal{K}_n)_1| = O(h_n^{-1}), \qquad r_n(R) = O(h_n^{-1}).$$

On the other hand, shape regularity gives face areas comparable to h_n^2, while the total surface area is fixed and positive on every nondegenerate component. Thus

$$|(\mathcal{K}_n)_2| = \Theta(h_n^{-2}).$$

Uniform incidence bounds give $|(\mathcal{K}_n)_0| + |(\mathcal{K}_n)_1| + |(\mathcal{K}_n)_2| = \Theta(|(\mathcal{K}_n)_2|)$, and hence

$$d_n = \dim \mathcal{H}(\mathcal{K}_n) = |(\mathcal{K}_n)_0| + |(\mathcal{K}_n)_1| + |(\mathcal{K}_n)_2| = \Theta(h_n^{-2}).$$

It follows that

$$\frac{r_n(R)}{d_n} = O(h_n) \longrightarrow 0.$$

Taking $R = R_{\mathrm{PL}}$ is legitimate because Theorem 2.8 supplies a single fixed collar radius over uniformly regular refinement families. □

6.23 Proof of Result 2.5

Proof of Theorem 2.5. Apply Lemma 2.9 with $W = \ell^2(N_{R_{\mathrm{PL}}}(\partial\mathcal{K}_n))$, divide by d_n, and take the supremum in t. For quasi-uniform fixed-surface two-dimensional refinements, Theorem 2.10 gives $r_n/d_n \to 0$; outside such geometric refinement regimes it should be checked as an explicit hypothesis. The statement about continuous test functions follows because the operators have uniformly bounded norm under uniform regularity, so the empirical spectral measures are supported in a common compact interval and the distribution-function estimate forces the same weak subsequential limits. □

6.24 Proof of Result 3.1

Proof of Theorem 3.1. Since D is self-adjoint and $\Im z \neq 0$, the operator $D - z$ is invertible. Standard block Gaussian elimination gives the Schur-complement formula. □

6.25 Proof of Result 3.2

Proof of Theorem 3.2. By Lemma 3.1,

$$P_{\text{orig}}(H_{\pm,M} - z)^{-1} P_{\text{orig}} = \big(A - z - K_M(z)\big)^{-1}, \qquad K_M(z) = C(D_M - z)^{-1} C^\dagger.$$

Hence

$$\|K_M(z)\| \leq \frac{\|C\|^2}{d_M(z)}.$$

Now set $R_0 = (A - z)^{-1}$ and $R_M = (A - z - K_M(z))^{-1}$. The resolvent identity gives

$$R_M - R_0 = R_M K_M(z) R_0.$$

Moreover R_M is the $(1,1)$ block of $(H_{\pm,M} - z)^{-1}$, so

$$\|R_M\| \leq \big\|(H_{\pm,M} - z)^{-1}\big\| \leq \frac{1}{|\Im z|}, \qquad \|R_0\| \leq \frac{1}{|\Im z|}.$$

Combining these bounds yields (3.1). Since

$$\operatorname{spec}(D_M) \subset [M - \|D\|, M + \|D\|],$$

we also have $d_M(z) \geq M - \|D\| - |\Re z|$ whenever $M > \|D\| + |\Re z|$, which gives (3). □

6.26 Proof of Result 3.3

Proof of Theorem 3.3. This is the first case in Section 6.28. □

6.27 Proof of Result 3.4

Proof of Theorem 3.4. This is the second case in Section 6.28. □

6.28 Routine Schur-complement corollaries

Proof. For Theorem 3.3, apply Theorem 3.2 with $A = H_{\pm}^{\partial}(\mathcal{K})$.

For Theorem 3.4, set $R_D(z) = (D_M - z)^{-1}$ and $S_M(z) = A - z - C R_D(z) C^\dagger$. The block inverse formula gives

$$(H_{\pm,M} - z)^{-1} = \begin{pmatrix} S_M(z)^{-1} & -S_M(z)^{-1} C R_D(z) \\ -R_D(z) C^\dagger S_M(z)^{-1} & R_D(z) + R_D(z) C^\dagger S_M(z)^{-1} C R_D(z) \end{pmatrix}.$$

The upper-left block is controlled by Theorem 3.2. The off-diagonal blocks have norm at most $\|C\| / (|\Im z|\, d_M(z))$, while the lower-right block is bounded by

$$\frac{1}{d_M(z)} + \frac{\|C\|^2}{d_M(z)^2 |\Im z|}.$$

Adding these four bounds gives Theorem 3.4. □

6.29 Proof of Result 3.5

Proof of Theorem 3.5. Write

$$D_M - z = M\Big(\mathbf{1} + \frac{D-z}{M}\Big).$$

If $M > \|D - z\|$, the Neumann series gives

$$\Big(\mathbf{1} + \frac{D-z}{M}\Big)^{-1} = \mathbf{1} - \frac{D-z}{M} + \Big(\frac{D-z}{M}\Big)^2 - \cdots,$$

which yields the stated expansion and remainder estimate. Multiplying on both sides by C and $C^\dagger$ gives the formula for $K_M(z)$.

Set

$$R_M = P_{\mathrm{orig}}(H_{\pm,M} - z)^{-1}P_{\mathrm{orig}} = (A - z - K_M(z))^{-1}, \qquad R_M^{(1)} = (A_M^{(1)} - z)^{-1}.$$

Then

$$R_M - R_M^{(1)} = R_M\,\widetilde{R}_M(z)\,R_M^{(1)}.$$

Since both $H_{\pm,M}$ and $A_M^{(1)}$ are self-adjoint,

$$\|R_M\| \le \frac{1}{|\Im z|}, \qquad \big\|R_M^{(1)}\big\| \le \frac{1}{|\Im z|}.$$

Combining these bounds with the estimate on $\widetilde{R}_M(z)$ gives (3.2). □

6.30 Proof of Result 3.6

Proof of Theorem 3.6. For $t \in [0,1]$ define the homotopy

$$H_{t,M} := \begin{pmatrix} A & tC \\ tC^\dagger & D_M \end{pmatrix}.$$

Fix $z \in \Gamma$. Since $\delta_D(\Gamma) > 0$, the operator $D_M - z$ is invertible. The Schur complement of $H_{t,M} - z$ with respect to the lower-right block is

$$S_{t,M}(z) = A - z - t^2 C(D_M - z)^{-1}C^\dagger.$$

Because $z \in \Gamma$, we have

$$\big\|(A-z)^{-1}\big\| \le \frac{1}{\delta_A(\Gamma)}, \qquad \big\|(D_M - z)^{-1}\big\| \le \frac{1}{\delta_D(\Gamma)}.$$

Hence

$$\Big\|(A-z)^{-1}t^2C(D_M-z)^{-1}C^\dagger\Big\| \le \frac{\|C\|^2}{\delta_A(\Gamma)\,\delta_D(\Gamma)} < 1$$

by the hypothesis $\delta_H(\Gamma) > 0$. Therefore

$$S_{t,M}(z) = (A-z)\Big[\mathbf{1} - (A-z)^{-1}t^2C(D_M-z)^{-1}C^\dagger\Big]$$

is invertible, because the bracketed term is invertible by the Neumann series. Since $D_M - z$ is invertible as well, the block matrix $H_{t,M} - z$ is invertible. This proves $\Gamma \subset \rho(H_{t,M})$ for all $t \in [0,1]$, and hence $\Gamma \subset \rho(H_{\pm,M})$.

The map $t \mapsto P_\Gamma(H_{t,M})$ is norm-continuous, so its rank is constant on the connected interval $[0,1]$. At $t=0$,

$$H_{0,M} = A \oplus D_M,$$

and therefore

$$P_\Gamma(H_{0,M}) = P_\Gamma(A) \oplus P_\Gamma(D_M).$$

This proves the rank identity.

For the norm estimate, abbreviate

$$K_M(z) := C(D_M - z)^{-1}C^\dagger, \qquad S_M(z) := A - z - K_M(z).$$

The Schur-complement inverse formula gives Set $R_D(z) = (D_M - z)^{-1}$. The block inverse formula gives

$$(H_{\pm,M} - z)^{-1} = \begin{pmatrix} S_M(z)^{-1} & -S_M(z)^{-1}CR_D(z) \\ -R_D(z)C^\dagger S_M(z)^{-1} & R_D(z) + R_D(z)C^\dagger S_M(z)^{-1}CR_D(z) \end{pmatrix}.$$

Because $\delta_H(\Gamma) > 0$,

$$\|S_M(z)^{-1}\| \le \frac{1}{\delta_H(\Gamma)}, \qquad \|(A-z)^{-1}\| \le \frac{1}{\delta_A(\Gamma)}, \qquad \|(D_M - z)^{-1}\| \le \frac{1}{\delta_D(\Gamma)}.$$

Also,

$$S_M(z)^{-1} - (A-z)^{-1} = S_M(z)^{-1}K_M(z)(A-z)^{-1},$$

so

$$\|S_M(z)^{-1} - (A-z)^{-1}\| \le \frac{\|C\|^2}{\delta_A(\Gamma)\,\delta_H(\Gamma)\,\delta_D(\Gamma)}.$$

The off-diagonal blocks satisfy

$$\|S_M(z)^{-1}C(D_M - z)^{-1}\| \le \frac{\|C\|}{\delta_H(\Gamma)\,\delta_D(\Gamma)},$$

and the extra lower-right block is bounded by

$$\left\|(D_M - z)^{-1}C^\dagger S_M(z)^{-1}C(D_M - z)^{-1}\right\| \le \frac{\|C\|^2}{\delta_H(\Gamma)\,\delta_D(\Gamma)^2}.$$

Subtracting the block-diagonal resolvent

$$\begin{pmatrix} (A-z)^{-1} & 0 \\ 0 & (D_M - z)^{-1} \end{pmatrix}$$

gives the pointwise bound (3). Integrating around Γ then gives (3.3). □

6.31 Proof of Result 3.7

Proof of Theorem 3.7. Let $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ be a normalized eigenvector with eigenvalue E. If $\psi_{\text{orig}} = 0$, then $(D_M)\psi_{\text{copy}} = E\psi_{\text{copy}}$, which is impossible because $|E| \le E_{\max}$ and $g_M(E_{\max}) > 0$. Thus $\psi_{\text{orig}} \neq 0$.

Solving the second block equation gives

$$\psi_{\text{copy}} = -(D_M - E)^{-1}C^\dagger \psi_{\text{orig}}.$$

Substituting into the original-block equation yields

$$\big(A - \Sigma_M(E)\big)\psi_{\text{orig}} = E\psi_{\text{orig}}, \qquad \Sigma_M(E) = C(D_M - E)^{-1}C^\dagger.$$

Hence $E \in \operatorname{spec}(A - \Sigma_M(E))$. Since

$$\|\Sigma_M(E)\| \le \|C\|^2 \, \big\|(D_M - E)^{-1}\big\| \le \frac{\|C\|^2}{g_M(E_{\max})},$$

Weyl's inequality implies (3). If $M > \|D\| + E_{\max}$, then

$$g_M(E_{\max}) \ge M - \|D\| - E_{\max},$$

which gives (3). □

6.32 Proof of Result 3.8

Proof of Theorem 3.8. By Theorem 2.8, the original-side block is $A = H_{\pm}^{\mathrm{PL}}(\mathcal{K}) + R_{\partial,\pm}$. The result is therefore Theorem 3.7 with this identification of A. □

6.33 Proof of Result 3.9

Proof of Theorem 3.9. Let $\psi = (\psi_{\text{orig}}, \psi_{\text{copy}})$ be a normalized eigenvector of $H_{\pm,M}$ with eigenvalue E, where $|E| \le E_{\max}$. As in the proof of Theorem 3.7,

$$\big(A - \Sigma_M(E)\big)\psi_{\text{orig}} = E\psi_{\text{orig}}, \qquad \Sigma_M(E) = C(D_M - E)^{-1}C^\dagger.$$

Now

$$(D_M - E)^{-1} - \frac{1}{M}\mathbf{1} = -\frac{1}{M}(D - E)(D_M - E)^{-1},$$

so

$$\left\|(D_M - E)^{-1} - \frac{1}{M}\mathbf{1}\right\| \le \frac{\|D - E\|}{M} \, \big\|(D_M - E)^{-1}\big\| \le \frac{\|D\| + E_{\max}}{M\, g_M(E_{\max})}.$$

Hence

$$\Sigma_M(E) = \frac{1}{M}CC^\dagger + \widetilde{\Sigma}_M(E), \qquad \left\|\widetilde{\Sigma}_M(E)\right\| \le \frac{\|C\|^2\,(\|D\| + E_{\max})}{M\, g_M(E_{\max})}.$$

Therefore

$$A - \Sigma_M(E) = A_M^{(1)} - \widetilde{\Sigma}_M(E).$$

Since $E \in \operatorname{spec}(A_M^{(1)} - \widetilde{\Sigma}_M(E))$, Weyl's inequality yields (3.4). The crude bound (3.5) follows from $g_M(E_{\max}) \ge M - \|D\| - E_{\max}$ when $M > \|D\| + E_{\max}$. □

6.34 Proof of Result 3.10

Proof of Theorem 3.10. From the second block equation,

$$(D_M - E)\psi_{\text{copy}} = -C^\dagger\psi_{\text{orig}}.$$

Because $g_M(E_{\max}) > 0$, the operator $D_M - E$ is invertible for every $|E| \le E_{\max}$, so

$$\|\psi_{\text{copy}}\| \le \big\|(D_M - E)^{-1}\big\| \, \|C\| \, \|\psi_{\text{orig}}\| \le \frac{\|C\|}{g_M(E_{\max})} \, \|\psi_{\text{orig}}\|.$$

If $M > \|D\| + E_{\max}$, then $g_M(E_{\max}) \ge M - \|D\| - E_{\max}$, which gives (3). □

6.35 Proof of Result 3.11

Proof of Theorem 3.11. Since $D_M > 0$, block Gaussian elimination gives the congruence

$$\begin{pmatrix} \mathbf{1} & -CD_M^{-1} \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} A & C \\ C^\dagger & D_M \end{pmatrix} \begin{pmatrix} \mathbf{1} & 0 \\ -D_M^{-1}C^\dagger & \mathbf{1} \end{pmatrix} = \begin{pmatrix} S_M(0) & 0 \\ 0 & D_M \end{pmatrix}.$$

Sylvester's law of inertia gives the first two identities because the lower-right block is positive. The norm estimate follows from $\|D_M^{-1}\| \le (M - \|D\|)^{-1}$. If this norm is strictly smaller than δ_A, then the straight-line path $A - tCD_M^{-1}C^\dagger$, $0 \le t \le 1$, remains invertible by Weyl's inequality, so its negative spectral count is constant. Combining this with the first identity proves the final claim. □

6.36 Proof of Result 3.12

Proof of Theorem 3.12. One has

$$H_{\pm,M_2} - H_{\pm,M_1} = (M_2 - M_1)P_{\text{copy}} \succeq 0.$$

The min-max principle therefore gives the monotonicity of each ordered eigenvalue. The spectral count statement is the same assertion written for distribution functions. □

6.37 Proof of Result 3.13

Proof of Theorem 3.13. Write $\psi = \psi_{\text{orig}} \oplus \psi_{\text{copy}}$. Since $\psi \in \text{Ran}Q_M$, the spectral theorem gives $\|H_{\pm,M}\psi\| \le E_{\max}\|\psi\|$. Taking the copy block of $H_{\pm,M}\psi$ yields

$$D_M\psi_{\text{copy}} + C^\dagger\psi_{\text{orig}} = P_{\text{copy}}H_{\pm,M}\psi.$$

Because $D_M \ge (M - \|D\|)\mathbf{1}$,

$$(M - \|D\|)\|\psi_{\text{copy}}\| \le E_{\max}\|\psi\| + \|C\|\ \|\psi_{\text{orig}}\|,$$

which proves the quantitative estimate. If $\psi_{\text{orig}} = 0$, the same block equation gives

$$(M - \|D\|)\|\psi_{\text{copy}}\| \le E_{\max}\|\psi_{\text{copy}}\|.$$

The strict inequality $M - \|D\| > E_{\max}$ forces $\psi_{\text{copy}} = 0$, so $\psi = 0$. Thus P_{orig} is injective on $\text{Ran}Q_M$, and the rank bound follows. □

6.38 Proof of Result 3.14

Proof of Theorem 3.14. Suppose, toward a contradiction, that $E \in \text{spec}(H_{\pm,M})$ and $|E| < g_A - \Delta_M$. Then $|E| \le g_A$, so Theorem 3.7 with $E_{\max} = g_A$ gives

$$\text{dist}(E, \text{spec}(A)) \le \Delta_M.$$

Since $\text{spec}(A) \cap (-g_A, g_A) = \emptyset$ and $|E| < g_A$, we also have

$$\text{dist}(E, \text{spec}(A)) \ge g_A - |E| > \Delta_M,$$

a contradiction. □

6.39 Proof of Result 3.15

Proof of Theorem 3.15. Apply Theorems 3.2, 3.5, 3.7, and 3.10 componentwise and take suprema over n, using the bounds $\|C_n\| \le \bar{C}$ and $\|D_n\| \le \bar{D}$. The full resolvent convergence follows similarly from Corollary 3.4. Since the resolvent functions $(t - z)^{-1}$ generate $C_0(\mathbb{R})$ as a C^*-algebra, the C_0-functional calculus convergence follows from the norm convergence of the resolvents. □

6.40 Proof of Result 3.16

Proof of Theorem 3.16. Set

$$\bar{C} = \sup_n \|C_{\pm,n}\|, \qquad \bar{D} = \sup_n \|D_{\pm,n}\|.$$

Since $M_n \to \infty$, the full block-resolvent estimate in Corollary 3.4, applied with the uniform bounds $\bar{C}, \bar{D}$, gives

$$\|(H_{\pm,n,M_n} - z)^{-1} - ((A_{\pm,n} - z)^{-1} \oplus 0)\| \longrightarrow 0$$

for every $z \in \mathbb{C} \setminus \mathbb{R}$. The algebra generated by the resolvent functions is dense in $C_0(\mathbb{R})$, so

$$\|f(H_{\pm,n,M_n}) - (f(A_{\pm,n}) \oplus 0)\| \longrightarrow 0 \qquad (f \in C_0(\mathbb{R})).$$

The doubled Hilbert space has dimension at most $2d_n$. Therefore

$$\left| \frac{1}{d_n} \operatorname{Tr} f(H_{\pm,n,M_n}) - \frac{1}{d_n} \operatorname{Tr} f(A_{\pm,n}) \right| \le 2 \, \|f(H_{\pm,n,M_n}) - (f(A_{\pm,n}) \oplus 0)\| \longrightarrow 0.$$

It remains to compare $A_{\pm,n}$ with $H_{\pm}^{\mathrm{PL}}(\mathcal{K}_n)$. By Theorem 2.8, their difference is $R_{\partial,\pm,n} = P_{W_n} R_{\partial,\pm,n} P_{W_n}$ for a collar subspace W_n of dimension r_n. Lemma 2.9 gives

$$\sup_{t \in \mathbb{R}} \left| \frac{1}{d_n} N_{A_{\pm,n}}(t) - \frac{1}{d_n} N_{H_{\pm}^{\mathrm{PL}}(\mathcal{K}_n)}(t) \right| \le \frac{r_n}{d_n} \longrightarrow 0.$$

The two sequences of operators have uniformly bounded norm by hypothesis, so their normalized spectral measures are supported in a common compact interval. Uniform convergence of the distribution functions up to an $o(1)$ error implies that the integrals of every continuous function on this compact interval have the same limit; applying this to the restriction of $f \in C_0(\mathbb{R})$ proves

$$\frac{1}{d_n} \operatorname{Tr} f(A_{\pm,n}) - \frac{1}{d_n} \operatorname{Tr} f(H_{\pm}^{\mathrm{PL}}(\mathcal{K}_n)) \longrightarrow 0.$$

Combining the two limits gives the claimed normalized-trace equivalence. □

6.41 Proof of Result 5.1

Proof of Theorem 5.1. Substituting the displayed matrices for $B(k)$ and $S(k)$ and expanding $\det(\lambda \mathbf{1} - H_{\pm}^{\triangle}(k))$ gives the stated factorization. Equivalently, with $z_j = e^{ik_j}$, clearing denominators first factors the characteristic polynomial into

$$4\lambda^2 - F(k)$$

times

$$4\lambda^4 - 26\lambda^2 + F(k) \pm 4\lambda G(k),$$

and division by the leading coefficient gives the monic form above. □

6.42 Proof of Result 5.2

Proof of Theorem 5.2. Proposition 5.1 gives

$$\det H_{\pm}^{\triangle}(k) = -\frac{1}{16}\left(-28 + 8\cos k_1 + 6\cos k_2 + 6\cos(k_1+k_2)\right)^2.$$

Since each cosine is at most 1,

$$-28 + 8\cos k_1 + 6\cos k_2 + 6\cos(k_1+k_2) \le -8,$$

so the determinant is at most -4. The absolute row sums of $H_{\pm}^{\triangle}(k)$ are bounded by 8, so $\|H_{\pm}^{\triangle}(k)\| \le 8$. If $\lambda_1, \dots, \lambda_6$ are the eigenvalues, then $|\lambda_j| \le 8$ and

$$4 \le |\det H_{\pm}^{\triangle}(k)| \le \operatorname{dist}(0, \operatorname{spec} H_{\pm}^{\triangle}(k))\, 8^5,$$

which gives the stated gap. At $k = 0$, the characteristic polynomial is

$$(\lambda^2 - 2)(2\lambda^4 - 13\lambda^2 + 4)/2,$$

so exactly three eigenvalues are negative. The gap prevents the rank from changing over $\mathbb{T}^2$.

The same matrices satisfy $B(k)S(k) + S(k)B(k)^* = 0$. The stronger anticommutator $(B(k) + B(k)^*)S(k) + S(k)(B(k) + B(k)^*)$ is not used here. □

6.43 Proof of Result 5.3

Proof of Theorem 5.3. Theorem 5.2 gives

$$\operatorname{dist}(0, \operatorname{spec} H_{\pm}^{\triangle}(k)) \ge 2^{-13} \qquad (k \in \mathbb{T}^2).$$

Weyl's inequality gives

$$\operatorname{dist}\big(0, \operatorname{spec}(H_{\pm}^{\triangle}(k) + V(k))\big) \ge 2^{-13} - \|V(k)\| > 0.$$

The negative spectral projection therefore varies continuously in k, and its rank is constant. At $V = 0$ this rank is 3, so it remains 3 along the straight-line homotopy $H_{\pm}^{\triangle}(k) + sV(k)$, $0 \le s \le 1$. □

6.44 Proof of Result 5.4

Proof of Theorem 5.4. The determinant identity is an exact computation from the displayed Bloch matrices. Since $c_1 + c_2 + c_{12} \le 3$ and $2 + c_2 + c_{12} \ge 0$, the bracket is nonnegative. If the first term vanishes, then $c_1 = c_2 = c_{12} = 1$, and the second term equals $8m^2 > 0$. Thus the bracket is positive for every $m > 0$, so the determinant never vanishes. The path $(m, k) \mapsto H_{\pm,m}^{\triangle}(k)$ is norm-continuous on each compact interval $m \in [m_0, m_1] \subset (0, \infty)$. □

6.45 Proof of Result 5.5

Proof of Theorem 5.5. The proof uses the orientation conventions fixed in Section 3. The Brillouin torus is oriented by $dk_1 \wedge dk_2$, the crossing parameter space is oriented by $dq_1 \wedge dq_2 \wedge d\mu$, and Chern numbers are computed as

$$\operatorname{Ch}(P) = \frac{1}{2\pi i}\int_{\mathbb{T}^2} \operatorname{tr}(P\, dP \wedge dP).$$

The parameter μ is the coefficient of S in

$$H_{\mu}^{\triangle}(k) = B(k) + B(k)^* + \mu S(k).$$

Proposition 5.4 shows that $H_{\mu}^{\triangle}(k)$ is gapped at zero for $\mu \neq 0$, so the Chern number can change only at the crossing $(k, \mu) = (0, 0)$.

Step 1: harmonic compression. At $k = 0$, the boundary matrices are

$$\partial_1(0) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}, \qquad \partial_2(0) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Thus $\operatorname{rank}\partial_1(0) = 0$ and $\operatorname{rank}\partial_2(0) = 1$. The kernel of the Hodge–Dirac operator $D_0 = B(0)+B(0)^*$ is the direct sum of the harmonic parts in degrees $0, 1, 2$: one vertex vector, the two-dimensional orthogonal complement of $\operatorname{im}\partial_2(0) \subset \mathbb{C}\langle a, b, c\rangle$, and the kernel of $\partial_2(0)$ in degree 2. With the ordered basis v, a, b, c, f, g, choose the orthonormal harmonic basis

$$h_0 = v, \qquad h_1 = \frac{a-b}{\sqrt{2}}, \qquad h_2 = \frac{a+b+2c}{\sqrt{6}}, \qquad h_3 = \frac{f+g}{\sqrt{2}}.$$

Let $U : \mathbb{C}^4 \to \mathcal{H}(0)$ be the isometry $Ue_j = h_{j-1}$, and let

$$\Pi_{\mathrm{h}} = UU^*$$

be the harmonic projector. The complement $(1 - \Pi_{\mathrm{h}})\mathcal{H}(0)$ is separated from zero for D_0, because D_0 is self-adjoint and has no kernel on that finite-dimensional complement.

Put $q_1 = k_1$, $q_2 = k_2$. Since $z_j = e^{iq_j} = 1 + iq_j + O(|q|^2)$, the first-order harmonic compression is

$$L(q_1, q_2, \mu) = \Pi_{\mathrm{h}}\big(q_1\partial_{q_1}H_0^{\triangle}(0) + q_2\partial_{q_2}H_0^{\triangle}(0) + \mu\partial_\mu H_\mu^{\triangle}(0)|_{\mu=0}\big)\Pi_{\mathrm{h}}\big|_{\mathcal{H}(0)}.$$

Equivalently,

$$\langle h_r, L(q_1, q_2, \mu)h_s\rangle = i\big(q_1X_1 + q_2X_2 + \mu X_3\big)_{r+1,s+1},$$

where the real skew-symmetric matrices are

$$X_1 = \begin{bmatrix} 0 & 1/\sqrt{2} & \sqrt{3/2} & 0 \\ -1/\sqrt{2} & 0 & 0 & -1/2 \\ -\sqrt{3/2} & 0 & 0 & 1/(2\sqrt{3}) \\ 0 & 1/2 & -1/(2\sqrt{3}) & 0 \end{bmatrix},$$

$$X_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & \sqrt{3/2} & 0 \\ 1/\sqrt{2} & 0 & 0 & -1/2 \\ -\sqrt{3/2} & 0 & 0 & -1/(2\sqrt{3}) \\ 0 & 1/2 & 1/(2\sqrt{3}) & 0 \end{bmatrix}, \qquad X_3 = \begin{bmatrix} 0 & 0 & 0 & -\sqrt{2} \\ 0 & 0 & -1/\sqrt{3} & 0 \\ 0 & 1/\sqrt{3} & 0 & 0 \\ \sqrt{2} & 0 & 0 & 0 \end{bmatrix}.$$

These entries are obtained by differentiating the displayed Bloch matrices in Section 5 and applying the matrix elements $\langle h_r, (\partial H)h_s\rangle$. The factor i is forced by Hermiticity: the derivatives of the unitary phases at 1 are imaginary, while X_j are real skew matrices.

It remains to justify that this compression has the same local Chern jump as the full crossing. Write the full local operator, relative to $\Pi_{\mathrm{h}}\mathcal{H} \oplus (1 - \Pi_{\mathrm{h}})\mathcal{H}$, as

$$\begin{pmatrix} A(x) & K(x) \\ K(x)^* & G(x) \end{pmatrix}, \qquad x = (q_1, q_2, \mu).$$

Here $A(x) = L(x) + O(|x|^2)$, $K(x) = O(|x|)$, and $G(0)$ is invertible. Shrinking to $|x| \le r$, $G(x)$ remains invertible and

$$A(x) - K(x)G(x)^{-1}K(x)^* = L(x) + O(|x|^2).$$

On the sphere $|x| = r$, the linear model below is invertible with a lower bound cr for some $c > 0$. Choose r so that the quadratic error has norm at most $cr/2$. Then the straight-line homotopy from the Schur complement to $L(x)$ is invertible on the sphere, and the complementary block remains invertible throughout. Therefore the full crossing and the displayed harmonic model have the same linking Chern number.

Step 2: self-dual and anti-self-dual splitting. Identify a real skew 4×4 matrix with a two-form by the ordered basis

$$e_{12}, e_{13}, e_{14}, e_{23}, e_{24}, e_{34}$$

for $\Lambda^2\mathbb{R}^4$, where $e_{rs} = e_r \wedge e_s$ and $e_1 \wedge e_2 \wedge e_3 \wedge e_4$ is the positive orientation. The Hodge involution is

$$*e_{12} = e_{34}, \quad *e_{13} = -e_{24}, \quad *e_{14} = e_{23},$$

together with the same equations read backwards. Hence the self-dual basis is

$$\omega_1^+ = \frac{e_{12} + e_{34}}{\sqrt{2}}, \quad \omega_2^+ = \frac{e_{13} - e_{24}}{\sqrt{2}}, \quad \omega_3^+ = \frac{e_{14} + e_{23}}{\sqrt{2}},$$

and the anti-self-dual basis is

$$\omega_1^- = \frac{e_{12} - e_{34}}{\sqrt{2}}, \quad \omega_2^- = \frac{e_{13} + e_{24}}{\sqrt{2}}, \quad \omega_3^- = \frac{e_{14} - e_{23}}{\sqrt{2}}.$$

For

$$X(q_1, q_2, \mu) = q_1 X_1 + q_2 X_2 + \mu X_3,$$

write $X = X_+(u) + X_-(\beta)$ in these bases. Direct projection onto the ω_j^+ coordinates gives

$$u(q_1, q_2, \mu) = \left(\frac{1 + 1/\sqrt{6}}{2}(q_1 - q_2), \left(\frac{\sqrt{3}}{2} + \frac{1}{2\sqrt{2}} \right)(q_1 + q_2), -(1 + 1/\sqrt{6})\mu \right).$$

The anti-self-dual vector is a scalar multiple of the same coordinate vector,

$$\beta(q_1, q_2, \mu) = \rho\, u(q_1, q_2, \mu), \qquad \rho = \frac{\sqrt{6} - 1}{\sqrt{6} + 1} < 1.$$

For a skew form with self-dual and anti-self-dual coordinates (α, β), the Hermitian matrix $iX(\alpha, \beta)$ has eigenvalues

$$\pm(|\alpha| + |\beta|), \qquad \pm(|\alpha| - |\beta|).$$

Along the homotopy $(\alpha, \beta) = (u, t\rho u)$, $0 \le t \le 1$, the smallest absolute eigenvalue is $(1 - t\rho)|u|$. This is positive on every linking sphere because $\rho < 1$ and $u \neq 0$ there. Thus the anti-self-dual part can be removed without changing the local Chern jump.

The self-dual coordinate map is linear with Jacobian

$$\frac{\partial u}{\partial(q_1, q_2, \mu)} = \begin{bmatrix} a & -a & 0 \\ b & b & 0 \\ 0 & 0 & -c \end{bmatrix}, \quad a = \frac{1 + 1/\sqrt{6}}{2}, \quad b = \frac{\sqrt{3}}{2} + \frac{1}{2\sqrt{2}}, \quad c = 1 + \frac{1}{\sqrt{6}}.$$

Therefore

$$\det \frac{\partial u}{\partial(q_1, q_2, \mu)} = -2abc < 0.$$

With the crossing orientation $dq_1 \wedge dq_2 \wedge d\mu$, the self-dual local model is orientation-reversing.

Step 3: Chern jump and FHS sign convention. Under the standard spinor identification of self-dual two-forms, $iX_+(u)$ is the direct sum of two identical Pauli Hamiltonians $u \cdot \sigma$. For the identity crossing $u = (q_1, q_2, \mu)$, the negative line over the linking sphere has Chern number $+1$ with the boundary orientation induced by $dq_1 \wedge dq_2 \wedge d\mu$. Pullback by a linear map multiplies this number by the degree of that map. Since $\det Du < 0$, one Pauli block contributes -1, and two identical blocks contribute

$$\mathrm{Ch}(E_-(H^{\triangle}_{\mu>0})) - \mathrm{Ch}(E_-(H^{\triangle}_{\mu<0})) = -2.$$

The FHS convention used in Tables 1 and 2 orders each plaquette by the positively oriented coordinate square in the k_1, k_2 directions. Its lattice integer therefore approximates the same $(2\pi i)^{-1} \int \mathrm{tr}(P\, dP \wedge dP)$ used here; no sign conversion is inserted between the proof and the tables.

Finally,

$$H^{\triangle}_{-\mu}(k) = \overline{H^{\triangle}_{\mu}(-k)}.$$

The base involution $k \mapsto -k$ has Jacobian determinant $+1$ on the two-torus, so it preserves the orientation $dk_1 \wedge dk_2$. Complex conjugation of a complex vector bundle reverses the first Chern class. Hence

$$\mathrm{Ch}(E_-(H^{\triangle}_{-\mu})) = -\,\mathrm{Ch}(E_-(H^{\triangle}_{\mu})).$$

Combining this symmetry with the jump -2 gives

$$\mathrm{Ch}(E_-(H^{\triangle}_{\mu})) = -1 \quad (\mu > 0), \qquad \mathrm{Ch}(E_-(H^{\triangle}_{\mu})) = +1 \quad (\mu < 0).$$

Taking $\mu = 1$ gives the stated signs for $H^{\triangle}_{+}$ and $H^{\triangle}_{-}$. □

6.46 Proof of Result 5.6

Proof of Theorem 5.6. The positive-mass statement follows from Proposition 5.4 and Theorem 5.5: changing $m > 0$ gives a norm-continuous gapped homotopy for $H^{\triangle}_{+,m}$, and $H^{\triangle}_{-,m} = H^{\triangle}_{-m}$ is handled by the same argument on the negative-mass side. Thus the Chern number is unchanged. The number $g_{\pm,m}$ is positive by compactness of $\mathbb{T}^2$ and the positive-mass gap. If $\|V\|_\infty < g_{\pm,m}$, Weyl's inequality keeps the straight-line homotopy $H^{\triangle}_{\pm,m} + sV$ gapped at zero, so the Chern number is unchanged along it.

For the uniform perturbation statement at $m = 1$, the straight-line homotopy

$$H^{\triangle}_{\pm,s}(k) = H^{\triangle}_{\pm}(k) + sV(k), \qquad 0 \le s \le 1,$$

is gapped by Theorem 5.2 and Weyl's inequality:

$$\mathrm{dist}\big(0, \mathrm{spec} H^{\triangle}_{\pm,s}(k)\big) \ge 2^{-13} - s \sup_{q \in \mathbb{T}^2} \|V(q)\| > 0.$$

The Chern number of the negative spectral projection is constant along this gapped homotopy. □

6.47 Proof of Result 6.21

Proof of Theorem 6.21. The map $p_A : \mathbb{T}^2 \to \mathbb{T}^2$, $p_A(\kappa) = A\kappa$, is a finite covering of degree $\det A$ with sign [15]. The pulled-back Hamiltonian has the same fiber spectra as the original one, so the gap is unchanged. Its negative Bloch bundle is $p_A^* E_-(H^{\triangle}_{\pm})$, and naturality of the Chern form gives

$$\mathrm{Ch}(p_A^* E) = \deg(p_A)\, \mathrm{Ch}(E) = \det A\ \mathrm{Ch}(E).$$

The displayed signs follow from Theorem 5.5; the positive-mass and small-perturbation statements follow in the same way from Corollary 5.6.

For a finite-index sublattice $\Lambda' \subset \mathbb{Z}^2$, the folded Bloch fiber over the smaller Brillouin torus is the direct sum of the original fibers over the finitely many momenta lying above it. The Chern form of this direct sum is the sum of the pulled-back Chern forms over the sheets, and integration over the smaller Brillouin torus gives exactly the integral over the original Brillouin torus. The gap is the minimum of the same original fiber gaps. Thus changing the period cell does not change the Chern number of the underlying periodic operator. □

6.48 Proof of Result 6.22

Proof of Theorem 6.22. The finite periodic torus diagonalizes by the discrete Bloch transform. Its fiber momenta are points of $\mathbb{T}^2$, so the gap bound follows from Theorem 5.2. The Chern data are thermodynamic Bloch-bundle data proved in Theorem 5.5 and numerically checked in Table 1; they are not finite-volume invariants of the discrete sample alone. □

6.49 Proof of Result 6.23

Proof of Theorem 6.23. The exact gap, finite-range symbol, and analytic Chern calculation provide the locality, norm, closed-bulk gap, and closed-bulk Chern values needed in Assumptions 5.8 and 5.13; here these inputs come from the periodic calculation rather than from Assumption 5.15. With the gap and sidewise inputs fixed, the sidewise and quotient-interface conclusions are exactly Theorem 5.35. For primitive straight periodic components, the final winding evaluation is the Toeplitz calculation of Theorem 5.7, with the periodic Chern numbers supplied by Theorem 5.5. □

6.50 Proof of Result 6.24

Proof of Theorem 6.24. For $H^{\triangle}_{\pm,m}$, the gap follows from Proposition 5.4 and the Chern signs from Corollary 5.6. The fixed-m perturbation case and the $m = 1$ explicit-radius perturbation case are also exactly Corollary 5.6. Finite-cover and finite-index cell choices are Corollary 6.21. Finite-range periodicity gives the same uniform locality and bounded-geometry hypotheses as in the unperturbed periodic annular family, after replacing the fixed collar by the enlarged collar in the statement. The proofs of Theorem 5.35, Corollary 6.38, and Theorem 6.37 then apply verbatim with $H^{\infty}_{\pm}$ replaced by the chosen finite-range periodic bulk family: the only inputs used there are uniform locality, boundedness, the closed-bulk gap, sidewise agreement away from the collar, and positivity of the copy-side comparison. For primitive straight periodic components, the final winding statement is the Toeplitz evaluation used in Corollary 6.23. □

6.51 Proof of Result 6.25

Proof of Theorem 6.25. The unperturbed case is Corollary 6.23. The positive-mass, small finite-range perturbation, finite-cover, and cell-enlargement cases are Corollary 6.24. Independence of the scalar jam scale is Corollary 6.38; bounded componentwise scalar jams and nonconstant positive copy-side confinements are covered by Theorem 6.37 and Corollary 6.36. Changing finitely many initial components changes the data by a finite-component perturbation, so Proposition 6.33 applies. Uniformly gapped sidewise homotopies are covered by Proposition 6.35. The last sentence is the straight-periodic Toeplitz evaluation already isolated in Corollaries 6.23 and 6.24. □

6.52 Proof of Result 5.7

Proof of Theorem 5.7. The proof is the standard Toeplitz calculation, included here so the periodic straight-edge family is evaluated internally. The extension above is the spatial tensor product of $C(S^1)$, in the tangential variable k_1, with the one-variable Toeplitz extension in the normal variable k_2. Hence its boundary map is a group homomorphism

$$K^0(\mathbb{T}^2) \longrightarrow K^1(S^1).$$

The group $K^0(\mathbb{T}^2)$ is generated by the trivial line bundle and one Bott class [18, 19]. A constant projection lifts to a constant projection in $\mathcal{T}_{\mathrm{per}}$, so the boundary map vanishes on the rank summand.

It remains to compute the Bott generator. Use the clutching description of a Chern-one line bundle: split the k_2-circle into upper and lower arcs, trivialize the bundle over the two cylinders $S^1_{k_1} \times I_{\pm}$, and take transition function $z(k_1) = e^{ik_1}$ on one overlap component and 1 on the other. The first Chern number is the winding difference of these two transition functions, hence 1 with orientation $dk_1 \wedge dk_2$. In the Toeplitz extension, lifting the projection to the two Toeplitz charts and comparing the lifts on the overlap gives exactly this boundary unitary z. Therefore

$$\partial_{\mathrm{T}}([\mathrm{Bott}]) = [z] \in K_1(C(S^1)), \qquad \mathrm{wind}(z) = 1.$$

By additivity and the standard $K^0(\mathbb{T}^2)$ decomposition, any projection class decomposes as

$$[P] = \mathrm{rank}(P)[1] + \mathrm{Ch}(P)[\mathrm{Bott}] \quad \text{in } K^0(\mathbb{T}^2),$$

and applying ∂_{T} gives

$$\partial_{\mathrm{T}}([P]) = \mathrm{Ch}(P)[z].$$

Pairing with the standard winding homomorphism on $K_1(C(S^1))$ gives the claimed equality. □

6.53 Proof of Result 5.8

Proof of Theorem 5.8. The map $(\kappa, \eta) \mapsto \kappa t + \eta n$ is a torus automorphism. Pulling H back by this automorphism preserves finite range, self-adjointness, and the zero gap, so Theorem 5.7 applies in the (κ, η)-coordinates. The first Chern number of the pulled-back projection is multiplied by the degree of the torus automorphism, namely $\det[t\ n]$. Therefore the coordinate Toeplitz winding equals $\varepsilon(t, n)\,\mathrm{Ch}(P)$. □

6.54 Proof of Result 6.26

Proof of Theorem 6.26. The map $p_A : \mathbb{T}^2 \to \mathbb{T}^2$, $p_A(\kappa) = A\kappa$, has signed degree $\det A$ [15], and $P_A = p_A^* P$. Hence

$$\mathrm{Ch}(P_A) = \det A\ \mathrm{Ch}(P).$$

Applying Corollary 5.8 to P_A in the primitive cover coordinates gives the displayed formula. The triangular values are Theorem 5.5 and Corollaries 5.6 and 6.21. □

6.55 Proof of Result 6.27

Proof of Theorem 6.27. The uniform gap gives a norm-continuous Riesz projection path $s \mapsto P_s$ in $C(\mathbb{T}^2, M_N)$. Hence $[P_s] \in K_0(C(\mathbb{T}^2))$ is constant. Naturality of the Toeplitz connecting map gives a constant class $\partial_{\mathrm{T},t,n}([P_s])$, and Corollary 5.8 evaluates its winding. The common finite hopping range is the locality condition needed to realize the whole homotopy by the same straight periodic edge extension. □

6.56 Proof of Result 6.28

Proof of Theorem 6.28. For a direct sum of extensions, the six-term exact sequence is natural and the connecting map is the direct sum of the connecting maps of the summands. Hence the boundary map for the split edge ideal sends the quotient projection class to

$$\bigoplus_{j=1}^{r} \partial_{\mathrm{T},t_j,n_j}([P]).$$

The j-th coordinate projection

$$K_1\left(\bigoplus_{m=1}^{r} C^*(Y^{(m)} \subset X)\right) \longrightarrow K_1(C^*(Y^{(j)} \subset X)) \cong K_1(C(S^1))$$

therefore recovers exactly the j-th Toeplitz class. Applying Corollary 5.8 gives the displayed component winding. Reversing the normal direction changes the sign of the Toeplitz extension for a fixed tangential coordinate; the induced boundary orientation reverses the tangential coordinate at the same time, which multiplies the winding homomorphism by the opposite sign and gives the final statement. □

6.57 Proof of Result 6.17

Proof of Theorem 6.17. Naturality of the six-term exact sequence for the quotient map of extensions [19] gives

$$q_*\partial_{\mathrm{poly}}([P]) = \partial_{\oplus\mathrm{T}}([P]),$$

where $\partial_{\oplus\mathrm{T}}$ is the connecting map for the quotient extension obtained after deleting the finite corner ideal. By hypothesis this quotient extension is the direct sum of the r primitive straight Toeplitz extensions, so

$$\partial_{\oplus\mathrm{T}}([P]) = \bigoplus_{j=1}^{r} \partial_{\mathrm{T},t_j,n_j}([P]).$$

Since $\mathcal{F}$ is finite-dimensional, $K_1(\mathcal{F}) = 0$. Exactness of the six-term sequence for

$$0 \to \mathcal{F} \to \mathcal{I}_{\mathrm{poly}} \xrightarrow{q} \bigoplus_j \mathcal{I}_j \to 0$$

therefore makes q_* injective on $K_1(\mathcal{I}_{\mathrm{poly}})$. Applying the j-th coordinate projection and then the standard winding homomorphism gives Corollary 5.8, hence the displayed formula. □

6.58 Proof of Result 6.18

Proof of Theorem 6.18. The preceding proof used finite-dimensionality only to obtain $K_1(\mathcal{F}) = 0$. With that condition assumed directly, exactness of

$$0 \to \mathcal{F} \to \mathcal{I}_{\mathrm{poly}} \xrightarrow{q} \bigoplus_j \mathcal{I}_j \to 0$$

again makes

$$q_* : K_1(\mathcal{I}_{\mathrm{poly}}) \to K_1\Big(\bigoplus_j \mathcal{I}_j\Big)$$

injective, and naturality identifies the quotient boundary class with the direct sum of the straight Toeplitz classes. Finite-dimensional algebras and c_0-sums of them have vanishing K_1. For a bounded product of finite-dimensional algebras, every unitary has a bounded self-adjoint logarithm with norm at most π in each finite-dimensional factor, so the unitary group is path connected after stabilization and K_1 again vanishes. □

6.59 Proof of Result 6.19

Proof of Theorem 6.19. For an algebraic finite-propagation operator T supported near Y, condition (ii) implies that all matrix coefficients coupling two different edge pieces are supported inside a bounded neighborhood of C. Modulo the corner ideal, T is therefore the direct sum of its edge restrictions. Condition (iii) identifies these restrictions with elements of the Toeplitz edge ideals and gives the quotient map q. Extension by zero from the separated edge collars gives surjectivity of q, again modulo corner-supported terms. If all edge restrictions vanish, then the support of T is contained in a bounded neighborhood of C, so the kernel is exactly $\mathcal{F}_C$ after taking norm closures. The final statement is Corollary 6.18. □

6.60 Proof of Result 5.9

Proof of Theorem 5.9. Corollary 6.25 identifies the doubled-and-jammed straight-interface class with the boundary image of the periodic bulk Fermi projection. For a straight periodic component, this boundary image is exactly the Toeplitz boundary map in Theorem 5.7; for a primitive direction it is the lattice-change version in Corollary 5.8. The Chern signs of the unperturbed triangular model, the positive-duality-strength models, and the allowed small periodic perturbations are supplied by Theorem 5.5 and Corollary 5.6; finite-cover and finite-index cell variants are Corollary 6.21. Substituting those signs into Theorem 5.7 gives the displayed winding values.

The invariance under scalar jams, componentwise scalar jams, positive copy-side confinement, finite initial-component changes, and uniformly gapped sidewise homotopies is exactly the robustness statement in Corollary 6.25; norm-vanishing component defects are Proposition 6.34. The split componentwise statement is Theorem 6.28; the finite-corner version is Theorem 6.17. □

6.61 Proof of Result 6.15

Proof of Theorem 6.15. Parts (a) and (b) preserve the closed-bulk Chern signs by Corollary 5.6. A fixed-collar perturbation in part (c) belongs to the straight-edge interface ideal, so Proposition 6.32 and Proposition 6.32 leave the interface class unchanged whenever the sidewise comparison gaps remain open. Part (d) is Theorem 6.37 together with Corollary 6.36, and part (e) is Propositions 6.33, 6.34, and 6.35. For part (f), Proposition 5.6 identifies the two fixed-collar ideals. For part (g), a finite tangential defect inside a fixed edge collar is finite rank, and every bounded finite-propagation perturbation supported in a fixed collar belongs to the algebraic interface ideal; the invariance follows from Propositions 6.32 and 6.32. Part (h) is Proposition 6.33 on the deleted components, followed by Theorem 6.28, Theorem 6.17, or Corollary 6.18 on the tail. The Toeplitz winding homomorphism, and each component coordinate winding in the split or finite-corner split case, depends only on the resulting K_1-class, so the winding values remain those of Corollary 5.9. □

6.62 Proof of Result 6.16

Proof of Theorem 6.16. The equality of ideals is Proposition 5.6. Since the sidewise differences belong to this common ideal, the quotient images of J_i and $T_i^{\pm}$ are the same. The proof of

Proposition 6.32 then gives the displayed equality of quotient Fermi projections and of p_{J_i}; applying the connecting map gives equality of interface classes. The class for the straight tail is evaluated by Corollary 5.8. A perturbation supported in a finite tangential interval and a fixed normal collar acts on only finitely many lattice degrees of freedom, hence is finite rank and in particular belongs to $\mathcal{I}$. An eventually periodic collar differs from its periodic tail exactly by such a transition defect. □

6.63 Proof of Result 6.29

Proof of Theorem 6.29. Deleting or changing finitely many initial components changes the sidewise data by an operator in the interface ideal by Proposition 6.33; hence the full interface class and the tail class have the same image under the quotient that discards those components. On the tail, the split case is Theorem 6.28. The finite-corner case is Theorem 6.17, and the K_1-trivial version is Corollary 6.18. In all cases, applying the j-th coordinate projection and the winding homomorphism gives $\det[t_j\ n_j]\,\mathrm{Ch}(P)$. The finite-cover formula is Corollary 6.26. □

6.64 Proof of Result 6.30

Proof of Theorem 6.30. The zero gap and Chern signs for (a)–(c) are Theorem 5.2, Theorem 5.5, and Corollary 5.6; finite covers and cell enlargements are Corollary 6.21. The straight-edge winding is Theorem 5.7 in coordinate directions and Corollary 5.8 in arbitrary primitive directions. Finite split sums are Theorem 6.28; finite-corner split sums are Theorem 6.17 and Corollary 6.18; the collar-separation criterion is Corollary 6.19. Bounded-distance and eventually periodic collars are Theorem 6.16; finite-defect and tail-split variants are Corollary 6.15 and Theorem 6.29. The scalar-jam, componentwise-jam, and positive copy-confinement invariance follows from Corollary 5.9 and Corollary 6.15. Norm-vanishing component defects are Proposition 6.34. The final sentence is the fixed-collar and homotopy part of Corollary 6.15. □

6.65 Proof of Result 6.3

Proof of Theorem 6.3. At $k = (0,0)$, $a_1 = a_2 = 0$ and $a_3^{\mathrm{hex}} = 1$. Thus

$$H_{\mathrm{hex}}(0,0;0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\det H_{\mathrm{hex}}(0,0;0) = 0$ and zero belongs to the spectrum. □

6.66 Proof of Result 6.4

Proof of Theorem 6.4. The trace of $H_{\mathrm{hex}}(k;m)$ is zero, and a direct 2×2 determinant calculation gives

$$\det H_{\mathrm{hex}}(k;m) = -\sin^2 k_1 - \sin^2 k_2 - m^2(\cos k_1 + \cos k_2 - 1)^2.$$

Set $s = \sin^2 k_1 + \sin^2 k_2$. If $s \geq 1/4$, then $|\det H_{\mathrm{hex}}(k;m)| \geq 1/4$. If $s < 1/4$, then $|\sin k_j| < 1/2$, hence $|\cos k_j| > \sqrt{3}/2$, for $j = 1,2$. Checking the four sign choices for $(\cos k_1, \cos k_2)$ gives

$$|\cos k_1 + \cos k_2 - 1| \geq \sqrt{3} - 1.$$

Therefore

$$|\det H_{\mathrm{hex}}(k;m)| \geq \min\left\{\frac{1}{4},\ (\sqrt{3}-1)^2 m^2\right\}.$$

Since the eigenvalues of a trace-zero 2×2 Hermitian matrix are $\pm\sqrt{|\det H_{\mathrm{hex}}(k;m)|}$, the stated spectral gap follows. □

6.67 Proof of Result 6.5

Proof of Theorem 6.5. For $m > 0$, the path replacing m by $(1-s)m + s$, $0 \le s \le 1$, stays in the positive-mass region and is gapped by Theorem 6.4. Hence the Chern number equals the value at $m = 1$. Write

$$d_{\mathrm{hex}}(k) = (\sin k_1, \sin k_2, \cos k_1 + \cos k_2 - 1), \qquad \widehat{d}_{\mathrm{hex}} = d_{\mathrm{hex}}/|d_{\mathrm{hex}}|.$$

For a two-band Hamiltonian $d \cdot \sigma$, the negative-energy Chern number is

$$\mathrm{Ch}(E_-) = -\frac{1}{4\pi} \int_{\mathbb{T}^2} \widehat{d} \cdot (\partial_{k_1} \widehat{d} \times \partial_{k_2} \widehat{d})\, dk_1\, dk_2 = -\deg(\widehat{d}),$$

with the same convention used in the Bloch-bundle formula; see [1, Chapters 4–5] and the Bellissard–van Elst–Schulz-Baldes formulation of the Chern character [2].

The north pole $(0, 0, 1)$ is a regular value. Its preimages satisfy $\sin k_1 = \sin k_2 = 0$ and $\cos k_1 + \cos k_2 - 1 > 0$. The only such point is $(0, 0)$. The local degree there is

$$\operatorname{sgn} \det \begin{pmatrix} \cos k_1 & 0 \\ 0 & \cos k_2 \end{pmatrix}_{(0,0)} = +1.$$

Thus $\deg(\widehat{d}_{\mathrm{hex}}) = 1$, and the negative-energy Chern number is -1. □

6.68 Proof of Result 6.6

Proof of Theorem 6.6. Let $U = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$. Since a_1 and a_2 are odd under $k \mapsto -k$, while a_3^{hex} is even,

$$H_{\mathrm{hex}}(k; -m) = U\, \overline{H_{\mathrm{hex}}(-k; m)}\, U^*.$$

The base involution $k \mapsto -k$ preserves the orientation $dk_1 \wedge dk_2$. Unitary conjugation preserves the Bloch line bundle, while complex conjugation changes the sign of its first Chern class. Therefore the negative-energy Chern number at $-m$ is the negative of the value at $m > 0$. By Theorem 6.5, it is $+1$. □

6.69 Proof of Result 6.7

Proof of Theorem 6.7. The entries of $H_{\mathrm{hex}}(k; m)$ are Laurent polynomials in e^{ik_1} and e^{ik_2}, so the symbol is finite range. By Theorem 6.4, it is gapped at zero for $m \neq 0$. The straight half-space Toeplitz extension therefore applies. The coordinate statement is Theorem 5.7, and the primitive-basis statement is Theorem 5.8. Substituting Theorems 6.5 and 6.6 gives the displayed signs. □

6.70 Proof of Result 6.8

Proof of Theorem 6.8. The gap estimate is Theorem 6.4. The Chern value is Theorem 6.5 for $m > 0$ and Theorem 6.6 for $m < 0$, which is the formula $-\operatorname{sgn}(m)$. The Toeplitz formula is Theorem 6.7. □

6.71 Proof of Result 6.10

Proof of Theorem 6.10. At $k = (0, 0)$, $b_1 = b_2 = 0$ and $b_3^{\square} = 3$. Hence

$$H_{\square}(0, 0; 0) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\det H_{\square}(0, 0; 0) = 0$. □

6.72 Proof of Result 6.11

Proof of Theorem 6.11. Again the matrix is trace zero, and

$$\det H_{\square}(k;m) = -\sin^2 k_1 - \sin^2 k_2 - m^2(1+\cos k_1+\cos k_2)^2.$$

If $s = \sin^2 k_1 + \sin^2 k_2 \geq 1/4$, then $|\det H_{\square}(k;m)| \geq 1/4$. If $s < 1/4$, then $|\cos k_j| > \sqrt{3}/2$, $j = 1, 2$, and the four possible sign choices give

$$|1+\cos k_1+\cos k_2| \geq \sqrt{3}-1.$$

This gives the lower bound by $c_{\square}(m)$. The eigenvalues are $\pm\sqrt{|\det H_{\square}(k;m)|}$, so the distance from zero to the spectrum is at least $\sqrt{c_{\square}(m)}$. □

6.73 Proof of Result 6.12

Proof of Theorem 6.12. For $m > 0$, the positive-mass path to $m = 1$ stays gapped by Theorem 6.11. At $m = 1$, set

$$d_{\square}(k) = (\sin k_1, \sin k_2, 1+\cos k_1+\cos k_2), \qquad \widehat{d}_{\square} = d_{\square}/|d_{\square}|.$$

As in the two-band Chern formula cited in the proof of Theorem 6.5, $\mathrm{Ch}(E_-) = -\deg(\widehat{d}_{\square})$. The north-pole preimages are

$$(0,0), \qquad (\pi,0), \qquad (0,\pi),$$

since $1+\cos k_1+\cos k_2 > 0$ at exactly those three points among the four points with $\sin k_1 = \sin k_2 = 0$. The local signs are respectively

$$\mathrm{sgn}(\cos k_1 \cos k_2) = +1,\ -1,\ -1.$$

Thus $\deg(\widehat{d}_{\square}) = -1$, and the negative-energy Chern number is $+1$. □

6.74 Proof of Result 6.13

Proof of Theorem 6.13. The entries of $H_{\square}(k;m)$ are Laurent polynomials, and Theorem 6.11 gives a zero gap for $m \neq 0$. The straight-edge Toeplitz index formula in Theorem 5.7 and the primitive-coordinate version Theorem 5.8 give

$$\mathrm{wind}(\partial_{\mathrm{T},t,n}([P_{\square,m}])) = \det[t\ n]\,\mathrm{Ch}(P_{\square,m}).$$

For $m > 0$, Theorem 6.12 gives $+1$. For $m < 0$, the same antiunitary identity used in Theorem 6.6, with $b_3^{\square}(-k) = b_3^{\square}(k)$, changes the Chern sign. This gives -1. □

6.75 Proof of Result 6.14

Proof of Theorem 6.14. Theorem 6.11 gives the zero gap. Theorem 6.12 gives the Chern number $+1$ for $m > 0$, and the antiunitary mass-reversal argument in the proof of Theorem 6.13 gives -1 for $m < 0$. The Toeplitz winding formula is Theorem 6.13. □

6.76 Proof of Result 5.4

Proof of Theorem 5.4. At the algebraic level, operators supported within a bounded neighborhood of $Y^{(R)}$ are stable under addition and adjoint. They are also stable under multiplication on either side by bounded finite-propagation operators in $C^*(X)$, because finite propagation enlarges the support neighborhood by at most a bounded amount. Taking norm closure gives a closed two-sided ideal. □

6.77 Proof of Result 5.5

Proof of Theorem 5.5. Assume $R' \geq R$. Then

$$Y^{(R)} \subset Y^{(R')} \subset N_{R'}(Y^{(R)}).$$

Thus an operator supported within a bounded neighborhood of $Y^{(R')}$ is also supported within a bounded neighborhood of $Y^{(R)}$, and the converse inclusion is immediate from $Y^{(R)} \subset Y^{(R')}$. The corresponding algebraic ideals therefore coincide, and taking norm closures gives the equality. The case $R > R'$ is symmetric. □

6.78 Proof of Result 5.6

Proof of Theorem 5.6. Let T be an algebraic Roe operator whose matrix support is contained in $N_s(Y_0) \times N_s(Y_0)$ for some $s < \infty$. Since $Y_0 \subset N_R(Y_1)$, the same matrix support is contained in $N_{s+R}(Y_1) \times N_{s+R}(Y_1)$. Hence the algebraic interface ideal for Y_0 is contained in the algebraic interface ideal for Y_1. The reverse inclusion is identical, using $Y_1 \subset N_R(Y_0)$. Taking norm closures gives the equality. □

6.79 Proof of Result 5.10

Proof of Theorem 5.10. The propagation bound is exactly Assumption 5.8(i). For the norm bound, every row sum and every column sum of absolute values is at most $N_S c_S$ by Assumption 5.8(ii)–(iii). The Schur test therefore gives

$$\|S_n\| \leq N_S c_S.$$

□

6.80 Proof of Result 5.11

Proof of Theorem 5.11. By the Higson–Prodan construction, $B + B^\dagger$ maps a simplex barycenter to barycenters of simplices that share an incidence relation. In the barycentric graph metric, every such pair is at distance 1. Hence $B + B^\dagger$ has propagation exactly 1 on every $\ell^2(X_n)$. The operator S is handled by Lemma 5.10. Since $H_{\pm,n} = (B + B^\dagger) \pm S$, the propagation satisfies

$$\mathrm{prop}(H_{\pm,n}) \leq \max\{1, R_S\},$$

which is bounded uniformly in n. This proves (a).

For (b), $\left\|B + B^\dagger\right\|$ is bounded by twice the maximum simplex degree in the barycentric graph. Under uniform regularity (Definition 5.2), the ball sizes $|B_{X_n}(x, 1)|$ are uniformly bounded, so the maximum degree is uniformly bounded. Similarly, $\|S_n\|$ is bounded uniformly in n by Lemma 5.10. Hence

$$\|H_{\pm,n}\| \leq \left\|B + B^\dagger\right\| + \|S_n\| \leq B_{\mathrm{bulk}}$$

for a constant B_{bulk} independent of n. □

6.81 Proof of Result 5.17

Proof of Theorem 5.17. Assumption 5.15 includes uniform regularity of $(\mathcal{K}_n)_n$, and its part (a) gives Assumption 5.8. Its part (b) gives the uniform norm bound and the uniform spectral gap in Assumption 5.13; the propagation bound follows from Proposition 5.11. The identification of $P_\pm$ with the Higson–Prodan Fermi projection and the Chern computation are precisely part (c) of the imported closed-bulk input. □

6.82 Proof of Result 5.18

Proof of Theorem 5.18. Boundedness and self-adjointness are immediate from the direct-sum construction and Assumption 5.13(b). The propagation bound in Assumption 5.13(a) passes to the direct sum, so $H_{\pm}^{\infty}$ has propagation at most R_{loc} on $\ell^2(X)$.

It remains to verify local compactness. Let $F \subset X$ be finite and write χ_F for multiplication by its characteristic function. Because F is finite and $H_{\pm}^{\infty}$ has propagation at most R_{loc}, every vector in the range of $\chi_F H_{\pm}^{\infty}$ is supported inside the finite set $N_{R_{\mathrm{loc}}}(F)$. Hence $\chi_F H_{\pm}^{\infty}$ has finite-dimensional range and is therefore finite rank. The same argument applied to the adjoint shows that $H_{\pm}^{\infty}\chi_F$ is finite rank as well. Thus $H_{\pm}^{\infty}$ is locally compact and belongs to the algebraic Roe algebra; therefore $H_{\pm}^{\infty} \in C^*(X)$ after taking norm closure.

By Assumption 5.13(b),

$$\mathrm{spec}_{\mathrm{op}}(H_{\pm}^{\infty}) \subset [-B_{\mathrm{bulk}}, B_{\mathrm{bulk}}],$$

and by Assumption 5.13(c),

$$\mathrm{spec}_{\mathrm{op}}(H_{\pm}^{\infty}) \subset (-\infty, -g_{\mathrm{bulk}}] \cup [g_{\mathrm{bulk}}, \infty).$$

Choose a compactly supported continuous function $f \in C_c(\mathbb{R}) \subset C_0(\mathbb{R})$ such that

$$f(t) = 1 \quad \text{for } t \in [-B_{\mathrm{bulk}}, -g_{\mathrm{bulk}}/2], \qquad f(t) = 0 \quad \text{for } t \in [g_{\mathrm{bulk}}/2, B_{\mathrm{bulk}}],$$

and, for example, $f(t) = 0$ for $|t| \geq B_{\mathrm{bulk}} + 1$. Then f agrees with $\chi_{(-\infty,0)}$ on $\mathrm{spec}_{\mathrm{op}}(H_{\pm}^{\infty})$ (which avoids the interval $(-g_{\mathrm{bulk}}, g_{\mathrm{bulk}})$), so the operator functional calculus on $\ell^2(X)$ gives

$$f(H_{\pm}^{\infty}) = \chi_{(-\infty,0)}(H_{\pm}^{\infty}) = P_{\pm}.$$

Since $H_{\pm}^{\infty} \in C^*(X)$ and $f \in C_0(\mathbb{R})$, continuous functional calculus in $C^*(X)$ gives $f(H_{\pm}^{\infty}) \in C^*(X)$. Hence $P_{\pm} \in C^*(X)$, and so $P_{\pm}$ defines a class

$$[P_{\pm}] \in K_0(C^*(X)).$$

□

6.83 Proof of Result 5.19

Proof of Theorem 5.19. Choose one function $f \in C_c(\mathbb{R})$ such that

$$f(t) = 1 \quad \text{for } t \in [-B, -g/2], \qquad f(t) = 0 \quad \text{for } t \in [g/2, B].$$

Then $P_s = f(H_s)$ for every s. Continuous functional calculus is norm-continuous for a fixed continuous function on a norm-continuous path of self-adjoint elements, so $s \mapsto P_s$ is a norm-continuous path of projections in $C^*(X)$. A norm-continuous path of projections represents a constant K_0-class. □

6.84 Proof of Result 5.20

Proof of Theorem 5.20. For $z \in \Gamma$,

$$\left\|(H - z)^{-1}\right\| \leq \delta_{\Gamma}^{-1}.$$

Since

$$\left\|V(H - z)^{-1}\right\| \leq \frac{\|V\|}{\delta_{\Gamma}} < 1,$$

the operator $H + V - z$ is invertible and

$$\left\|(H + V - z)^{-1}\right\| \leq \frac{1}{\delta_\Gamma - \|V\|}.$$

The resolvent identity gives

$$(H + V - z)^{-1} - (H - z)^{-1} = -(H + V - z)^{-1} V (H - z)^{-1}.$$

Integrating this identity around Γ gives the stated projection estimate. □

6.85 Proof of Result 6.31

Proof of Theorem 6.31. For $t \in [0, 1]$, set $H_t := H + tV$. Weyl's spectral-stability estimate gives

$$\mathrm{spec}_{\mathrm{op}}(H_t) \cap \big(-(g - t\,\|V\|),\, g - t\,\|V\|\big) = \emptyset.$$

Since $g - \|V\| > 0$, the path H_t is uniformly gapped at zero. Proposition 5.19 therefore applies, with spectral bound $B + \|V\|$ and gap $g - \|V\|$, and gives equality of the endpoint Fermi-projection classes. In the interface case, the path

$$s \longmapsto \chi_{(-\infty,0)}(H_\pm^\infty + sV)$$

is a norm-continuous path of projections joining $P_\pm$ to P_V. Applying Proposition 5.32 gives the stated equality. □

6.86 Proof of Result 5.21

Proof of Theorem 5.21. The difference $H_{\pm,n} - H_{\pm,n}^{\mathrm{split}}$ is exactly the sum of the two off-diagonal blocks

$$P_{\mathrm{orig},n} H_{\pm,n} P_{\mathrm{copy},n} + P_{\mathrm{copy},n} H_{\pm,n} P_{\mathrm{orig},n}.$$

Thus it suffices to prove that every nonzero matrix element between the original side and the strict copy side is supported inside Y_n.

A vertex of the barycentric graph is the barycenter of a simplex of $D(\mathcal{K}_n)$, and an edge joins two vertices precisely when the corresponding simplices are incident by inclusion. Let

$$U_{n,\mathrm{orig}} := X_{n,\mathrm{orig}} \setminus \Gamma_n, \qquad U_{n,\mathrm{copy}} := X_{n,\mathrm{copy}} \setminus \Gamma_n.$$

We claim that no graph edge joins a vertex of $U_{n,\mathrm{orig}}$ directly to a vertex of $U_{n,\mathrm{copy}}$. An edge in the barycentric graph corresponds to a pair of simplices $\sigma \subset \tau$ or $\tau \subset \sigma$. If one of these simplices lies strictly on the original side and the other lies strictly on the copy side, then they cannot be nested: a simplex containing vertices from both sides necessarily meets the glued boundary, and a boundary simplex itself already belongs to Γ_n. Hence every adjacency from the original side to the copy side must pass through a simplex meeting the glued boundary.

As a consequence, every graph path from $X_{n,\mathrm{orig}}$ to $X_{n,\mathrm{copy}}$ must pass through Γ_n. Now let

$$x \in X_{n,\mathrm{orig}}, \qquad y \in X_{n,\mathrm{copy}}, \qquad \langle x, H_{\pm,n} y\rangle \neq 0.$$

Since $H_{\pm,n}$ has propagation at most R_{loc}, we have

$$d_{X_n}(x, y) \leq R_{\mathrm{loc}}.$$

Choose a geodesic path from x to y in the barycentric graph. By the previous paragraph, this path meets Γ_n at some vertex z. Because the path has total length at most R_{loc}, we have

$$d_{X_n}(x,z) \le R_{\mathrm{loc}}, \qquad d_{X_n}(y,z) \le R_{\mathrm{loc}}.$$

Hence both x and y belong to $N_{R_{\mathrm{loc}}}(\Gamma_n) = Y_n$. Therefore every original–copy matrix coefficient of $H_{\pm,n}$ vanishes after compressing to $X_n \setminus Y_n$, which is exactly (5.1).

For (5.2), observe that the jam term is already diagonal with respect to the decomposition $\mathcal{H}_{\mathrm{orig},n} \oplus \mathcal{H}_{\mathrm{copy},n}$. Consequently,

$$H_{\pm,M,n} - \left(H_{\pm,n}^{\mathrm{split}} + MP_{\mathrm{copy},n}\right) = H_{\pm,n} - H_{\pm,n}^{\mathrm{split}},$$

and (5.2) follows from (5.1). □

6.87 Proof of Result 5.22

Proof of Theorem 5.22. By Proposition 5.21, each summand of either difference is supported inside $Y_n \times Y_n$. In both differences the jam term is absent or cancels exactly, so their propagations are uniformly bounded by R_{loc} and their norms are uniformly bounded by the uniform bound on $(H_{\pm,n})_n$. Hence the direct sums are bounded operators on $\ell^2(X)$ with uniformly finite propagation and support contained in a bounded neighborhood of Y.

Because X has bounded geometry and we work with scalar coefficients, such operators are locally compact: if $F \subset X$ is finite, then finite propagation implies that both $T\chi_F$ and $\chi_F T$ have range contained in ℓ^2 of a finite neighborhood of F, hence are finite rank. Therefore the two direct sums belong to the algebraic interface ideal, and thus to its norm closure $C^*(Y \subset X)$. □

6.88 Proof of Result 5.23

Proof of Theorem 5.23. The operator $\widetilde{T}^{-}_{\pm,M}$ is a direct sum of bounded local self-adjoint operators, so it is bounded, self-adjoint, and local. On the original-side sector it equals the identity, hence has spectral value 1.

On the copy-side sector it equals $H^{\infty}_{\pm,\mathrm{copy}} + MP^{\infty}_{\mathrm{copy}}$. For each n every spectral value λ of $H^{\mathrm{copy}}_{\pm,n} + M\mathbf{1}$ satisfies

$$\lambda \ge M + \lambda_{\min}(H^{\mathrm{copy}}_{\pm,n}).$$

By the definition of $B_{\mathrm{copy},-}$, the right-hand side is strictly positive when $M > B_{\mathrm{copy},-}$. Thus the whole operator spectrum of $\widetilde{T}^{-}_{\pm,M}$ is contained in $\{1\} \cup [M - B_{\mathrm{copy},-}, \infty)$, which gives the stated spectral-distance bound. In particular, the spectrum is contained in $(0,\infty)$, so the negative operator spectral projection vanishes. □

6.89 Proof of Result 5.25

Proof of Theorem 5.25. Fix $r > 0$. By the definition of coarse disjoint union, there exists N_r such that

$$\mathrm{dist}(X_n, X_m) > 2r \qquad \text{whenever } n \ne m \text{ and } \max\{n,m\} > N_r.$$

Let

$$x \in N_r(X^+) \cap N_r(X^-).$$

Choose $x_+ \in X^+$ and $x_- \in X^-$ with

$$d(x,x_+) \le r, \qquad d(x,x_-) \le r.$$

Then $d(x_+, x_-) \le 2r$. Suppose first that either x_+ or x_- lies in a component X_n with $n > N_r$. The separation condition forces both points to lie in that same component. If either point lies in Y_n, then $x \in N_r(Y)$. Otherwise

$$x_+ \in X_{n,\mathrm{orig}} \setminus Y_n, \qquad x_- \in X_{n,\mathrm{copy}} \setminus Y_n.$$

By the same separation argument used in Proposition 5.21, every geodesic in the barycentric graph from x_+ to x_- meets $\Gamma_n \subset Y_n$. Therefore $d(x_+, Y_n) \le 2r$, and consequently

$$d(x, Y) \le d(x, x_+) + d(x_+, Y_n) \le 3r.$$

Thus every such x lies in $N_{3r}(Y)$.

It remains to treat the case in which neither x_+ nor x_- lies in a component of index larger than N_r. Let

$$F_r := \bigsqcup_{n=1}^{N_r} X_n.$$

This is a finite set. Because each $\mathcal{K}_n$ has nonempty boundary, every Y_n is nonempty, so

$$c_r := \max_{x \in N_r(F_r)} d(x, Y)$$

is finite. Therefore

$$N_r(X^+) \cap N_r(X^-) \subset N_{s(r)}(Y), \qquad s(r) := \max\{3r, c_r\},$$

which is exactly coarse excision. □

6.90 Proof of Result 5.27

Proof of Theorem 5.27. For $Q_Y T$ and $T Q_Y$, finite propagation implies

$$Q_Y T = \chi_Y \, T \, \chi_{N_R(Y)}, \qquad T Q_Y = \chi_{N_R(Y)} \, T \, \chi_Y,$$

so both are supported in $N_R(Y) \times N_R(Y)$ and hence belong to $\mathcal{I}$.

Now consider $Q_+ T Q_-$. If

$$\langle x, Q_+ T Q_- \, y \rangle \neq 0,$$

then $x \in Z^+$, $y \in Z^-$, and $d(x, y) \le R$. Therefore

$$x \in N_R(X^-), \qquad y \in N_R(X^+).$$

Since also $x \in X^+$ and $y \in X^-$, Proposition 5.25 gives

$$x, y \in N_{s(R)}(Y).$$

Hence $Q_+ T Q_-$ is supported in $N_{s(R)}(Y) \times N_{s(R)}(Y)$, so $Q_+ T Q_- \in \mathcal{I}$. The same argument gives $Q_- T Q_+ \in \mathcal{I}$.

Because $\mathcal{I}$ is closed and algebraic Roe operators are dense in $\mathcal{A}$, the same containments hold for every $T \in \mathcal{A}$.

We also record that multiplication by Q_+, Q_-, or Q_Y preserves $\mathcal{I}$: algebraic Roe operators supported near Y remain supported near Y after such left or right multiplication, and the claim follows by norm closure.

We now prove part (ii). Since $Q_Y \in \mathcal{I}$ by part (i), with $T = \mathbf{1}$, we have

$$e_+ + e_- = \pi(Q_+ + Q_-) = \pi(\mathbf{1} - Q_Y) = \mathbf{1}_{\mathcal{A}/\mathcal{I}}.$$

Orthogonality is immediate from $Q_+Q_- = 0$. The centrality identity proved below shows that $e_\pm$ are central projections.

We next prove part (iii). If $\pi(a) = \pi(a')$, then $a - a' \in \mathcal{I}$, hence

$$Q_\pm(a - a') \in \mathcal{I}.$$

Therefore $E_\pm$ are well-defined.

Idempotence and orthogonality are immediate:

$$E_\pm^2(\pi(a)) = \pi(Q_\pm Q_\pm a) = E_\pm(\pi(a)), \qquad E_+E_-(\pi(a)) = \pi(Q_+Q_-a) = 0,$$

and similarly for E_-E_+. Also,

$$\pi(a) = \pi\big((Q_+ + Q_- + Q_Y)a\big) = E_+(\pi(a)) + E_-(\pi(a)),$$

because $Q_Y a \in \mathcal{I}$. Thus $E_+ + E_- = \mathbf{1}_{\mathcal{A}/\mathcal{I}}$.

To prove multiplicativity for E_+, let $a, b \in \mathcal{A}$. Then

$$E_+(\pi(a))E_+(\pi(b)) = \pi(Q_+a)\,\pi(Q_+b) = \pi(Q_+aQ_+b).$$

Now

$$Q_+ab - Q_+aQ_+b = Q_+a(Q_- + Q_Y)b.$$

The term Q_+aQ_-b lies in $\mathcal{I}$ because $Q_+aQ_- \in \mathcal{I}$ and $\mathcal{I}$ is an ideal. The term Q_+aQ_Yb also lies in $\mathcal{I}$ because $aQ_Y \in \mathcal{I}$. Hence

$$\pi(Q_+aQ_+b) = \pi(Q_+ab) = E_+(\pi(ab)).$$

So E_+ is multiplicative. The same proof applies to E_-. The $*$-property is immediate:

$$E_+(\pi(a))^* = \pi((Q_+a)^*) = \pi(a^*Q_+) = E_+(\pi(a^*)),$$

because

$$a^*Q_+ - Q_+a^* = (Q_+a - aQ_+)^* \in \mathcal{I},$$

as shown next.

To prove the centrality identity, let $a, b \in \mathcal{A}$. Then

$$E_+(\pi(a))\,\pi(b) - \pi(a)\,E_+(\pi(b)) = \pi\big((Q_+a - aQ_+)b\big).$$

Now

$$Q_+a - aQ_+ = Q_+a(Q_- + Q_Y) - (Q_- + Q_Y)aQ_+.$$

Each term on the right belongs to $\mathcal{I}$: the Q_+aQ_- and Q_-aQ_+ terms are in $\mathcal{I}$ by part (i), and the terms involving Q_Y are in $\mathcal{I}$ because aQ_Y and Q_Ya are in $\mathcal{I}$ and $\mathcal{I}$ is stable under multiplication by Q_+. Hence $Q_+a - aQ_+ \in \mathcal{I}$, so

$$E_+(\pi(a))\,\pi(b) = \pi(a)\,E_+(\pi(b)).$$

Thus E_+ is central in the sense stated. The same holds for E_-.

Finally, every $b \in \mathcal{A}/\mathcal{I}$ decomposes as

$$b = E_+(b) + E_-(b),$$

and orthogonality $E_+E_- = E_-E_+ = 0$ implies uniqueness of this decomposition. Since $E_\pm(\mathcal{A}/\mathcal{I})$ are closed two-sided ideals, this proves the direct-sum decomposition. □

6.91 Proof of Result 5.28

Proof of Theorem 5.28. The operators J, T^+, T^- are locally compact: for every finite $F \subset X$, the operators $\chi_F T$ and $T\chi_F$ have finite-dimensional range, with T equal to J, T^+, or T^-. Hence

$$J, T^+, T^- \in \mathcal{A} = C^*(X),$$

so $\pi(J)$ and $\pi(T^\pm)$ are well-defined.

Since 0 lies in an operator spectral gap of $T^\pm$, choose compactly supported continuous functions $h_\pm$ that agree with $\chi_{(-\infty,0)}$ on $\mathrm{spec}_{\mathrm{op}}(T^\pm)$. Continuous functional calculus in $\mathcal{A}$ gives

$$P^\pm = h_\pm(T^\pm) \in \mathcal{A}.$$

Thus $\pi(P^\pm)$ is well-defined.

Set

$$D := J - Q_+ T^+ Q_+ - Q_- T^- Q_-.$$

We prove $D \in \mathcal{I}$.

Expand J using $Q_+ + Q_- + Q_Y = \mathbf{1}$ on $\ell^2(X)$. Every term in this expansion containing Q_Y lies in $\mathcal{I}$ by Proposition 5.27(i). The same is true for the cross terms $Q_+ J Q_-$ and $Q_- J Q_+$.

It remains to treat the diagonal difference terms

$$Q_+(J - T^+)Q_+, \qquad Q_-(J - T^-)Q_-.$$

Let

$$S_+ := \mathrm{prop}(J - T^+).$$

Suppose

$$\langle x, Q_+(J - T^+)Q_+\, y\rangle \neq 0.$$

Then $x, y \in Z^+$ and $d(x, y) \leq S_+$. If $x \notin N_{R_{\mathrm{ag}}+S_+}(Y)$, then

$$d(x, Y) > R_{\mathrm{ag}} + S_+,$$

so

$$d(y, Y) \geq d(x, Y) - d(x, y) > R_{\mathrm{ag}}.$$

Hence both x and y lie in $X^+ \setminus N_{R_{\mathrm{ag}}}(Y)$, and the assumed sidewise agreement forces the matrix coefficient to vanish, a contradiction. Thus every nonzero matrix coefficient of $Q_+(J - T^+)Q_+$ has

$$x \in N_{R_{\mathrm{ag}}+S_+}(Y).$$

By symmetry the same holds for y. Therefore

$$Q_+(J - T^+)Q_+ \in \mathcal{I}.$$

The same argument gives

$$Q_-(J - T^-)Q_- \in \mathcal{I}.$$

Hence $D \in \mathcal{I}$.

Applying π gives

$$\pi(J) = \pi(Q_+ T^+ Q_+) + \pi(Q_- T^- Q_-).$$

By Proposition 5.27,

$$\pi(Q_+T^+Q_+) = E_+(\pi(T^+)), \qquad \pi(Q_-T^-Q_-) = E_-(\pi(T^-)),$$

which proves (i).

Now fix $0 < \gamma < \min\{\gamma_+, \gamma_-\}$, choose $B \geq \max\{\|T^+\|, \|T^-\|\}$, and let h be as in part (ii). Because

$$\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I}),$$

continuous functional calculus is componentwise on this direct sum. Therefore part (i) implies

$$h(\pi(J)) = E_+\big(h(\pi(T^+))\big) + E_-\big(h(\pi(T^-))\big).$$

Since π is a $*$-homomorphism, it commutes with continuous functional calculus:

$$h(\pi(T^\pm)) = \pi(h(T^\pm)).$$

On the operator spectrum of $T^\pm$, the function h agrees with $\chi_{(-\infty,0)}$ because $\mathrm{spec}_{\mathrm{op}}(T^\pm) \subset [-B, B]$ and $\mathrm{spec}_{\mathrm{op}}(T^\pm) \cap (-\gamma, \gamma) = \emptyset$. Hence the operator functional calculus on $\ell^2(X)$ gives

$$h(T^\pm) = \chi_{(-\infty,0)}(T^\pm) = P^\pm.$$

Consequently,

$$h(\pi(J)) = E_+(\pi(P^+)) + E_-(\pi(P^-)),$$

which proves (ii).

For part (iii), each $\pi(P^\pm)$ is a projection, $E_\pm$ are $*$-homomorphisms, and their ranges are orthogonal ideals. Therefore

$$p_J^2 = E_+(\pi(P^+))^2 + E_-(\pi(P^-))^2 = p_J, \qquad p_J^* = p_J.$$

So p_J is a projection. The identity $p_J = h(\pi(J)) = \pi(h(J))$ follows from part (ii) and the continuous functional calculus. Since the right-hand side is the same for every admissible cutoff h, the projection p_J is independent of the choice of γ and h. □

6.92 Proof of Result 5.31

Proof of Theorem 5.31. Because $\widetilde{p}$ is a self-adjoint lift of the projection p,

$$\pi\big(\exp(2\pi i\,\widetilde{p})\big) = \exp(2\pi i\,p) = \mathbf{1}.$$

Hence

$$\exp(2\pi i\,\widetilde{p}) - \mathbf{1} \in M_n(\mathcal{I}),$$

so $\exp(2\pi i\,\widetilde{p})$ is a unitary in $\mathbf{1} + M_n(\mathcal{I})$ and therefore defines a class in $K_1(\mathcal{I})$.

If $\widetilde{p}_0$ and $\widetilde{p}_1$ are two self-adjoint lifts of the same projection p, then

$$\widetilde{p}_t := (1-t)\widetilde{p}_0 + t\widetilde{p}_1 \qquad (0 \leq t \leq 1)$$

is a norm-continuous path of self-adjoint lifts of p. Therefore

$$t \longmapsto \exp(2\pi i\,\widetilde{p}_t)$$

is a homotopy of unitaries in $\mathbf{1} + M_n(\mathcal{I})$, so the K_1-class is independent of the chosen lift.

For the last statement, because $\mathcal{A}$ is unital, $h(a) \in M_n(\mathcal{A})$ for every continuous h. Functional calculus in $\mathcal{A}/\mathcal{I}$ gives

$$\pi(h(a)) = h(\pi(a)).$$

Since h agrees with $\chi_{(-\infty,0)}$ on $\operatorname{spec}(\pi(a))$, the element $h(\pi(a))$ is a projection. Applying the first part with lift $h(a)$ proves

$$\partial_Y\big(\pi(h(a))\big) = \big[\exp(2\pi i\, h(a))\big].$$

□

6.93 Proof of Result 5.32

Proof of Theorem 5.32. The map

$$s \longmapsto E_+(\pi(P_s))$$

is a norm-continuous path of projections in $\mathcal{A}/\mathcal{I}$, because π and E_+ are bounded $*$-homomorphisms. Hence the $K_0(\mathcal{A}/\mathcal{I})$-class of $E_+(\pi(P_s))$ is constant. Applying the connecting map

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \longrightarrow K_1(\mathcal{I})$$

gives the claim. □

6.94 Proof of Result 6.32

Proof of Theorem 6.32. Since $V, V^{\pm} \in \mathcal{I}$, the quotient images satisfy

$$\pi(J') = \pi(J), \qquad \pi(T'^{\pm}) = \pi(T^{\pm}).$$

Let

$$P^{\pm} := \chi_{(-\infty,0)}(T^{\pm}), \qquad P'^{\pm} := \chi_{(-\infty,0)}(T'^{\pm}).$$

Choose continuous compactly supported cutoffs $h^{\pm}$ and $h'^{\pm}$ agreeing with $\chi_{(-\infty,0)}$ on the operator spectra of $T^{\pm}$ and $T'^{\pm}$, respectively. Since

$$\operatorname{spec}(\pi(T^{\pm})) = \operatorname{spec}(\pi(T'^{\pm}))$$

and this quotient spectrum is contained in both operator spectra, both cutoffs give the same functional calculus element in the quotient:

$$\pi(P'^{\pm}) = h'^{\pm}(\pi(T'^{\pm})) = h'^{\pm}(\pi(T^{\pm})) = h^{\pm}(\pi(T^{\pm})) = \pi(P^{\pm}).$$

Therefore

$$p_{J'} = E_+(\pi(P'^{+})) + E_-(\pi(P'^{-})) = E_+(\pi(P^{+})) + E_-(\pi(P^{-})) = p_J.$$

Applying ∂_Y gives equality of the interface classes. □

6.95 Proof of Result 6.33

Proof of Theorem 6.33. Let

$$F := \bigsqcup_{n=1}^{N} X_n.$$

This is finite. Since each Y_n is nonempty,

$$R_F := \max_{x \in F} \operatorname{dist}(x, Y)$$

is finite, so $F \subset N_{R_F}(Y)$. An operator supported on $F \times F$ is therefore supported inside $N_{R_F}(Y) \times N_{R_F}(Y)$. Since V is bounded and finite-propagation, it belongs to the algebraic interface ideal, hence to $C^*(Y \subset X)$. Applying this observation to each finite-component change in the sidewise data, the final statement follows from Proposition 6.32. □

6.96 Proof of Result 6.34

Proof of Theorem 6.34. Let $V^{(N)} := \bigoplus_{n \le N} V_n$. This operator is supported on the finite set $\bigsqcup_{n \le N} X_n$. Since each Y_n is nonempty, the proof of Proposition 6.33 shows that $V^{(N)} \in C^*(Y \subset X)$. Also

$$\|V - V^{(N)}\| = \sup_{n > N} \|V_n\| \longrightarrow 0.$$

The interface ideal is norm closed, so $V \in C^*(Y \subset X)$. The final statement is Proposition 6.32. □

6.97 Proof of Result 6.35

Proof of Theorem 6.35. Let

$$B := \sup_{s \in [0,1]} \max\{\|T_s^+\|, \|T_s^-\|\} < \infty.$$

Choose one compactly supported continuous cutoff h that equals 1 on $[-B, -\gamma/2]$ and 0 on $[\gamma/2, B]$. Then h agrees with $\chi_{(-\infty,0)}$ on the operator spectra of all $T_s^{\pm}$, and

$$P_s^{\pm} := h(T_s^{\pm})$$

are norm-continuous paths of projections in $\mathcal{A}$. Hence

$$p_s := E_+(\pi(P_s^+)) + E_-(\pi(P_s^-))$$

is a norm-continuous path of projections in $\mathcal{A}/\mathcal{I}$, so its K_0-class is constant. Applying the boundary map

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \to K_1(\mathcal{I})$$

shows that the interface class is independent of s. □

6.98 Proof of Result 6.36

Proof of Theorem 6.36. The positive-side comparison is fixed and has the uniform bulk gap from Assumption 5.13. On the copy sector,

$$H^{\infty}_{\pm,\mathrm{copy}} + V_s \ge (M_0 - B_{\mathrm{copy},-})\mathbf{1},$$

while on the original sector T_s^- equals $\mathbf{1}$. Hence the negative-side comparison has a uniform positive gap at zero. The collar-locality argument is unchanged: away from the glued boundary, J_s agrees with T_s^+ on the positive side and with T_s^- on the negative side. Thus $(J_s; T_s^+, T_s^-)$ is a norm-continuous family of sidewise class-A interfaces with a uniform comparison gap, and Proposition 6.35 applies. □

6.99 Proof of Result 6.37

Proof of Theorem 6.37. The operators are bounded, self-adjoint, finite-propagation Roe operators. The positive comparison $T^+ = H^\infty_\pm$ has the bulk gap and Fermi projection $P_\pm$ by Proposition 5.18. On the original sector, T^-_V is the identity. On the copy sector,

$$H^\infty_{\pm,\mathrm{copy}} + V \geq (M_0 - B_{\mathrm{copy},-})\mathbf{1} > 0,$$

so

$$\operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^-_V)\big) \geq \min\{1, M_0 - B_{\mathrm{copy},-}\}.$$

Thus T^-_V is strictly positive and its negative Fermi projection is zero.

It remains to check sidewise agreement. On the positive side outside Y we are strictly in the original sector, so V vanishes and $J_V = T^+$. On the negative side outside Y, the original sector projection vanishes, and Proposition 5.21 kills the original–copy cross block. Thus $J_V = T^-_V$ there. Therefore Proposition 5.28 applies, and since the negative-side Fermi projection is zero it gives

$$p_{J_V} = E_+(\pi(P_\pm)).$$

The formula for the interface class is Definition 5.33. □

6.100 Proof of Result 5.35

Proof of Theorem 5.35. The operators $J^\infty_{\pm,M}$, $T^+ = H^\infty_\pm$, and $T^- = \widetilde{T}^-_{\pm,M}$ are bounded self-adjoint local operators on the scalar bounded-geometry space X, hence belong to $\mathcal{A} = C^*(X)$.

By Proposition 5.18,

$$T^+ = H^\infty_\pm$$

is bounded, self-adjoint, local, has an operator gap at 0, and has operator Fermi projection

$$P^+ = P_\pm \in C^*(X).$$

By Proposition 5.23,

$$T^- = \widetilde{T}^-_{\pm,M}$$

is bounded, self-adjoint, local, and has

$$\operatorname{dist}\big(0, \operatorname{spec}_{\mathrm{op}}(T^-)\big) \geq \min\{1, M - B_{\mathrm{copy},-}\}$$

when $M > B_{\mathrm{copy},-}$. Therefore

$$P^- := \chi_{(-\infty,0)}(T^-) = 0.$$

It remains to verify the sidewise agreement conditions. On the positive side outside Y we are strictly on the original side, so the jam term vanishes and

$$\chi_{X^+\setminus Y}\big(J^\infty_{\pm,M} - T^+\big)\chi_{X^+\setminus Y} = 0.$$

On the negative side outside Y we are strictly on the copy side. There P^∞_{orig} vanishes, and Proposition 5.21 shows that all original–copy matrix coefficients vanish there. Hence

$$\chi_{X^-\setminus Y}\big(J^\infty_{\pm,M} - T^-\big)\chi_{X^-\setminus Y} = 0.$$

Thus $(J^\infty_{\pm,M}; T^+, T^-)$ is a sidewise class-A interface along Y.

Now Proposition 5.28 applies. Since $P^- = 0$, it yields

$$p_{J^\infty_{\pm,M}} = E_+\big(\pi(P^+)\big) + E_-\big(\pi(P^-)\big) = E_+\big(\pi(P_\pm)\big),$$

which is (5.3). The equivalent cutoff statement follows from the same proposition.

Finally, Definition 5.33 gives

$$\mathrm{Ind}_{\mathrm{int}}(J^\infty_{\pm,M}; T^+, T^-) = \partial_Y\big(p_{J^\infty_{\pm,M}}\big) = \partial_Y\big(E_+(\pi(P_\pm))\big),$$

which is (5.4). The statement about the jammed side is exactly the vanishing of P^- and hence of the negative-side quotient contribution. □

6.101 Proof of Result 6.38

Proof of Theorem 6.38. The right-hand side depends only on the bulk projection $P_\pm$ and the positive quotient endomorphism E_+, not on M. Apply Theorem 5.35. □

6.102 Proof of Result 5.36

Proof of Theorem 5.36. Each $H_{\pm,n} = (B_n + B_n^\dagger) \pm S_n$ is the closed-case Higson–Prodan Hamiltonian on $D(\mathcal{K}_n)$, so the direct-sum family $H^\infty_\pm$ is the Higson–Prodan family for the refinement sequence of closed doubles. The bulk projection $P_\pm$ constructed in Proposition 5.18 is therefore the Higson–Prodan Fermi projection, and the Chern-number computation is the imported content of the Higson–Prodan construction and its Higson–Roe analytic-surgery input [8, 9, 10, 11, 12]. □

6.103 Proof of Result 6.39

Proof of Theorem 6.39. The identity is Theorem 5.35 under the additional Chern-number input recorded in Theorem 5.36. The independence of the large scalar jam scale is Corollary 6.38. This gives exactly the displayed formula and the stated identification of the bulk projection. □

6.104 Proof of Result 6.40

Proof of Theorem 6.40. Theorem 5.17 gives Assumptions 5.8 and 5.13. The interface-class formula is then Theorem 5.35, and the Chern-number statement is the Higson–Prodan bulk theorem as recorded in Theorem 5.36. For the original/copy partition, the winding statement is Theorem 4.9. □

6.105 Proof of Result 6.41

Proof of Theorem 6.41. The closed-bulk input gives the positive-side model and its Chern pairing. Proposition 5.25 proves that the original/copy decomposition gives a coarse partition with interface Y. Corollary 5.22 gives sidewise agreement away from Y: the positive side agrees with the closed Higson–Prodan model and the copy side agrees with the jammed comparison model. Proposition 5.23 shows that this comparison model is trivial on the negative side once $M > B_{\mathrm{copy},-}$. These are the geometric and sidewise pieces of Definition 5.37; the Ewert–Meyer/Roe identification is Theorem 4.9. None of these steps uses a boundary gap. □

6.106 Proof of Result 6.42

Proof of Theorem 6.42. The double of a full-boundary triangulated oriented surface is a closed oriented triangulated surface by Proposition 2.4. The listed refinement schemes have uniformly bounded incidence under the stated hypotheses, and doubling preserves that bounded geometry. The HP-covered condition supplies the controlled cap-product operator, the closed-bulk gap, and the Chern pairing $\mathrm{Ch}([P_\pm]) = \pm 1$. Proposition 5.25 supplies the original/copy coarse partition, Proposition 6.41 records the Ewert–Meyer convention, and Theorem 4.9 gives the K_1-class identity. □

6.107 Proof of Result 6.43

Proof of Theorem 6.43. Changing or deleting finitely many components gives an operator supported on a finite subset of the coarse disjoint union. Since each interface component is nonempty, such an operator belongs to the interface ideal by Proposition 6.33. The sidewise interface K_1-class is therefore unchanged by Proposition 6.32. In the straight periodic Toeplitz case (Theorem 4.10) the integer winding, as a K_1-homomorphism, has the same value on the unchanged K_1-class. □

6.108 Proof of Result 5.38

Proof of Theorem 5.38. The Higson–Prodan closed-bulk theorem gives Assumptions 5.8 and 5.13 through Theorem 5.17. It also identifies $P_\pm$ as the Higson–Prodan Fermi projection with $\mathrm{Ch}([P_\pm]) = \pm 1$. Theorem 5.35 gives the sidewise interface realization and the formula

$$\mathrm{Ind}_{\mathrm{int}}(J^{\infty}_{\pm,M}; T^+, T^-) = \partial_Y(E_+(\pi(P_\pm))).$$

The negative comparison operator has no negative-energy Fermi projection by Proposition 5.23, so the negative side contributes Chern number 0. The K_1-class identity is therefore unconditional modulo HP. In the straight periodic Toeplitz case (Theorem 4.10), the Toeplitz integer pairing applied to this class yields ± 1 internally. For general partitions the integer pairing is recorded in Theorem 4.11, conditional on the existence of an integer-valued Chern-jump pairing on $K_1(\mathcal{I})$. Independence of M is Corollary 6.38. □

6.109 Proof of Result 6.44

Proof of Theorem 6.44. An orientation-preserving boundary-compatible simplicial isomorphism induces a unitary identification of the simplex Hilbert spaces. It conjugates the boundary operator, cap-product operator, original/copy projections, collar ideals, and quotient maps, so the corresponding K-classes and pairings are unchanged. Reversing orientation changes the sign of the fundamental and relative fundamental classes. Hence the Poincaré-duality operator S changes sign, up to the same unitary relabeling, and $H_+ = (B + B^\dagger) + S$ is identified with the old H_-. The Higson–Prodan Chern number and the compatible interface winding therefore change sign. □

6.110 Proof of Result 6.45

Proof of Theorem 6.45. The scalar jam-scale statement is Corollary 6.38. Positive copy-side confinements are covered by Theorem 6.37 and Corollary 6.36. Interface-supported changes are Proposition 6.32, finite-component changes are Proposition 6.33, norm-vanishing component defects are Proposition 6.34, and norm-continuous sidewise homotopies are Proposition 6.35. Each operation preserves the K_1-class $\partial_Y(E_+(\pi(P_\pm)))$. In the straight periodic Toeplitz case (Theorem 4.10) the integer winding, as a K_1-homomorphism, takes the same value on the preserved K_1-class. □

6.111 Appendix proofs for Section 4

6.112 Proof of Result 4.1

Proof of Theorem 4.1. The construction of S_K is the Higson–Prodan cap-product construction [8, 9]. The boundary matrix B has entries only between incident simplices, so $B + B^\dagger$ is self-adjoint and has propagation 1 in the barycentric graph. The operator S_K is Hermitian by construction and is supported on the uniformly bounded barycentric neighborhood prescribed by the controlled cap-product formula. Therefore $H_\pm(K)$ is self-adjoint and finite propagation.

For a bounded-geometry family, the number of nonzero entries in each row of B, $B^\dagger$, and S_K is uniformly bounded, and the matrix coefficients belong to a fixed finite set of incidence signs and cap-product coefficients. The Schur test gives a uniform norm bound for each summand, hence for $H_\pm(K)$. □

6.113 Proof of Result 4.4

Proof (citation). This is the closed universal-model theorem of Higson–Prodan, *Universal Chern Model on Arbitrary Triangulations*, Phys. Rev. Lett. **136** (2026), 096602, and arXiv:2510.20862 [8, 9]. Their theorem is stated for the same universal Hamiltonians built from the simplicial boundary operator and the controlled Poincaré-duality operator on bounded-geometry closed triangulations. The closed doubles $D(\mathcal{K}_n)$ are closed oriented triangulations with the bounded-geometry control required in that theorem, so its uniform clean Fermi-gap conclusion applies. No part of the Higson–Prodan gap theorem is reproved here. □

6.114 Proof of Result 4.5

Proof (citation). The same Higson–Prodan closed universal-model theorem identifies the Fermi projection of $H_\pm = (B + B^\dagger) \pm S$ with the nontrivial closed-bulk class and computes its Chern pairing as ± 1, with the sign fixed by the two choices H_+, H_- and the orientation convention [8, 9]. The headline theorem uses exactly that closed-bulk class on the closed doubles. Thus $P_\pm$ has $\mathrm{Ch}([P_\pm]) = \pm 1$. The proof is purely by citation to the published closed theorem. □

6.115 Proof of Result 4.6

Proof of Theorem 4.6. The identity operator on $\ell^2(X)$ has propagation 0. On a discrete bounded-geometry scalar Roe space, finite-support compressions of the identity are finite-rank, so the identity is locally compact and $C^*(X)$ is unital.

By Lemmas 4.1 and 4.4, the direct sum $H_\pm^\infty = \bigoplus_n H_\pm(D(\mathcal{K}_n))$ is a bounded self-adjoint finite-propagation element of $C^*(X)$ with a uniform spectral gap about 0. Choose $f \in C_c(\mathbb{R})$ that equals 1 on the negative spectral window and 0 on the positive spectral window. Functional calculus in the unital Roe algebra gives

$$P_\pm = f(H_\pm^\infty) = \chi_{(-\infty,0)}(H_\pm^\infty) \in C^*(X).$$

This is the same argument as Proposition 5.18, now with the gap supplied by Lemma 4.4. □

6.116 Proof of Result 4.2

Proof of Theorem 4.2. Fix a component X_n and a basepoint $w \in X_n$. Let R be a uniform propagation bound for $H_{\pm,n}$, let B_0 be a uniform norm bound, and let

$$W_a \delta_x = e^{ad(x,w)} \delta_x$$

for $a > 0$. The conjugated operator $H_a = W_a H_{\pm,n} W_a^{-1}$ has the same matrix support as $H_{\pm,n}$, and each matrix coefficient changes by a factor between e^{-aR} and e^{aR}. The Schur-test constants from bounded geometry therefore give

$$\|H_a - H_{\pm,n}\| \le C_0(e^{aR} - 1)$$

with C_0 independent of n. Choose $a > 0$ so that this norm is at most $g_{\mathrm{HP}}/4$.

Let Γ be a fixed contour enclosing the negative spectral window and staying at distance at least $g_{\mathrm{HP}}/2$ from $\mathrm{spec}(H_{\pm,n})$. The previous estimate and the resolvent identity imply that, for $z \in \Gamma$,

$$\|(H_a - z)^{-1}\| \le 4/g_{\mathrm{HP}}$$

after decreasing a if necessary. Since

$$(H_{\pm,n} - z)^{-1} = W_a^{-1}(H_a - z)^{-1} W_a,$$

we obtain, for all $v, w \in X_n$,

$$\left|\langle \delta_v, (H_{\pm,n} - z)^{-1}\delta_w \rangle\right| \le (4/g_{\mathrm{HP}}) e^{-a d(v,w)}.$$

The Riesz formula

$$P_{\pm,n} = \frac{1}{2\pi i}\int_\Gamma (z - H_{\pm,n})^{-1}\, dz$$

then gives the stated bound with $c = a$ and $C = |\Gamma|(4/g_{\mathrm{HP}})/(2\pi)$. □

6.117 Proof of Result 4.7

Proof of Theorem 4.7. The ideal property is Lemma 5.4. Hence the inclusion $\mathcal{I} \hookrightarrow \mathcal{A}$ and quotient map $\pi : \mathcal{A} \to \mathcal{A}/\mathcal{I}$ form a short exact sequence of C^*-algebras. The six-term exact sequence in K-theory for a short exact sequence of C^*-algebras gives the connecting map

$$\partial_Y : K_0(\mathcal{A}/\mathcal{I}) \to K_1(\mathcal{I})$$

with the sign convention fixed in Proposition 5.31; see, for example, Blackadar's account of C^*-algebra K-theory [19]. The coarse geometric meaning of the same extension is Roe's interface ideal construction [35]. □

6.118 Proof of Result 4.3

Proof of Theorem 4.3. By Definition 4.8, $h(J^\infty_{\pm,M})$ is a self-adjoint lift of the quotient projection p_J. The exponential formula for the even-to-odd connecting map gives

$$\partial_Y([p_J]) = \left[\exp(2\pi i\, h(J^\infty_{\pm,M}))\right].$$

Because $\pi(h(J^\infty_{\pm,M})) = p_J$, applying π to the exponential gives $\exp(2\pi i p_J) = \mathbf{1}$; hence the exponential differs from the identity by an element of $\mathcal{I}$. If h_0 and h_1 are two admissible cutoffs, the straight-line path $(1-t)h_0(J^\infty_{\pm,M}) + t h_1(J^\infty_{\pm,M})$ consists of self-adjoint lifts of the same projection p_J, because their quotient images are both p_J. Exponentiating gives a homotopy in $\mathbf{1} + \mathcal{I}$, so the K_1-class is independent of the cutoff. □

6.119 Computational details for Theorem 4.9

The body proof of Theorem 4.9 is given immediately after the theorem statement in Section 4; see the four-step *Body proof of Theorem 4.9*. This appendix subsection retains the explicit computational verifications referenced from the four body steps. It is intended for the reader who wishes to inspect (i) the cutoff-functional-calculus identity used to lift the quotient projection to a self-adjoint element of $\mathcal{A}$, (ii) the explicit quotient identification of the comparison-operator pair at finite jam scale, and (iii) the homotopy verification underlying the admissible-cutoff invariance.

Computational verifications for Theorem 4.9. Inputs. Lemma 4.4 gives the Higson–Prodan closed-doubled-bulk gap, and Lemma 4.5 gives $\mathrm{Ch}([P_\pm]) = \pm 1$. Lemma 4.6 places $P_\pm$ in $C^*(X)$. These are the inputs to the body proof's Step (3) and Definition 4.8, recorded here for ease of cross-reference.

Quotient identification of the comparison-operator pair. The finite-M sidewise realization is the one already constructed in Theorem 5.35. Its proof uses only collar locality, the quotient splitting, and the positivity of the auxiliary copy-side model. Specifically, for $M > B_{\mathrm{copy},-}$, the negative comparison operator $\widetilde{T}^-_{\pm,M}$ is strictly positive by Proposition 5.23, so its negative Fermi projection is 0. On the positive side, the comparison operator is $T^+ = H^\infty_\pm$ with Fermi projection $P_\pm$. The quotient-side functional calculus of Proposition 5.28 therefore gives

$$p_J = E_+(\pi(P_\pm)) + E_-(\pi(0)) = E_+(\pi(P_\pm)).$$

This is the explicit form of the quotient identification used in the body proof's Step (4).

Cutoff-functional-calculus identity. The same proposition gives $p_J = \pi(h(J^\infty_{\pm,M}))$ for every admissible cutoff h. Lemma 4.3 then identifies the Roe boundary map with the cutoff unitary:

$$\partial_Y([p_J]) = [\exp(2\pi i\, h(J^\infty_{\pm,M}))].$$

This is exactly the interface class $\mathrm{Ind}_{\mathrm{int}}(J^\infty_{\pm,M}; T^+, T^-)$ of Definition 5.33, and is the cutoff-unitary representative used in the body proof's Step (1).

Ewert–Meyer compatibility for the original/copy partition. The original/copy decomposition is coarsely excisive by Proposition 5.25, so it is represented by the same Mayer–Vietoris extension

$$0 \to \mathcal{I} \to \mathcal{A} \to \mathcal{A}/\mathcal{I} \to 0$$

used to define ∂_Y; see also the top horizontal row of Figure 1. Ewert–Meyer's coarse bulk–interface construction pairs the boundary class of a quotient bulk projection with the interface cocycle for this partition [29]. In the present doubled-jammed model, the quotient splitting

$$\mathcal{A}/\mathcal{I} = E_+(\mathcal{A}/\mathcal{I}) \oplus E_-(\mathcal{A}/\mathcal{I})$$

of Proposition 5.27 identifies the paired class as the Chern jump between the positive and negative comparison projections. The positive projection is $P_\pm$, and the negative projection is 0. Therefore

$$\langle \mathrm{EM}_Y, [p_J] \rangle = \mathrm{Ch}([P_\pm]) - \mathrm{Ch}([0]) = \pm 1.$$

This is the detailed Ewert–Meyer/Roe identification used in the body proof's Step (2); the abstract naturality argument given in the body uses only that the same six-term exact sequence underlies both boundary maps with the orientation choice fixed in Convention (C3).

Cutoff homotopy and M-invariance. For two admissible cutoffs h_0, h_1 and the straight-line interpolation $(1-t)h_0(J^\infty_{\pm,M}) + t h_1(J^\infty_{\pm,M})$, both endpoints are self-adjoint lifts of p_J. Their quotient images coincide, so the difference is a self-adjoint element of $\mathcal{I}$; the homotopy $t \mapsto \exp(2\pi i((1-t)h_0 + t h_1)(J^\infty_{\pm,M}))$ takes values in $\mathbf{1} + \mathcal{I}$ and gives a continuous path of unitaries representing the same K_1-class. Independence from M along admissible jam scales is Corollary 6.38; this completes the K_1-limit step of the body proof's Step (4). □

6.120 Proof of Result 4.10

Proof of Theorem 4.10. Suppose $Y \subset X$ is the straight periodic interface of the periodic triangular class with primitive tangential/positive-normal basis (t, n). By Theorem 4.9 the interface class is $\partial_Y(E_+(\pi(P_\pm))) \in K_1(\mathcal{I})$. Compressing the n-direction to a half-space realizes $\mathcal{I}$ as the kernel of the canonical Toeplitz extension recalled in Theorem 5.7, whose connecting map $\partial_{\mathrm{T},t,n} : K_0(C(\mathbb{T}^2)) \to K_1(C(S^1)) \cong \mathbb{Z}$ satisfies $\mathrm{wind}(\partial_{\mathrm{T},t,n}([P])) = \varepsilon(t,n)\ \mathrm{Ch}(P)$. For the original/copy partition the Fermi projection P_+ of the positive-side closed-bulk operator $H_\pm^{\triangle}$ is gapped by Theorem 5.2, and the trivial-jam statement (Proposition 5.23) gives $[P_-] = [0]$. The Chern signs $\mathrm{Ch}(E_-(H_\pm^{\triangle})) = \mp 1$ of Theorem 5.5 therefore determine the integer

$$\mathrm{wind}\big(\partial_Y(E_+(\pi(P_\pm)))\big) = \varepsilon(t,n)\ \mathrm{Ch}(E_-(H_\pm^{\triangle})) = \mp\,\varepsilon(t,n) = \pm 1.$$

Every step is internal to this paper: the Toeplitz identification of $K_1(\mathcal{I})$ with $\mathbb{Z}$ and the integer pairing are recalled in Theorem 5.7. □

6.121 Proof of Result 4.11

Proof of Theorem 4.11. Assume an integer-valued Chern-jump-compatible pairing $\langle \cdot, \cdot \rangle : K_0(\mathcal{A}/\mathcal{I}) \times K_1(\mathcal{I}) \to \mathbb{Z}$ exists, in the sense made precise in the statement. By Theorem 4.9 the class $\partial_Y([p_J])$ coincides with the K_1-class $[\exp(2\pi i\, h(J^{\infty}_{\pm,M}))]$. The hypothesized pairing then evaluates $\langle [p_J], \partial_Y([p_J]) \rangle = \mathrm{Ch}([P_+]) - \mathrm{Ch}([P_-])$. The trivial-jam statement (Proposition 5.23) gives $[P_-] = [0]$, and the closed-bulk Chern part of Higson–Prodan [8, 9] gives $\mathrm{Ch}([P_+]) = \pm 1$. Hence the pairing yields ± 1, conditional on the existence of the integer-valued Chern-jump pairing on the general partition. In the periodic triangular Toeplitz case the pairing exists internally (Corollary 4.10); for general partitions the construction of the integer pairing is not carried out in this paper and is flagged in the corollary statement. □

6.122 Proof of Result 4.12

Proof of Theorem 4.12. For $M > B_{\mathrm{copy},-}$ the quotient projection $p_J = E_+(\pi(P_\pm))$ is independent of M by Proposition 5.28: the jam scale only enters through the copy-side block, whose negative spectrum is empty above the trivial-jam threshold and whose quotient image is trivial. The connecting class $\partial_Y([p_J]) \in K_1(\mathcal{I})$ depends only on the quotient projection, hence is independent of M. Cutoff-homotopy independence is the statement of Corollary 6.38. □

6.123 Proof of Result 4.13

Proof of Theorem 4.13. Because $V \in \mathcal{I} = C^*(Y \subset X)$, the quotient image of the perturbed operator is the same as the quotient image of the original operator. Thus the quotient projection p_J is unchanged. The perturbation is not required to be small in norm; the only spectral requirement is that the comparison gaps used to choose the sidewise Fermi cutoffs remain open. With those gaps open, the same cutoff-functional-calculus argument applies and the K_1-class $\partial_Y([p_J])$ is unchanged. In the straight periodic Toeplitz case the integer winding (Theorem 4.10) is a K_1-homomorphism of this class, hence is unchanged as well. □

6.124 Proof of Result 4.14

Proof of Theorem 4.14. Let H_s, $0 \le s \le 1$, be the deformed closed doubled Hamiltonian path. The preserved uniform gap gives one continuous cutoff f with $P_s = f(H_s)$. Hence $s \mapsto P_s$ is a

norm-continuous path of projections in $C^*(X)$. The quotient projections

$$E_+(\pi(P_s))$$

form a norm-continuous path in $\mathcal{A}/\mathcal{I}$, so their K_0-class is constant. Applying ∂_Y gives a constant interface K_1-class. Any K_1-homomorphism (in particular the Toeplitz integer pairing in the straight periodic case, Theorem 4.10) takes the same value at the two endpoints. □

6.125 Proof of Result 4.15

Proof of Theorem 4.15. The Higson–Prodan gap bound is uniform along the refinement family, and $M_n > B_{\mathrm{copy},-} + 1$ gives a uniform positive gap for the auxiliary negative comparison operators. Passing to a cofinal subsequence changes the coarse disjoint union only by deleting or coarsely reindexing components. Deleting finitely many components produces an operator supported on a finite subset of X, hence an element of $C^*(Y \subset X)$ by Proposition 6.33. Cofinal reindexing gives the same quotient projection and therefore the same boundary class. Since $M_n \to \infty$, the finite-M_n correction terms vanish in the Schur-complement estimates, so the limiting interface class is eventually constant. □

6.126 Proof of Result 4.16

Proof of Theorem 4.16. For the periodic triangular model, Theorem 5.2 gives the closed zero-gap input and Theorem 5.5 gives the Chern signs directly. Theorem 4.9 uses the same positive closed-bulk Fermi projection and the same trivial negative comparison projection. When the interface is primitive and straight, Theorem 5.7 identifies the Ewert–Meyer interface cocycle with the Toeplitz winding. Thus Theorem 4.9 and the internal periodic calculation evaluate the same K_1-class and give the same sign. □

6.127 Proof of Result 4.17

Proof of Theorem 4.17. Fixed-collar defects are elements of the interface ideal, so Lemma 4.13 applies. Defects supported in finitely many coarse components are also in the interface ideal by Proposition 6.33. Therefore the quotient projection p_J and the K_1-class $\partial_Y(E_+(\pi(P_\pm)))$ agree with the finite-defect robustness statements; in the straight periodic Toeplitz case the integer winding also agrees via Theorem 4.10. □

6.128 Proof of Result 4.18

Proof of Theorem 4.18. A gapped homotopy between a hexagonal or square appendix symbol and the corresponding closed-bulk class gives a norm-continuous path of Fermi projections. Lemma 4.14 keeps the K_1-class of Theorem 4.9 fixed along that path. The endpoint Chern sign is the one computed in Lemmas 6.5, 6.6, and 6.12, with the relevant orientation convention. In the straight periodic Toeplitz case the integer winding agrees with the appendix benchmark via Theorem 4.10. □

6.129 Proof of Result 4.19

Proof of Theorem 4.19. The tail-split and annular constructions decompose the interface ideal into the same component quotient coordinates used by the Toeplitz maps in Appendix 6.7. The interface class of Theorem 4.9 is $\partial_Y(E_+(\pi(P_\pm)))$. Projecting this class to a component quotient commutes

with the boundary map because both operations are induced by quotient homomorphisms of the same Roe extension. Each component therefore receives exactly the Toeplitz K_1-coordinate used in Theorem 6.29. □

7 Numerical experiments and reproducibility

The numerical experiments record the finite matrices used to test the mechanisms in the paper: doubled bulk gaps on sampled meshes, the intrinsic Poincaré–Lefschetz/compression comparison, boundary localization after jamming, finite-size flux insertion, chirality reversal, collar support, and the M^{-1}/M^{-2} decoupling laws. The HP theorem or the periodic Bloch calculation supplies the closed-bulk gap in the theorems; the numerics do not define separate K-theoretic indices for individual boundary components. The localization, transport, flux, disorder, and refinement-scaling figures are included to document finite-matrix behavior. They are checks of the mechanisms proved above, not inputs to the proofs.

Pipeline and examples. The companion code generates boundary triangulations from a subdivided icosphere, verifies the doubling identities, assembles $B, S, H_{\pm}(D(\mathcal{K}))$, forms $H_{\pm,M}$, and evaluates the localization, transport, and decoupling quantities below. The reported computations correspond to the linked repository snapshot `1c888c900cdbbfff078f2b55b445ee01f2b4bd17`, accessed 16 May 2026.

Table 3: Mesh sizes for the three boundary triangulations $\mathcal{K}$ and their doubles $D(\mathcal{K})$. The counts agree with the identities in Theorem 2.5.

ex	$\#\partial$	$\lvert\partial\mathcal{K}_0\rvert$	$\lvert\partial\mathcal{K}_1\rvert$	$\lvert\mathcal{K}_0\rvert$	$\lvert\mathcal{K}_1\rvert$	$\lvert\mathcal{K}_2\rvert$	$\dim\mathcal{H}(\mathcal{K})$	$\lvert D_0\rvert$	$\lvert D_1\rvert$	$\lvert D_2\rvert$	$\dim\mathcal{H}(D)$	g
ex1	1	22	22	73	194	122	389	124	366	244	734	0.5666
ex2	2	88	88	272	728	456	1456	456	1368	912	2736	0.3380
ex3	3	78	78	329	912	582	1823	580	1746	1164	3490	0.3272

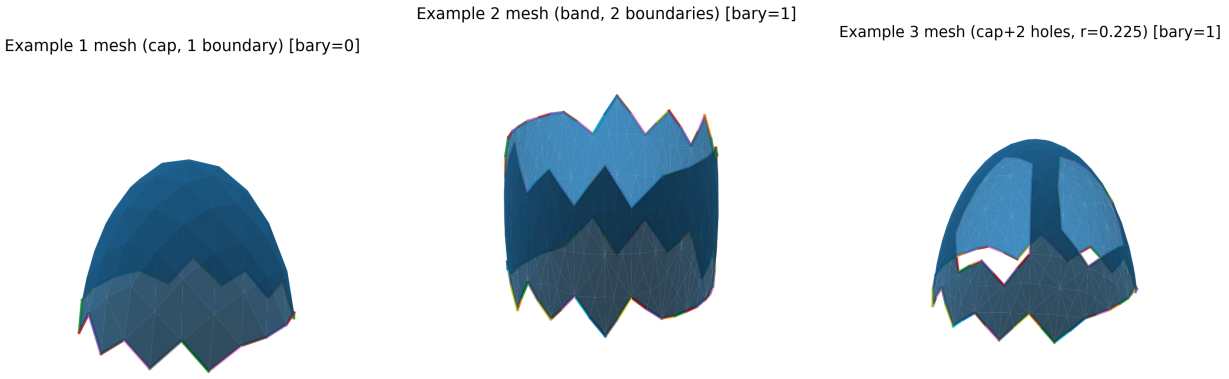
Table 4: Selected numerical outputs: the measured doubled bulk half-gap g for $H_+(D(\mathcal{K}))$, the number of jammed eigenvalues selected by $|E| < g$ at $M = 50$, and fitted drift slopes when reported.

ex	g	# in-gap ($\lvert E\rvert < g$)	v_+	v_-
ex1	0.5666	5	-0.890	0.893
ex2	0.3380	14	–	–
ex3	0.3272	12	–	–

Table 5: Finite-M checks. The measured slopes are fitted from the log-log data generated by the companion script.

Quantity	Theoretical scale	Observed check
Compressed resolvent error	M^{-1}	slope -0.943
First-order resolvent error	M^{-2}	slope -1.942
Mean squared copy leakage	M^{-2}	slope -1.965
First-order eigenvalue localization	M^{-2}	slope -1.901
Contour margin $\delta_H(\Gamma)$	positive	positive on sampled M-range
Window separation $g_M(E_{\max})$	positive for large M	linear growth after threshold
Original–copy coupling support	bounded collar	all nonzero entries inside measured collar

Meshes, doubled gaps, and jammed spectra. Figure 1 shows the boundary triangulations and Figure 2 shows the sorted spectra of $H_+(D(\mathcal{K}))$. In the displayed finite matrices all three doubled models have a positive measured gap at zero. Figure 3 then shows the spectra after adding $M\mathbf{P}_{\text{copy}}$ with $M = 50$. The high-energy cluster is copy-supported; the low-energy levels inside $|E| < g$ are the finite-size interface candidates.

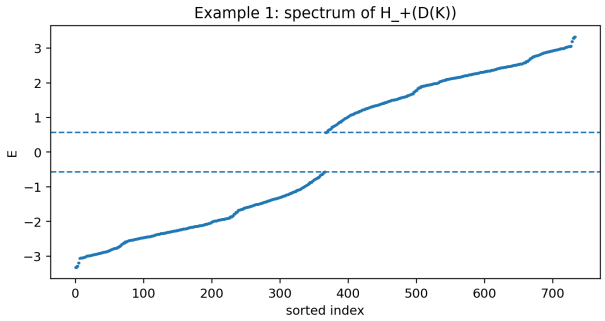


(a) Disk-like patch (1 boundary).

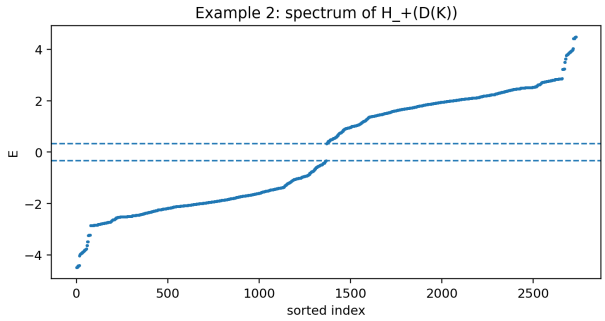
(b) Annulus-like band (2 boundaries).

(c) Three-boundary patch (disk with two holes).

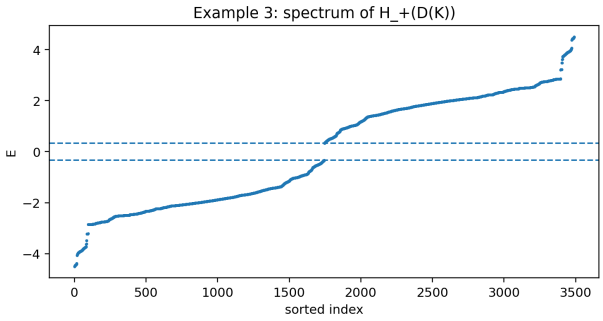
Figure 1: Boundary triangulations used in the experiments. Boundary edges are highlighted.



(a) Example 1.

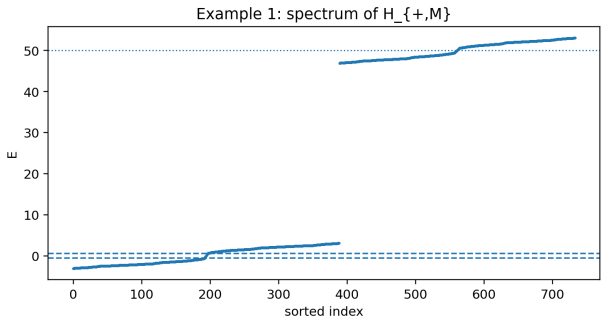


(b) Example 2.

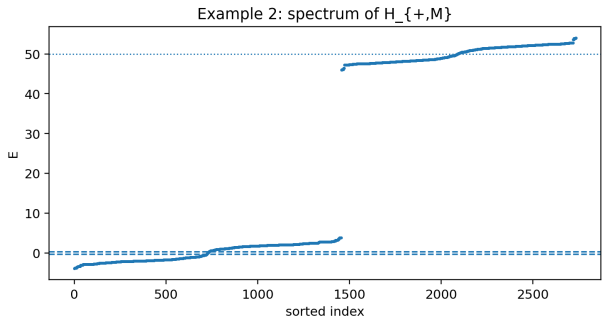


(c) Example 3.

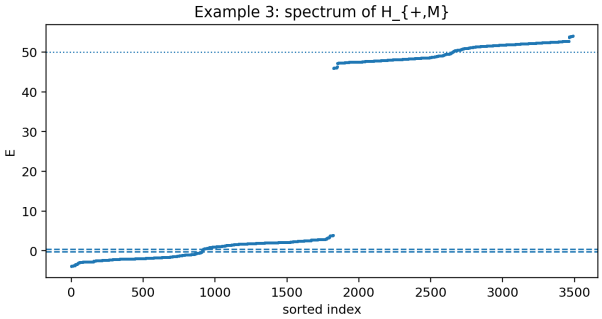
Figure 2: Sorted spectra of $H_+(D(\mathcal{K}))$. Dashed lines mark the measured half-gap $\pm g$, where $g = \min |\mathrm{spec}(H_+(D(\mathcal{K})))|$.



(a) Example 1.



(b) Example 2.



(c) Example 3.

Figure 3: Spectra of $H_{+,M}$ at $M = 50$. Dashed lines indicate the measured doubled bulk gap scale inherited from $H_+(D(\mathcal{K}))$.

Boundary localization. Figure 4 plots the boundary weight $w_\partial(\psi) = \|\mathbf{P}_\partial \psi\|^2$ for Example 1. Figure 5 repeats this check component by component for the two- and three-boundary examples. The in-gap states have enhanced boundary weight, while the high-energy copy cluster has negligible original-boundary weight.

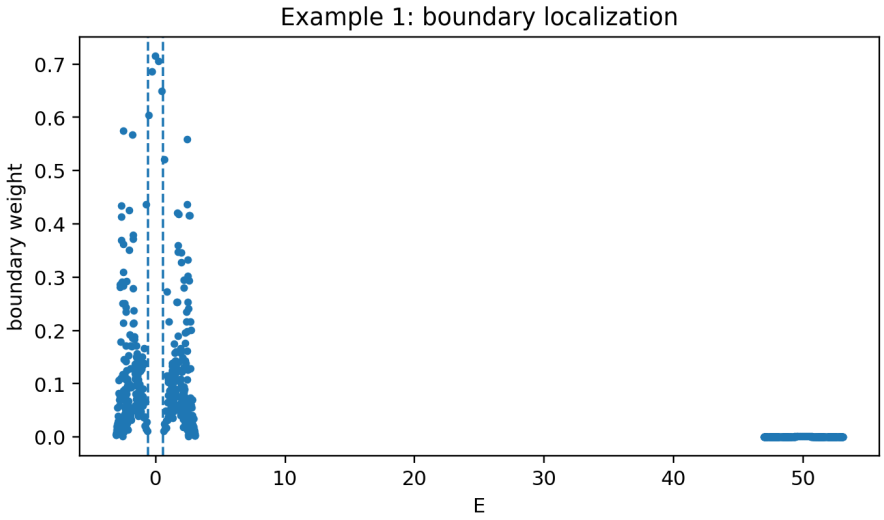


Figure 4: Example 1: boundary weight versus eigenvalue. States selected by $|E| < g$ have enhanced boundary weight.

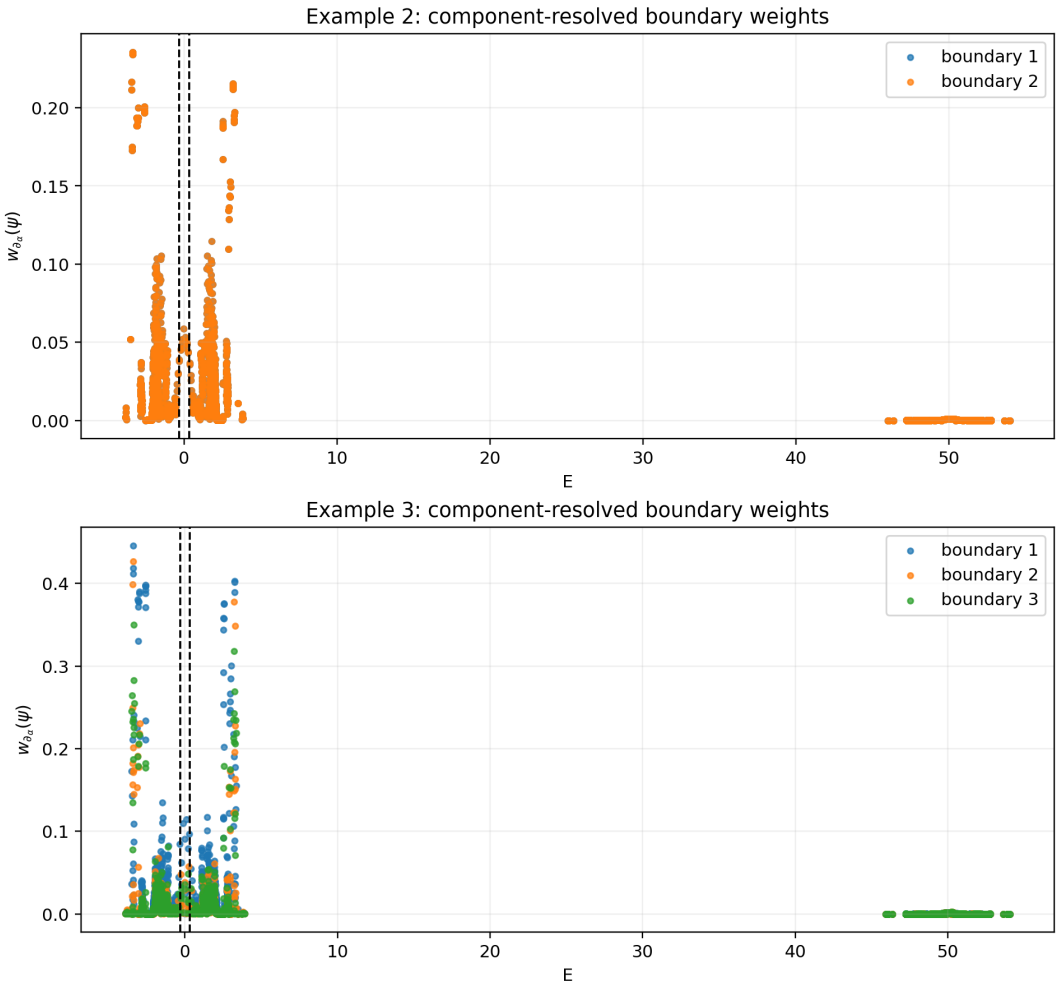


Figure 5: Examples 2 and 3: component-resolved boundary weights. The dashed vertical lines mark the finite-size gap window $|E| < g$.

Intrinsic boundary comparison. Figure 6 and table 6 compare the intrinsic Poincaré–Lefschetz boundary Hamiltonian $H_+^{\mathrm{PL}}(\mathcal{K})$, the doubled original-side compression $H_+^{\partial}(\mathcal{K})$, and the finite-$M$ jammed model. This is the numerical check most directly tied to Theorem 2.8: the discrepancy $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\mathrm{PL}}(\mathcal{K})$ is visible but confined to the boundary collar, and the jammed low-energy spectrum tracks the compressed boundary model.

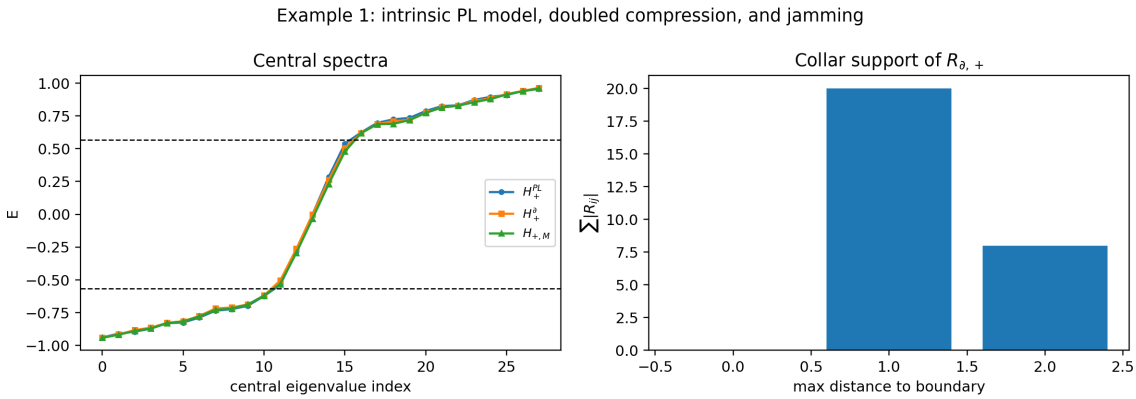
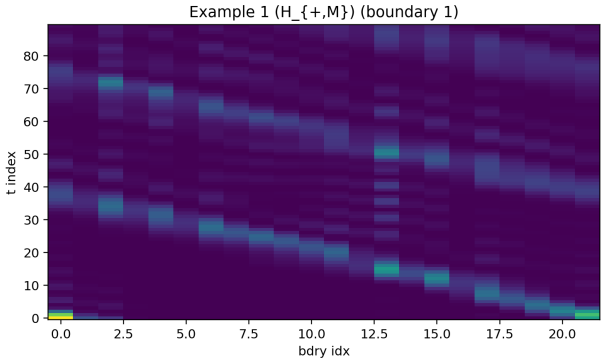


Figure 6: Example 1: central spectra of $H_+^{\mathrm{PL}}(\mathcal{K})$, $H_+^{\partial}(\mathcal{K})$, and $H_{+,M}$, together with the measured support distribution of the collar correction $R_{\partial,+}$. The correction is not norm-small, but it is boundary-collar supported.

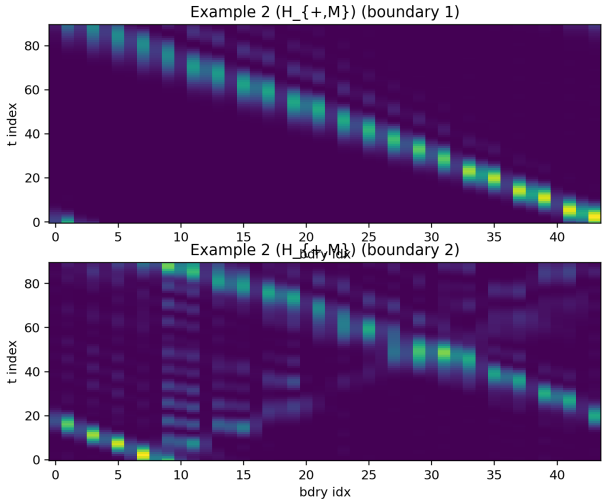
Table 6: Example 1: numerical size and support of $R_{\partial,+} = H_+^{\partial}(\mathcal{K}) - H_+^{\mathrm{PL}}(\mathcal{K})$.

example	$\dim \mathcal{H}(\mathcal{K})$	$\operatorname{rank} R$	support dim.	support radius	$\|R\|$	$\|R\|_F$
ex1	389	32	44	2	1.0000	3.7417

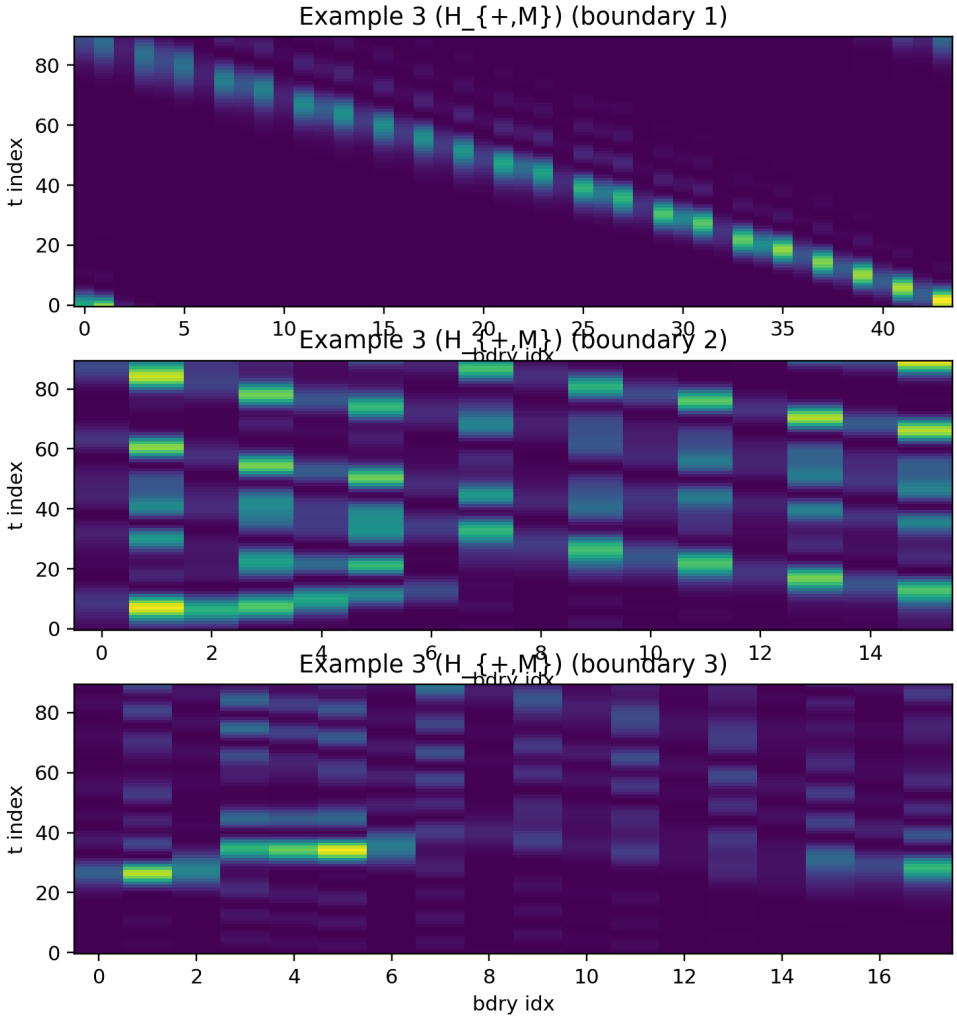
Boundary transport and chirality. We evolve a wavepacket projected to the finite in-gap subspace and order the boundary vertices cyclically along each component. Diagonal ridges in Figure 7 show one-way boundary motion. Figure 8 and Figure 9 show that the signed drift reverses under $H_{+,M} \leftrightarrow H_{-,M}$. These are finite-matrix transport checks; the global K_1-class remains the one in Theorem 4.9.



Example 1.



Example 2.



Example 3.

Figure 7: Boundary transport heatmaps for wavepackets projected to the finite-size in-gap subspace.

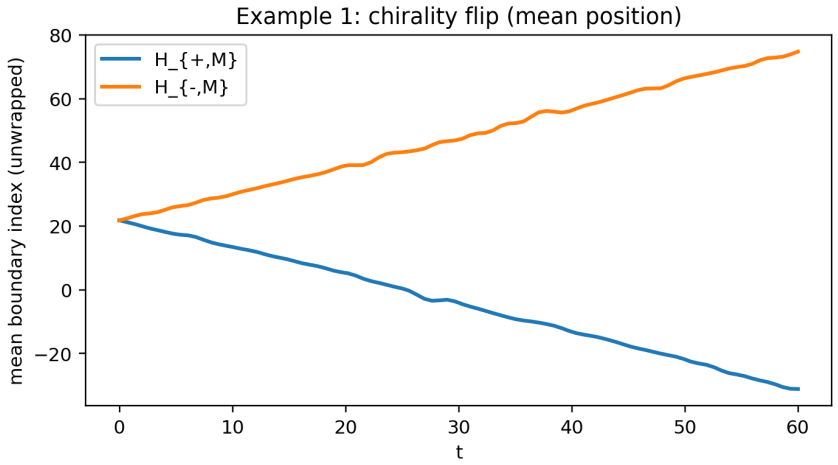


Figure 8: Example 1: opposite signed drift for $H_{+,M}$ and $H_{-,M}$.

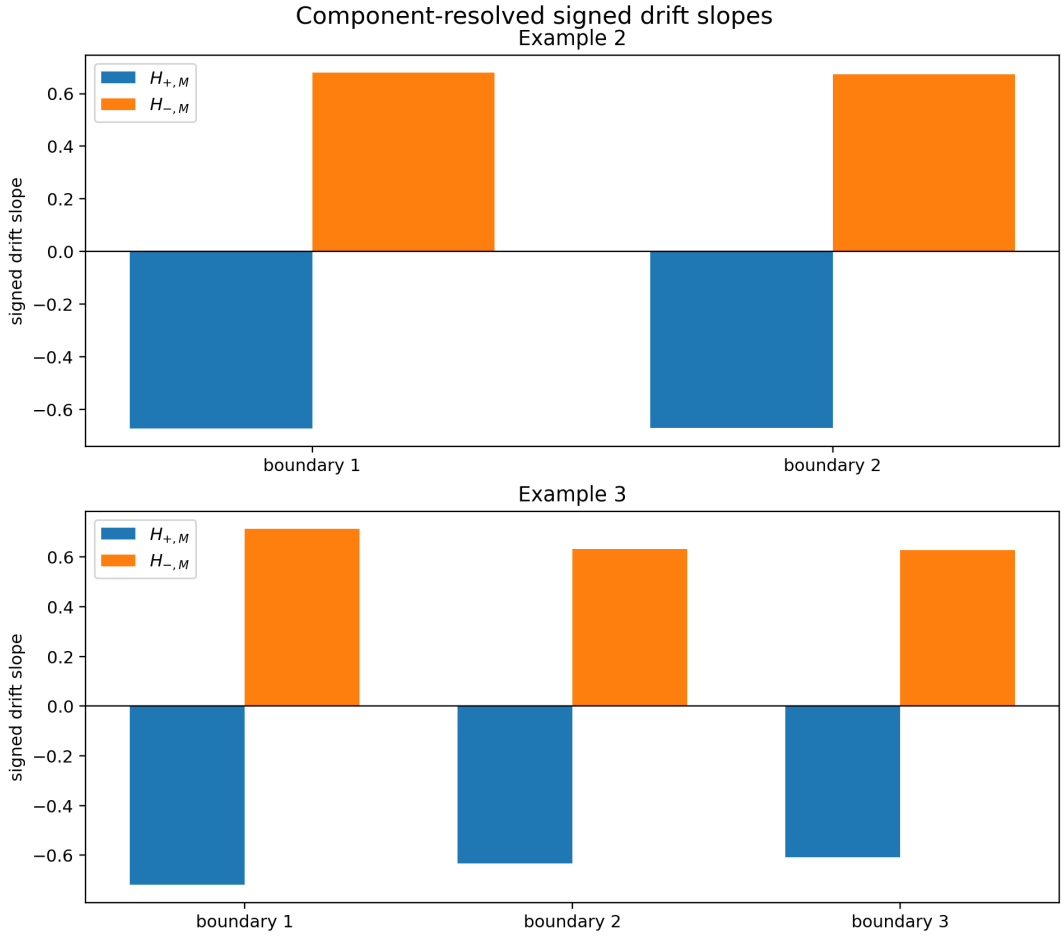


Figure 9: Component-resolved signed drift slopes for the two- and three-boundary examples. The sign reverses under $H_{+,M} \leftrightarrow H_{-,M}$ on every sampled boundary component.

Flux insertion. For the annular example we also insert a $U(1)$ twist across a fixed angular cut

and recompute near-zero eigenvalues for $\theta \in [0, 2\pi]$. Figure 10 is a finite-size spectral-flow check for the same interface class appearing in Theorem 4.9: edge branches move through the bulk gap under flux, with the sign-changing events summarized in Table 7. This figure is evidence for the interface mechanism, not a replacement for the coarse bulk–interface pairing theorem.

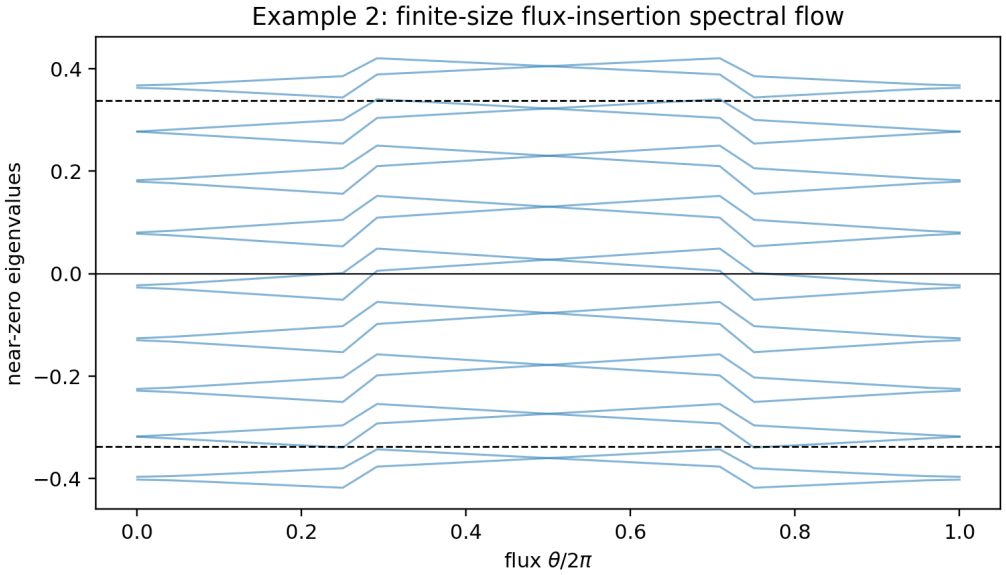
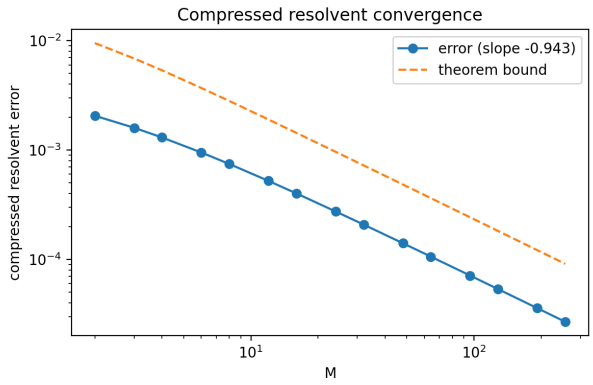


Figure 10: Example 2: finite-size flux-insertion spectral flow for $H_{+,M}$. The curves are the computed near-zero eigenvalues as the twist angle runs from 0 to 2π.

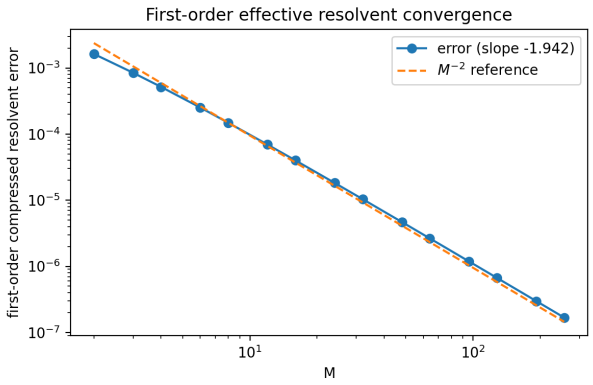
Table 7: Example 2: flux-insertion check. Sign changes count crossings between consecutive sampled flux values among the near-zero eigenvalues computed by sparse shift-invert solves.

example	flux samples	eigenvalues/sample	sign changes	min $\lvert E\rvert$
ex2	25	18	4	0.0011

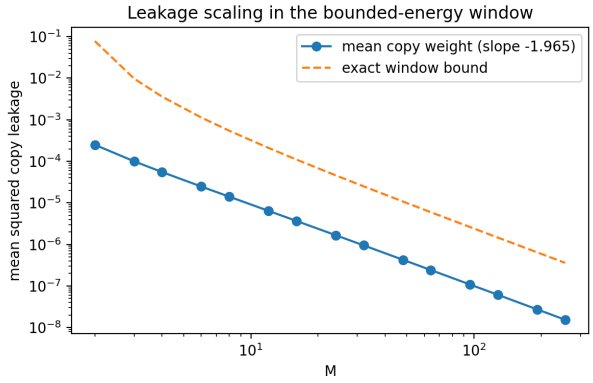
Finite-M decoupling checks. The next figures test the quantitative estimates in Section 3. The compressed resolvent error follows the predicted M^{-1} behavior from Theorem 3.2. The first-order resolvent correction, squared copy leakage, and first-order eigenvalue localization follow the predicted M^{-2} behavior from Theorems 3.5, 3.10, and 3.9.



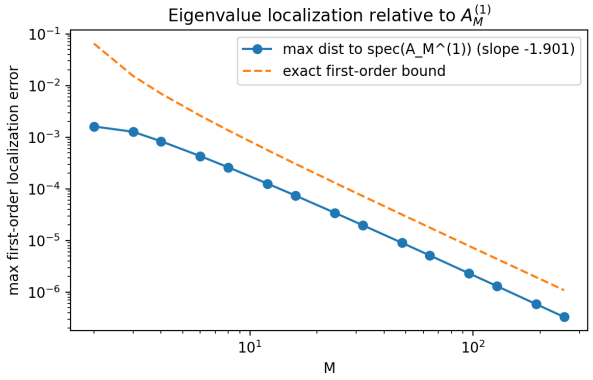
(a) Compressed resolvent.



(b) First-order resolvent.

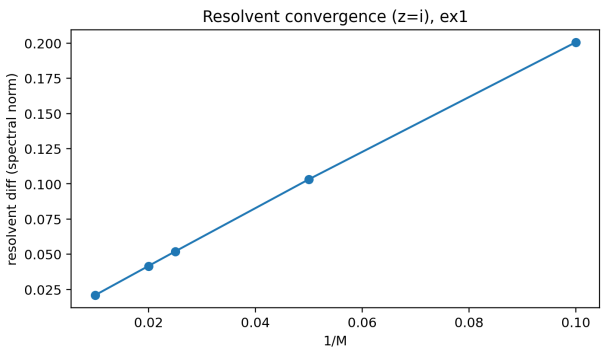


(c) Copy leakage.

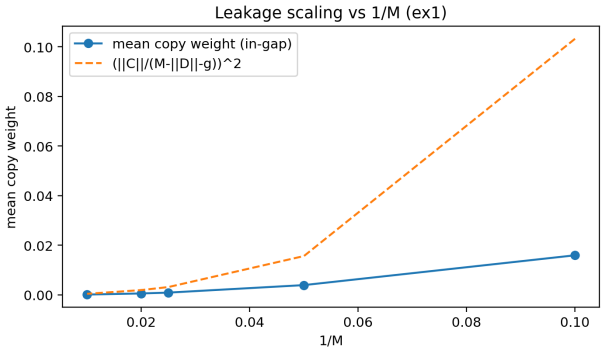


(d) First-order eigenvalue localization.

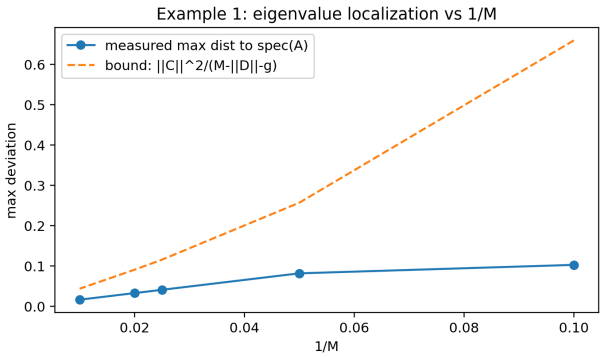
Figure 11: Log-log finite-M checks. The fitted slopes are reported in Table 5.



(a) Compressed resolvent.



(b) Copy leakage.



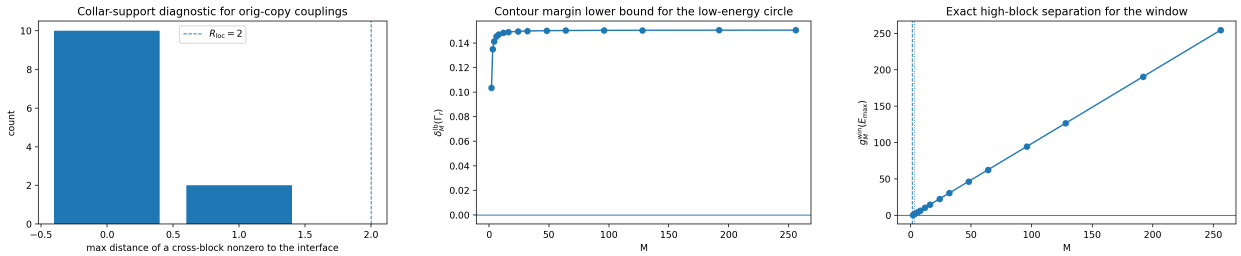
(c) Eigenvalue localization.

Figure 12: Example 1 finite-M decoupling quantities with explicit comparison bounds.

Table 8: Example 1: numerical eigenvalue-localization summary for several M values.

M	$1/M$	# in-gap	$\max_{\lvert E\rvert<g} \operatorname{dist}(E, \operatorname{spec}(A))$	bound
10.0	0.1000	4	0.1027	0.6591
20.0	0.0500	4	0.0817	0.2568
40.0	0.0250	5	0.0409	0.1156
50.0	0.0200	5	0.0328	0.0907
100.0	0.0100	5	0.0164	0.0437

Collar support, contour separation, refinement, and robustness. The collar and contour checks in Figure 13 test hypotheses used by the contour projection and interface arguments. The refinement plot in Figure 14 records the measured bulk half-gap across the sampled subdivision levels; the theorem-level gap input comes from HP or from the periodic proof. Figure 15 records a separate refinement check for the relative-boundary comparison: the support fraction of $R_{\partial,+}$ decreases with subdivision and remains in a radius-two boundary collar in the sampled family. Figure 16 adds weak onsite disorder and shows that the qualitative boundary transport ridge persists. Figure 17 tests the robustness of the low-energy boundary sector under a nonconstant positive copy-side confining potential.



(a) Collar support. (b) Contour margin. (c) Window separation.

Figure 13: Numerical checks of collar support and separation quantities used in the finite-M projection estimates.

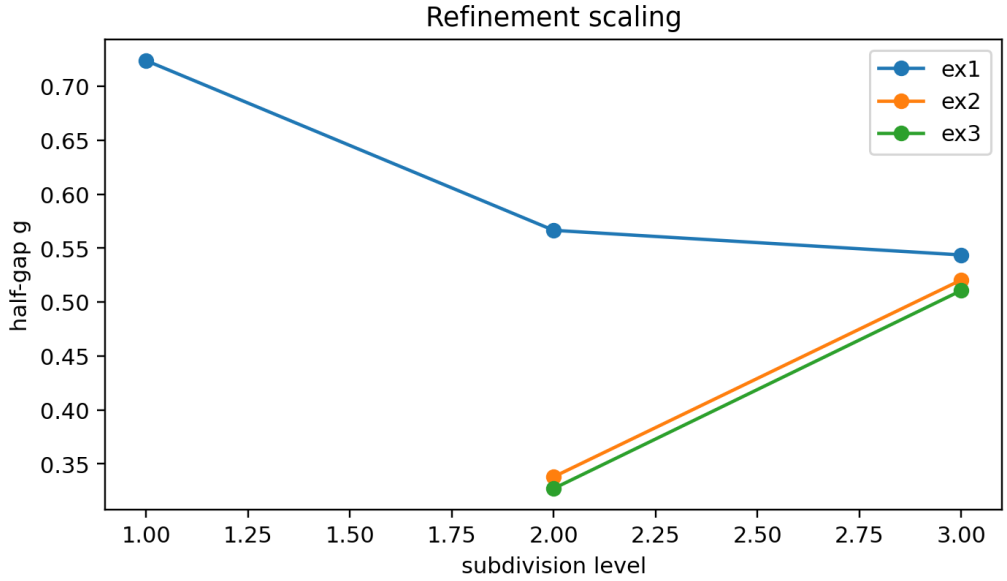


Figure 14: Measured bulk half-gap g_{gap} versus subdivision level for the three example families. This finite-mesh check documents the sampled matrices; theorem-level gap input comes from HP or from the periodic proof.

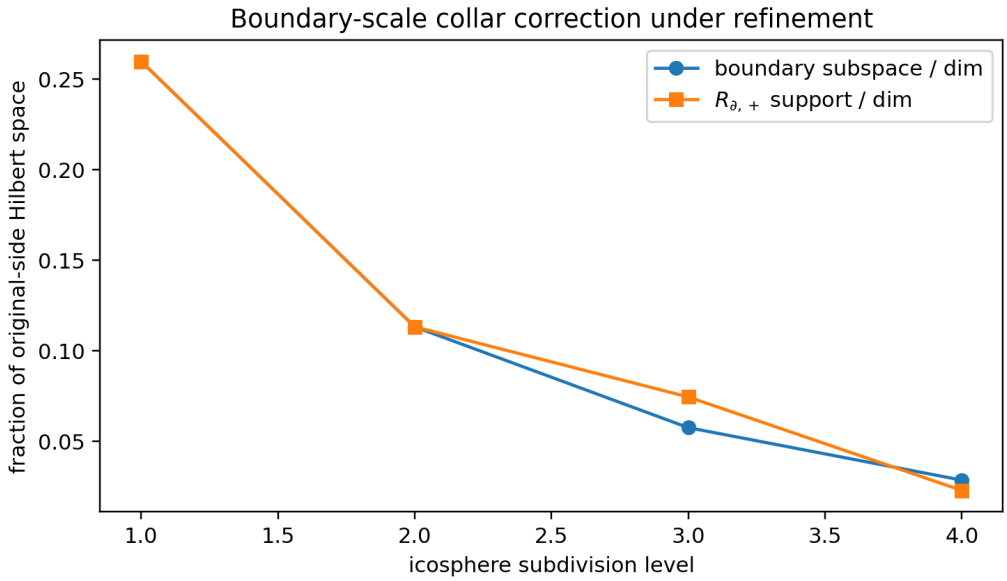


Figure 15: Example 1 refinement check for the relative-boundary comparison. The measured support fraction of the collar correction $R_{\partial,+}$ decreases with refinement, consistent with the boundary-rank interpretation in Lemma 2.9.

Table 9: Example 1: subdivision scaling of the support of $R_{\partial,+} = H^{\partial}_{+}(\mathcal{K}) - H^{\mathrm{PL}}_{+}(\mathcal{K})$.

subdiv.	bary rounds	$\dim \mathcal{H}(\mathcal{K})$	boundary dim.	support dim.	radius	support/dim
1	0	77	20	20	2	0.2597
2	0	389	44	44	2	0.1131
3	0	1533	88	114	2	0.0744
4	0	6337	180	144	2	0.0227

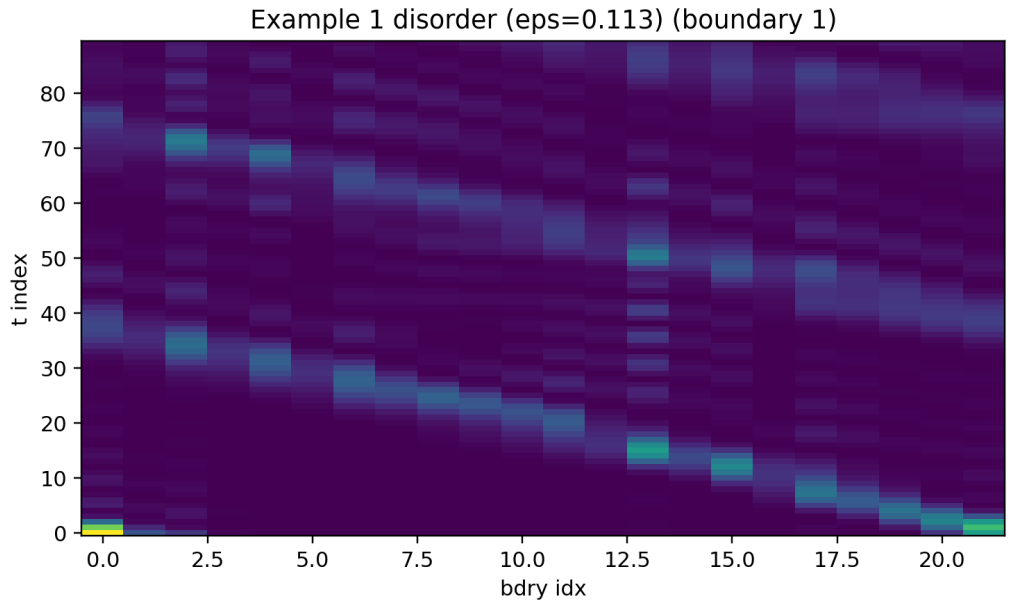


Figure 16: Example 1 with weak onsite disorder: the qualitative one-way boundary transport remains visible in the finite in-gap subspace.

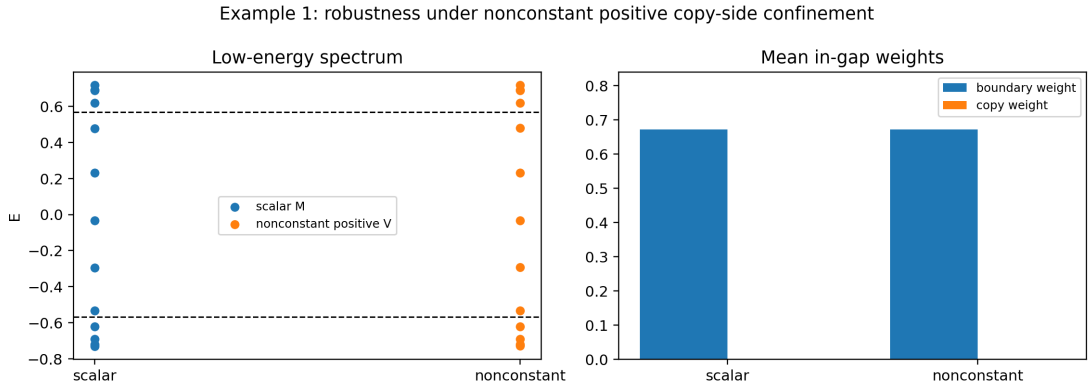


Figure 17: Example 1: scalar jamming compared with a nonconstant positive copy-side confining potential. The in-gap count and mean boundary weight are unchanged at the displayed precision, while the copy weight remains small.

Table 10: Example 1: robustness under nonconstant positive copy-side confinement. The nonconstant potential has entries in the displayed range on the copy sector.

copy potential	# in-gap	mean boundary weight	mean copy weight	potential range
scalar M	5	0.6721	5.982×10^{-4}	$[50.0, 50.0]$
nonconstant V	5	0.6722	5.942×10^{-4}	$[32.7, 67.1]$

7.1 Reproducibility appendix

The repository includes the rendered figures and summary tables. The companion code is available in the linked project repository; the computations reported here correspond to main-branch snapshot `1c888c900cdbbfff078f2b55b445ee01f2b4bd17`, accessed 16 May 2026. The repository URL, snapshot hash, and included rendered artifacts are the availability record for this manuscript.

The script fixes the random seeds used for disorder and nonconstant copy-side confinement. It also exports the finite matrices, the barycentric adjacency graph, and the boundary-collar data used above. The dependency versions used for the local regeneration check were Python 3.14.3, numpy 2.4.3, scipy 1.17.1, and matplotlib 3.10.8. Matplotlib artifacts are PNG files.

7.2 Minimal reproduction script

After installing the packages listed in `requirements.txt`, the key periodic triangular certificate tables, including Tables 1 and 2, are regenerated with this command:

```
python periodic_triangular_bulk_certificate.py --check --write-table
```

The finite-matrix figure panels and data-backed tables in Appendix 7 are regenerated with this command:

```
python run_companion_numerics_with_robustness.py
```

The static theorem-map and notation tables are generated by this LaTeX source.

The local source tree expected in the GitHub repository contains the following numerical artifacts:

```
periodic_triangular_bulk_certificate.py
paper_outputs/periodic_triangular_certificate_table.tex
paper_outputs/periodic_triangular_extension_certificate_table.tex
run_companion_numerics_with_robustness.py
requirements.txt
paper_outputs/mesh_stats_table.tex
paper_outputs/numerics_summary.tex
paper_outputs/pl_boundary_comparison_table.tex
paper_outputs/pl_collar_scaling_table.tex
paper_outputs/flux_spectral_flow_table.tex
paper_outputs/eig_localization_summary.tex
paper_outputs/copy_confinement_robustness_table.tex
paper_outputs/fig_ex1_mesh.png
paper_outputs/fig_ex2_mesh.png
paper_outputs/fig_ex3_mesh.png
paper_outputs/fig_ex1_bulk_spectrum.png
paper_outputs/fig_ex2_bulk_spectrum.png
paper_outputs/fig_ex3_bulk_spectrum.png
paper_outputs/fig_ex1_jammed_spectrum.png
paper_outputs/fig_ex2_jammed_spectrum.png
paper_outputs/fig_ex3_jammed_spectrum.png
paper_outputs/fig_ex1_boundary_localization.png
paper_outputs/fig_component_localization_ex23.png
```

paper_outputs/fig_pl_boundary_comparison.png
paper_outputs/fig_ex1_edge_propagation.png
paper_outputs/fig_ex2_edge_propagation.png
paper_outputs/fig_ex3_edge_propagation.png
paper_outputs/fig_ex1_chirality_flip.png
paper_outputs/fig_multiboundary_component_drifts.png
paper_outputs/fig_flux_spectral_flow_ex2.png
paper_outputs/fig_resolvent_loglog.png
paper_outputs/fig_first_order_resolvent_loglog.png
paper_outputs/fig_leakage_loglog.png
paper_outputs/fig_first_order_localization_loglog.png
paper_outputs/fig_resolvent_convergence.png
paper_outputs/fig_ex1_leakage_vs_M.png
paper_outputs/fig_ex1_eig_localization_vs_M.png
paper_outputs/fig_collar_support_histogram.png
paper_outputs/fig_contour_margin_lower_bound.png
paper_outputs/fig_window_separation_thresholds.png
paper_outputs/fig_refinement_scaling.png
paper_outputs/fig_pl_collar_scaling.png
paper_outputs/fig_ex1_edge_propagation_disorder.png
paper_outputs/fig_copy_confinement_robustness.png

Acknowledgments and funding

The research presented in this paper was supported by the European Research Council (ERC) under the European Union's Horizon Europe research and innovation programme (grant agreement No. 101041711), by the Simons Foundation as part of the Collaboration on the Mathematical and Scientific Foundations of Deep Learning, by Heights Labs, by Convex Nexus Capital, by the Israel Science Foundation (grant number 2258/19), and by the Israel Science Foundation (ISF Grant 4101/25). Computational resources were provided in part by Google Cloud through its education credits program (credit 530550103).

This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [36] and working together inside the Brain framework [38], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [37], the Math Brain repository [36] and the Brain framework repository [38].

Code and data availability

The code and data for this paper are at https://github.com/yspennstate/Doubling-and-Jamming-A-Universal-Chern-Hamiltonian-on-Triangulated-Surfaces-with-Boundary. The paper is part of the Math Brain project, whose public repository is https://github.com/yspennstate/math-brain-public [36].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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